Cross-fitting and learned regression for few-item benchmark prediction

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Abstract

Historical item-level responses can reduce the cost of evaluating a new language model, but representative selection, score reconstruction and model ranking are different objectives. We study a cross-fitted generalized-difference estimator of a finite binary benchmark mean. Independent residual sampling gives exact design-unbiasedness, without assuming source–target distribution agreement; this does not imply low predictive risk. We compare eighteen estimators and selection adapters on thirteen public matrices at budgets of 10, 30 and 100 items, with source-only tuning, twenty paired item draws and smaller-source and name-proxy exclusion checks. Crossfit reduces equally weighted benchmark MAE relative to random means, but exceeds random-subset score ridge by 3.030, 1.074 and 0.292 percentage points at the three budgets. At 30 items its excess is 1.074 [0.999, 1.150] points, using a conditional 95% paired interval. Ranking also depends on the metric: at this budget crossfit slightly increases Spearman correlation over random means while reducing Kendall correlation. Learned anchors and item-response adapters provide further useful but heterogeneous comparisons. The evidence favors separating an expectation guarantee from score and ranking performance; it does not establish a universally best method or transfer to future model families.

1 Introduction

Evaluating a model on every item of a benchmark can be expensive when many checkpoints or inference configurations must be compared. If historical models have already answered every item, a new target model may need to answer only a subset. The historical responses can guide which items to retain, how to reconstruct the full score, or both.

These choices should be distinguished. A representative subset can produce useful rankings while shifting absolute scores. A learned score predictor can reduce error within a familiar collection but depend on relationships that change for new models. Even an unbiased estimator can have large realized errors. The practical question is therefore how much each use of historical data improves score error and within-benchmark ranking at a fixed target-item cost, and where it fails.

This study evaluates a cross-fitted generalized-difference proposal against simple and relevant existing methods. Residual correction and source-response imputation are established prior art [Zhang et al., 2025]; we make no novelty claim for those ingredients. The candidate instead motivates a precise finite-population expectation identity and a comparison of its variance and prediction-error costs. Its imputer uses half the observed target responses in each direction. That can sacrifice predictive information even when the expectation identity is valid.

We ask three questions: whether source assistance and learned selection improve equal-cost prediction; whether cross-fitting's guarantee justifies its realized error relative to close correction and

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

regression comparators; and how conclusions change with less history, name-proxy family exclusion and response collisions. All thirteen supplied matrices and all planned conditions were evaluated. The candidate reduces macro MAE relative to random means, yet learned-score ridge has lower macro MAE and better rank correlations at every budget. Matched-imputer controls show that residual correction can help at larger budgets without overcoming the stronger learned controls. These conditional and heterogeneous findings, the explicit finite-population derivation, and reproducible comparisons are the contribution; new residual-correction ingredients and universal superiority are not claimed.

2 Related work

Anchor Points selects representatives using correlations of source-model class confidences and uses cluster-weighted confidence scores or item prediction [Vivek et al., 2024]. Binary correctness clustering is a distinct adaptation explicitly described in tinyBenchmarks [Maia Polo et al., 2024]. That work also selects items from multidimensional item-response representations, preserves observed responses in p-IRT, and blends imputed and observed estimates in gp-IRT. Selection and estimation are thus separate ingredients; a correctness-anchor baseline is not an exact reproduction of confidence-based Anchor Points.

The permitted study of Zhang et al. [2025] already compares random-subset ridge score prediction and same-sample ridge AIPW. Its AIPW fits target correctness across sampled items using source responses as features and corrects the aggregate using sampled residual information. Our comparison distinguishes its paper formula from the return expression in the supplied pinned implementation. It also distinguishes that item-regression orientation from historical-model regression, avoiding attribution of every crossfit–AIPW difference solely to splitting. Published result values are not reused as our measurements; all empirical comparisons below use the supplied matrices and delivered execution artifacts.

The bounded experiment concerns fixed reusable subsets of one binary matrix. It does not evaluate every efficient-evaluation family or establish superiority over response-adaptive, text-dependent or continuous-score approaches. The source dossier records broader permitted context and the exact versions and reading scopes; the present method discussion relies on the three directly consulted papers above.

3 Problem and methods

3.1 Finite-matrix estimand

Let $X \in \{0,1\}^{M\times N}$ contain complete historical-model responses. A new target has fixed binary responses $y_1,\dots,y_N$ and score $\mu = N^{-1}\sum_{j=1}^{N} y_j$. A method observes y_S at k distinct items and predicts μ using X and those responses. Full target outcomes are judging data, not training or tuning inputs. The complete matrices are physically available for audit; the separation is controlled estimator use, not inaccessible labels. The budget counts unique target observations per benchmark. Historical fitting costs are reported separately.

The estimand is the unweighted mean of supplied matrix columns. Without verified subject/subtask weights, this does not reconstruct an official macro-averaged benchmark or multi-benchmark leaderboard score. At $k = 100$, the observed fraction ranges from 0.83% to 22.88%; this budget is not always a tiny subset. Appendix B gives every benchmark fraction.

3.2 Simple and learned estimates

For a uniform sample S without replacement, the random mean is $\bar{y}_S$. Let $\bar{x}_j = M^{-1}\sum_m X_{mj}$ and $\bar{x} = N^{-1}\sum_j \bar{x}_j$. A fixed source-difference control is

$$\widehat{\mu}_{\mathrm{SD}} = \bar{x} + \frac{1}{k}\sum_{j\in S}(y_j - \bar{x}_j). \tag{1}$$

It is design-unbiased because $\bar{x}_j$ is fixed before sampling. Improvement over the random mean depends on the across-item variance of $y_j - \bar{x}_j$, which can increase under mismatch. The zero-

observation historical mean is a separate control; its constant predictions can have undefined ranking correlations.

Random-subset ridge regresses historical full scores on responses at S, then inserts y_S. Ridge minimizes a sum of squared residuals plus α times squared coefficients, with an unpenalized intercept and no feature standardization. Eligible $\alpha \in \{0.1, 1, 10, 100, 1000\}$ minimizes mean development MAE over five independent item draws before historical refitting. The primary predictions are not clipped. A full-sample imputation control instead regresses the historical mean on unsampled items on the selected responses, and includes observed target items exactly.

3.3 Cross-fitted generalized difference

Draw a uniformly ordered sample of k distinct items and partition it into A and B, of sizes $\lfloor k/2 \rfloor$ and $\lceil k/2 \rceil$. Conditional on A, B is a uniform sample from $U = \{1, \ldots, N\} \setminus A$. An imputer fitted using X and y_A supplies $f_j(A, y_A)$ on U. Define

$$\widehat{\mu}_{A\to B} = \frac{1}{N}\left[\sum_{j\in A} y_j + \sum_{j\in U} f_j + \frac{N-|A|}{|B|}\sum_{j\in B}(y_j - f_j)\right], \qquad \widehat{\mu}_{\mathrm{CF}} = \frac{\widehat{\mu}_{A\to B} + \widehat{\mu}_{B\to A}}{2}. \quad (2)$$

Proposition 1. *For fixed finite y, X, positive fold sizes, and an imputer independent of the residual sample conditional on its fitting fold, $\mathbb{E}[\widehat{\mu}_{\mathrm{CF}}] = \mu$ over the declared sampling design. No agreement of source and target response distributions is required.*

Proof. Conditional on A, y_A and the fixed imputer, each $j \in U$ belongs to B with probability $|B|/|U|$. The expected expanded residual sum is therefore $\sum_{j\in U}(y_j - f_j)$. Substitution in Eq. (2) leaves $N^{-1}\sum_j y_j$. The exchanged direction has the same property, so averaging preserves the expectation. □

The implementation fits multi-output historical-model ridge from responses at A to other items and inserts y_A. Linearity permits computing aggregate predictions without materializing an item-by-item output matrix. Although the optimized aggregate subtracts the fitted mean at B, it is algebraically identical to evaluating a previously fixed full imputer at a random residual sample. This computation does not fit that imputer using B.

Tuning on the same realized residual-item identities or outcomes could invalidate the proposition. Regularization is fixed using independent development draws before final sampling. A source-only choice fixed before sampling is allowed. The guarantee concerns expected signed error, not MAE, RMSE, ordering or confidence-interval coverage. Predictions may leave $[0, 1]$; clipped predictions are a separately named biased ablation. Controls remove residual correction with the same half-sample imputer (both matched and independently selected regularization), and compare full-sample same-orientation imputation.

Figure 2 summarizes the sampling and information flow; its historical matrix H is X above. Both directions reuse the same k observed target responses.

3.4 Close same-sample AIPW comparators

AIPW regresses target responses across sampled items, using source-model responses as features, whereas the candidate regresses across historical models using sampled items as features. Let m_S, m_U be the fitted means on sampled and unsampled items. The paper-formula adapter [Zhang et al., 2025, Eq. 3] and supplied pinned-code reproduction are respectively

$$\widehat{\mu}_{\mathrm{paper}} = \bar{y}_S + (1 - k/N)(m_U - m_S), \qquad \widehat{\mu}_{\mathrm{code}} = m_U + (1 - k/N)(\bar{y}_S - m_S). \quad (3)$$

The code uses fixed $\alpha = 10$ and an unpenalized intercept; the paper adapter uses development-selected regularization. The primary variants differ in regularization as well as return formula, so their empirical contrast cannot isolate the effect of the formula discrepancy. At identical fits their difference is $(k/N)(m_U - \bar{y}_S)$. The intercept makes the sampled residual mean numerically zero: the paper form equals the fitted population mean and the code form the fitted unsampled mean. Independent analysis checks compare the optimized forms with the exact pinned predict body and direct scikit-learn ridge. The word "AIPW" here identifies the prior comparator; no crossfit design guarantee is asserted for these same-sample fits.

3.5 Selections and item-response adapters

Correctness anchors cluster $X_{\cdot j}$ with k-means, select the nearest real member to each centroid and weight its observed outcome by cluster mass [Maia Polo et al., 2024]. A ridge score estimator on the same selected items separates estimation from selection. Selection uses historical responses only, with three initializations, at most 100 iterations, and frozen seeds.

MAP-MIRT fits $p_{mj} = \mathrm{sigmoid}(\theta_m^\top a_j + b_j)$, with a fixed Gaussian MAP penalty and development-selected dimension in $\{2, 5, 10, 15\}$. The p-style estimate preserves observed outcomes exactly and imputes only unobserved probabilities. The gp-style adapter blends it with the observed estimate using a development-selected mixture weight. IRT anchors cluster $(a_j, -b_j)$, retaining cluster weights. Random and IRT-anchor comparisons share the selected dimension. These are explicit MAP/inference/validation-mixture adapters, not exact tinyBenchmarks variational hierarchical fitting. Appendix C gives all optimization, continuation and mixing rules.

4 Experimental design

4.1 Audit and separation

All thirteen supplied matrices were acquired at the fixed upstream revision recorded in the reproduction instructions. The measured audit finds finite binary entries and no missing cells (Appendix A). Exact repeated OpenLLM names have identical response vectors and are collapsed before splitting. Distinct names with identical responses remain distinct in the primary evaluation. All HELM rows use opaque row IDs because matching lengths do not establish name alignment; LegalBench has fewer rows than the supplied name list.

The original development pilot's locked target membership is preserved, with deterministic seed 20261103. Approximately 30% of deduplicated rows are targets; historical rows are divided approximately 80/20 into fitting/development pools. Exact primary counts are in Table 3; row IDs and identity maps are committed. No pilot-evaluated development model becomes a final target. Item selection and all dimension, penalty and mixture choices use only historical data; eligible components are refitted on all history after development.

4.2 Matched budgets, ablations and stress conditions

An independent prospective assessment accepted the comparison and robustness scope before target execution; its decision and continuing conditions are recorded in `study/execution_gate.md`. A subsequent pre-target scoring-only amendment pairs macro rank differences on common-defined benchmark cells; it leaves estimators, tuning, partitions and item draws unchanged. The feasibility owner extended its decision to the amended scorer. The original authorization concerned the unchanged predictor scope; the formal exact amended-scorer-hash extension was published after prediction launch. Original timestamps are preserved (Appendix D). The primary protocol covers all matrices at $k \in \{10, 30, 100\}$ with twenty final draws, distinct from the five development draws. Budgets are nested within each independently drawn ordering. Uniform methods share the same draw across targets, allowing within-benchmark ranking and paired method comparisons. Source-selected anchors are fixed across final repeats; copied anchor outputs do not create independent fit uncertainty. Target predictors receive only their declared items through an observation-accounting guard. A separate scoring step computes full target truth only after every required prediction exists. The guard prevents accidental budget violations; it is not a cryptographic isolation mechanism.

Selection controls compare mean versus ridge on correctness anchors, random versus IRT anchors with the same MIRT estimator, and observed versus p-/gp-style estimation on IRT anchors. Correction controls include source difference, the uncorrected half-fit imputer, full-sample imputation and clipped crossfit. Source-size sensitivity uses nested historical fractions $.25, .5, 1$ at $k = 30$ with the original targets. Name-proxy family exclusion is restricted to verified per-benchmark OpenLLM lists: the largest eligible target proxy group is selected from names/counts alone, requiring at least ten target members and forty remaining historical rows. It compares the same target cohort with ordinary and proxy-excluded history. It does not verify ancestry. Metadata eligibility is frozen before scoring (Table 4).

After predictions are fixed, exact source–target response collisions and target-pattern-weighted errors are retrospective diagnostics. They do not determine training data or items. Full-response deduplication would change the population and can leak target information into fitting, so it is not substituted for the primary experiment.

4.3 Metrics, uncertainty and failures

Primary metrics are per-benchmark MAE/RMSE in percentage points; secondary metrics are within-benchmark Spearman and Kendall τ_b, retaining ties and explicit undefined counts. Signed error, tail error, out-of-range rates and close-pair inversions are diagnostics. True ties are distinct from the predefined one-percentage-point close-pair bin. No pooled cross-benchmark rank correlation substitutes for within-benchmark ranking. Final rank/tie calculations round truth and predictions to twelve decimals; errors retain raw predictions. This dated post-scoring repair, triggered by a violated mathematical rank invariant, is disclosed with its nonzero precision sensitivity in Appendix D.

Primary 95% paired t intervals for the mean of twenty independent item-draw differences quantify Monte Carlo uncertainty in expected performance conditional on the fixed source/target pools. Repeats are not independent new models; these intervals are not prediction intervals for a future model or draw. A prespecified sensitivity uses 5,000 paired draw-bootstrap replicates and percentile endpoints, with seed 817. Macro summaries weight the thirteen finite matrices equally within each repeat before computing intervals; undefined correlations retain their denominators. A method–reference macro contrast first takes per-benchmark paired differences where both metrics are defined, then averages; differing defined subsets are never subtracted as if they were the same population. Each contrast records the common-defined counts, and a rank summary of fewer than thirteen benchmarks is labeled as that subset. Any benchmark-bootstrap summary would be descriptive of these heterogeneous matrices. Per-benchmark outcomes and unfavorable candidate comparisons must accompany macro-averages. Numerical failures, capped fits, continuation attempts, timing and memory are retained, with no target-error-selected fallback.

5 Results

All 45 source/target scenarios completed: thirteen main matrices at three budgets, twenty-six reduced-source cases and six proxy-exclusion cases. There are 1,611,440 saved predictions in 22,360 method cells. Access/cell audits and independent full-truth, baseline and error checks passed. All 219 MIRT source fits met the declared relative-objective stopping criterion; thirteen needed the allowed continuation. This certifies optimizer stopping, not global optimality: the largest recorded gradient component is 1.250. No primary fit or inference failure was replaced by another method.

5.1 Source assistance and selection

Table 1 reports all eighteen methods at $k = 30$. Figure 1 retains all thirteen per-benchmark contrasts against four references, and Appendix F supplies the complete all-budget tables and diagnostics. At $k = 10, 30, 100$, crossfit's macro MAE is respectively 8.563, 4.836 and 2.498 percentage points (pp), compared with 10.616, 5.989 and 3.248 for random means. The paired crossfit-minus-random differences are -2.054 $[-2.353, -1.754]$, -1.153 $[-1.325, -0.981]$ and -0.750 $[-0.849, -0.651]$ pp. These and subsequent brackets are conditional 95% paired t intervals. Source difference also improves score error, but adds a common constant to every target's random mean within a draw, so its rankings are exactly the random-mean rankings.

Stronger learned comparators change the conclusion. Random-subset score ridge has macro MAE 5.533, 3.761 and 2.206 pp. Crossfit's excess is 3.030 [2.885, 3.175], 1.074 [0.999, 1.150] and 0.292 [0.242, 0.341] pp; macro RMSE is also larger at each budget. The macro result hides benchmark differences. At $k = 30$, crossfit-minus-ridge MAE ranges from -0.558 pp on HELM Math $[-0.775, -0.341]$ to $+3.831$ pp on GPQA [3.554, 4.107]. This is not a claim that ridge wins every benchmark or that a tally of unadjusted intervals establishes family-wide significance.

Selection and estimation both matter. At $k = 30$, correctness-anchor mean MAE is 4.225 pp and ridge on those same items is 3.242 pp, compared with 3.761 pp for random-subset ridge. The IRT-anchor mean, p- and gp-style adapters give 4.759, 4.553 and 3.479 pp; random p-/gp-style adapters give 4.847 and 4.084 pp. These controls keep the representation/dimension or selected items shared

Table 1: Absolute score and rank accuracy at $k = 30$: equal-weight point means over 13 benchmarks and 20 item orderings. MAE/RMSE are percentage points (lower is better); within-benchmark Spearman and Kendall correlations are higher-is-better. All 18 methods are retained. The source prior observes zero items and has undefined rank on every benchmark/draw. Paired effect displays and complete tables supply conditional item-draw intervals; these points alone do not establish a winner. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	MAE	RMSE	ρ	τ_b
Source prior (0 items)	11.13	13.78	undefined	undefined
Random mean	5.99	7.40	0.793	0.665
Source difference	5.00	6.31	0.793	0.665
Random-subset ridge	3.76	4.84	0.847	0.695
Paper AIPW (tuned)	4.54	5.80	0.811	0.664
Code AIPW (alpha 10)	4.56	5.78	0.810	0.663
Crossfit difference	4.84	6.11	0.799	0.652
Crossfit clipped	4.81	6.09	0.799	0.652
Half-fit impute (tuned)	3.81	4.83	0.848	0.695
Half-fit impute (matched)	4.97	6.26	0.856	0.706
Full-fit impute	3.72	4.77	0.849	0.697
Correctness-anchor mean	4.23	5.58	0.851	0.725
Correctness-anchor ridge	3.24	4.25	0.899	0.756
Random MIRT p	4.85	5.82	0.841	0.694
Random MIRT gp	4.08	5.13	0.839	0.693
IRT-anchor mean	4.76	5.88	0.818	0.674
IRT-anchor MIRT p	4.55	5.50	0.862	0.710
IRT-anchor MIRT gp	3.48	4.40	0.868	0.721

while changing the other ingredient. At $k = 100$, IRT-anchor gp has macro MAE 2.027 pp versus correctness-anchor ridge's 2.250 pp; the latter has higher Spearman correlation (0.943 versus 0.935). These are point comparisons of the fixed selections, not selection-seed confidence statements or a universal winner claim.

5.2 The error cost of cross-fitting

Against development-tuned paper-AIPW, crossfit-minus-reference MAE is 0.094 $[-0.189, 0.377]$ pp at $k = 10$, 0.295 [0.207, 0.384] at $k = 30$, and 0.070 [0.040, 0.100] at $k = 100$. The lowest budget gives no resolved macro MAE advantage in either direction. The item-regression orientation and regularization differ from crossfit, so these contrasts do not isolate splitting.

The matched-half-imputer control is more diagnostic. Adding crossfit's residual correction changes MAE by $+0.738$ [0.575, 0.900] pp at $k = 10$, then -0.130 $[-0.215, -0.046]$ and -0.335 $[-0.394, -0.277]$ at $k = 30, 100$. Correction helps the same imputer at the larger budgets, but the separately tuned uncorrected half-fit aggregate remains stronger: crossfit exceeds it by 2.706, 1.027 and 0.190 pp. Full-fit ridge imputation is also stronger, at 5.532, 3.720 and 2.160 pp. Regularization and using all observed items thus remain relevant explanations; design-unbiasedness alone is insufficient.

Clipping lowers crossfit's MAE by 0.112 [0.077, 0.147] pp at $k = 10$, 0.025 [0.013, 0.037] at $k = 30$, and 0.002 [0.001, 0.003] at $k = 100$. It changes the estimator and its expectation guarantee. The saved per-target twenty-draw variance/squared-mean-error decomposition is descriptive: finite Monte Carlo squared mean error is not a population-design-bias estimate.

Ranking does not follow MAE. At $k = 30$, crossfit has macro Spearman 0.799 and Kendall $\tau_b = 0.652$, versus 0.793 and 0.665 for random means. Its paired Spearman gain is 0.0058 [0.0007, 0.0109], but its Kendall change is -0.0134 $[-0.0185, -0.0083]$. Relative to ridge, both decrease: -0.0477 $[-0.0573, -0.0382]$ and -0.0428 $[-0.0498, -0.0357]$. The corresponding primary contrasts use all thirteen common-defined benchmarks and twenty draws. Constant zero-observation

source-prior predictions have undefined correlations and are retained as such. Ties, close-pair inversions and every other undefined cell remain in the full results.

An additive retrospective same-fit AIPW diagnostic, prompted by independent review after primary predictions were fixed, reconstructs paper-$\alpha 10 = (1 - k/N)\,\mathrm{code}_{\alpha 10} + (k/N)\bar{y}_S$; 55,520 actual preselected-$\alpha 10$ predictions agree with maximum discrepancy 4.44×10^{-16}. Code-minus-paper MAE at this matched fit is -0.0062 [$-0.0083, -0.0042$], -0.0022 [$-0.0061, 0.0016$] and $+0.0498$ [0.0392, 0.0604] pp. The macro MAE effect is small and budget-dependent here; it is distinct from comparing the differently tuned primary variants. Individual prediction differences reach 2.569 pp at $k = 100$, so the small macro effect does not establish individual equivalence.

5.3 Less history, proxy exclusion and response collisions

At $k = 30$, reducing historical rows to one quarter increases crossfit MAE by 0.236 [0.196, 0.276] pp and ridge MAE by 0.648 [0.527, 0.768] pp, macro-averaged over thirteen matrices. With half the history, the increases are 0.181 [0.135, 0.227] and 0.248 [0.175, 0.322] pp. Excluding the selected name-proxy family increases MAE by 0.067 [0.014, 0.121] pp for crossfit and 0.220 [0.174, 0.266] for ridge over the six eligible fixed target cohorts. Smaller degradation does not establish lower resulting error or robustness to unseen lineages. Random mean is unchanged because targets and item draws are shared.

Retrospective collisions are concentrated unevenly: OpenLLM Math has 129 target rows, 119 distinct response patterns and thirteen source-matching targets; GPQA and MMLU-Pro each have 129 rows and 127 patterns. On OpenLLM Math at $k = 30$, equal weighting of target patterns changes crossfit MAE from 2.156 to 2.281 pp and ridge MAE from 1.770 to 1.820 pp. Restricting to the 116 targets without an exact historical response match gives 2.313 and 1.839 pp. These descriptive populations do not remove related training or verify ancestry, and were not used for fitting or selection.

6 Discussion and limitations

The expectation identity separates a sampling guarantee from predictive usefulness. Learned imputation can reduce variance only when residual structure is favorable; splitting also reduces the observed information available to each fit. The completed comparisons show this distinction in practice: macro MAE improves over random means but remains above stronger learned estimators. Neither the sign of the expectation bias nor one ranking statistic substitutes for the requested score and ordering comparisons.

This retrospective study predicts frozen binary responses. It does not evaluate changes of prompt, decoding, benchmark version or scoring. Models in a collection may be related; identity deduplication does not remove shared training or ancestry. HELM identity alignment remains unresolved, and name-proxy exclusion is a bounded stress condition. Conditional item-draw uncertainty does not establish transfer to future model families. Matrix means need not match official benchmark aggregates. No new LLM evaluations or API costs are measured, so reduced observation count alone cannot establish a measured end-to-end inference saving.

Efficient evaluation could encourage wider, cheaper checks, but inaccurate reconstructed scores can mislead model selection or overstate small differences. Full evaluation remains appropriate when decisions require precision unsupported by subset estimates. This work does not establish contamination resistance, fairness across applications or deployment safety. Agent-based tools participate in literature inspection, method development, implementation, experimental planning, writing and review in this research workflow; their scientific contributions require the same evidence checks as the rest of the study. No human-subject experiment or new model training is performed.

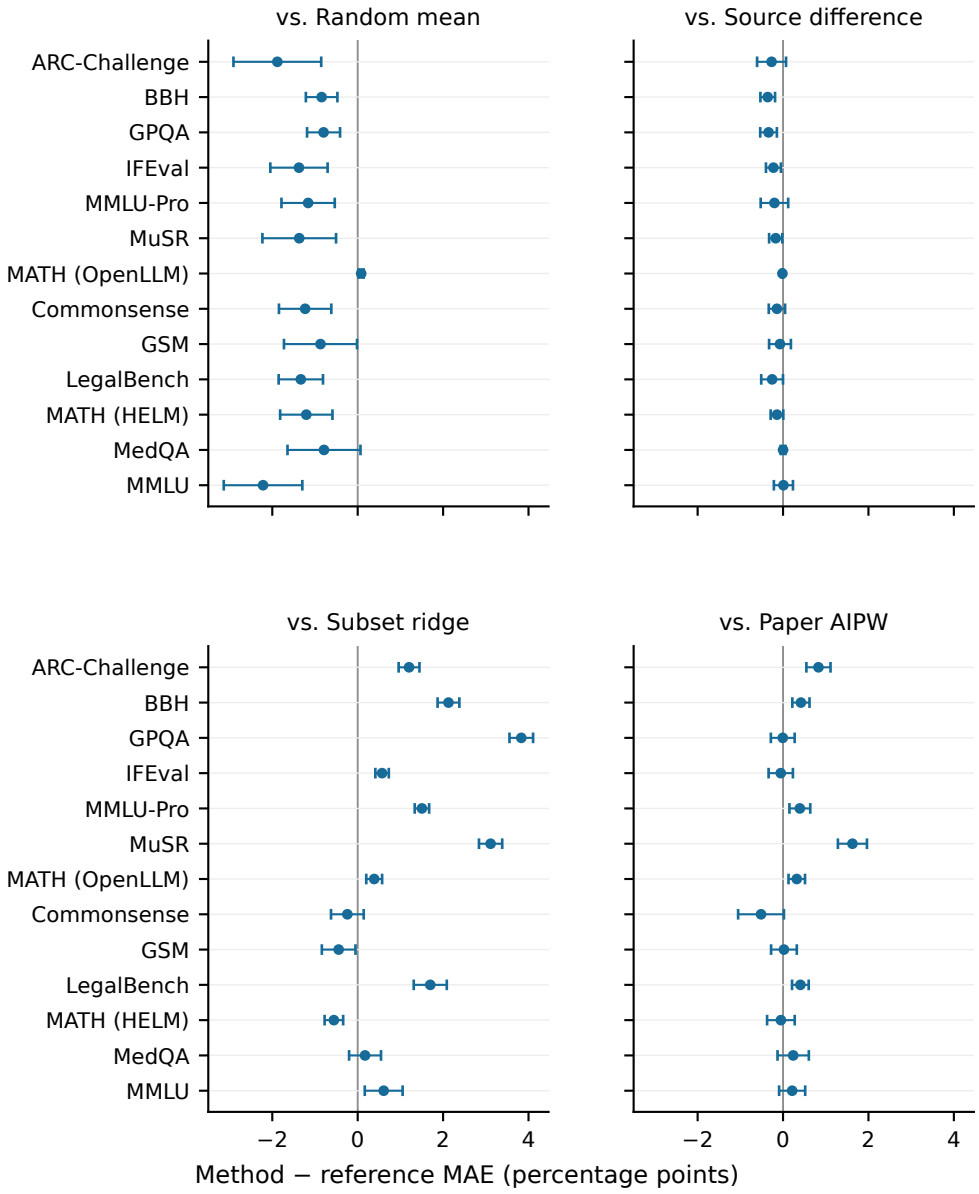


Figure 1: Per-benchmark crossfit-minus-reference MAE at $k = 30$. Negative differences favor crossfit; each reference shares the locked target pool and uniform item ordering. All 13 benchmarks and four core references are retained, including unfavorable outcomes. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. All four panels share one horizontal scale. Item fractions are in the frozen-pool table.

References

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A Frozen data audit

Table 2: Frozen binary matrices before identity deduplication or model splitting. Raw rows count model records; unique rows count distinct complete response vectors, not independent identities. Name repeats are extra occurrences in the per-benchmark OpenLLM lists (first seven rows). Dashes mark unverified HELM name alignment: LegalBench has 79 rows versus 83 global names; the other five have matching lengths only. Const. items have identical responses across all raw rows. For a budget of $k = 100$ items, the fraction k/N of the full N items reaches 22.9%. All matrices are complete and finite binary. These are data-audit counts, not prediction-performance results.

Benchmark	Raw rows	Items	Unique rows	Name repeats	Const. items	k/N (%)
ARC-Challenge	218	1,172	209	6	32	8.5
BBH	446	5,761	425	19	0	1.7
GPQA	447	1,192	424	19	0	8.4
IFEval	448	541	427	19	7	18.5
MMLU-Pro	448	12,032	425	19	1	0.8
MuSR	448	756	427	19	10	13.2
MATH (OpenLLM)	447	894	375	19	119	11.2
Commonsense	83	500	82	–	0	20.0
GSM	83	1,000	83	–	3	10.0
LegalBench	79	2,047	79	–	53	4.9
MATH (HELM)	83	437	83	–	1	22.9
MedQA	83	1,000	82	–	2	10.0
MMLU	83	567	82	–	1	17.6

B Frozen model pools and proxy cohorts

Table 3: Frozen primary model pools and item budgets. Fit and Dev. are the historical fitting and development pools; Hist. is their union used for allowed final source refits. Target is the reserved evaluation pool. Exact repeated OpenLLM names were removed before splitting; all six HELM sets use opaque row IDs because identity alignment is unverified. Distinct names with identical responses remain in the primary pools. The study specifies $k = 10, 30, 100$ unique items per target; the last column gives $100/N$ as a percentage of full item count N. These are pre-target configuration counts, not evidence that the experiments completed.

Benchmark	Fit	Dev.	Hist.	Target	N	$100/N$ (%)
ARC-Challenge	118	30	148	64	1,172	8.5
BBH	238	60	298	129	5,761	1.7
GPQA	239	60	299	129	1,192	8.4
IFEval	240	60	300	129	541	18.5
MMLU-Pro	240	60	300	129	12,032	0.8
MuSR	240	60	300	129	756	13.2
MATH (OpenLLM)	239	60	299	129	894	11.2
Commonsense	46	12	58	25	500	20.0
GSM	46	12	58	25	1,000	10.0
LegalBench	44	11	55	24	2,047	4.9
MATH (HELM)	46	12	58	25	437	22.9
MedQA	46	12	58	25	1,000	10.0
MMLU	46	12	58	25	567	17.6

Table 4: Pre-target eligibility for the $k = 30$ name-proxy source-exclusion stress test. The ordinary and excluded-history comparisons use exactly the same selected locked target cohort and item draws. The selected proxy has the largest eligible target count, requiring at least 10 targets and 40 remaining historical rows; it is a documented string proxy, not verified model ancestry. ARC-Challenge has no eligible proxy; dashes denote an absent planned comparison, not zero error. The six HELM sets are outside this test because their name alignment is unverified. History counts describe available rows, not completed fits or performance.

Benchmark	Proxy	Targets	Ordinary history	Excluded history
ARC-Challenge	Ineligible	–	–	–
BBH	llama	26	298	260
GPQA	llama	18	299	252
IFEval	llama	19	300	255
MMLU-Pro	qwen	24	300	257
MuSR	llama	17	300	252
MATH (OpenLLM)	llama	20	299	256

C Implementation and sampling detail

MAP-MIRT fits $p_{mj} = \mathrm{sigmoid}(\theta_m^\top a_j + b_j)$ with Gaussian penalty one on all parameters. Dimensions $\{2, 5, 10, 15\}$ are selected by development random-p-IRT MAE; dimensions ineligible for a small fitting pool are recorded. Randomized-SVD initialization is deterministic. L-BFGS-B starts with 500 iterations and continues once for at most 1500 more upon nonconvergence. Unconverged development dimensions are recorded and excluded; absence of an eligible converged fit or a failed selected historical refit stops the run for investigation.

Target ability uses fixed item parameters and MAP fitting with penalty one, zero initialization and at most 200 L-BFGS-B iterations. The p-style estimate is $N^{-1}(\sum_{j\in S} y_j + \sum_{j\notin S} \widehat{p}_j)$. A gp-style convex mixture assigns observed-estimate weight in $\{0, .25, .5, .75, 1\}$ by development MAE. IRT anchors cluster $(a_j, -b_j)$ and use cluster masses. The random and IRT-anchor mean/p-/gp-style comparisons share the selected latent dimension. These are explicit adapters: MAP, fixed priors, target regularization and validation mixing differ from tinyBenchmarks' variational hierarchical fitting and mixing heuristic. They are not an exact end-to-end reproduction.

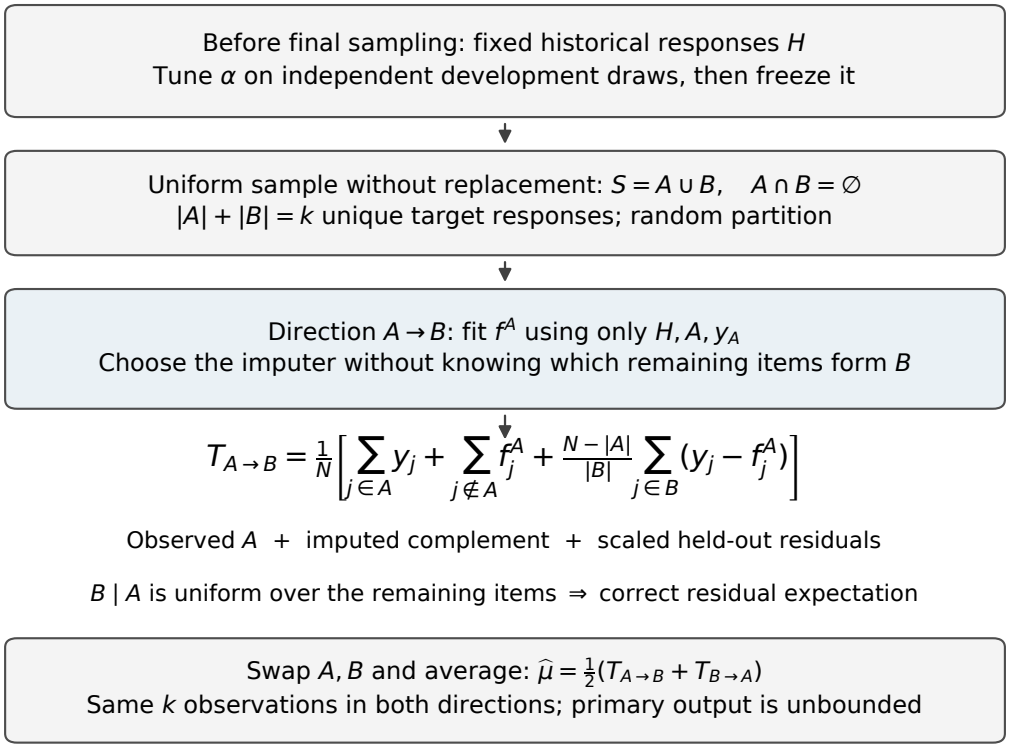


Figure 2: Sampling and information flow for crossfit difference. Historical responses H are fixed; regularization α is selected using development models and item draws independent of the final sample. One uniform ordered sample is partitioned into disjoint A, B, totaling k unique target observations. In the first direction, the imputer uses H, A, y_A and is fixed without knowing which remaining items form B; only the explicit residual correction uses y_B. Conditional uniform sampling of B from the complement of A makes the scaled residual total unbiased. Swap A, B and average the two directions without extra observations. The identity concerns signed expectation over item sampling for fixed model pools and target outcomes; it does not guarantee smaller MAE, RMSE or better ranks. The primary estimate is unbounded; clipping is a separate biased ablation. This is an estimator diagram, not a plot of measured performance; derivation is supplied by analysis.

D Numerical metrics and execution chronology

After initial scoring on 2026-11-05, exact binary-float ranking fragmented mathematical ties: random mean and source difference, which differ only by a target-independent constant, disagreed in 95 of 780 primary draw cells. The fixed operational repair rounds truth and predictions to twelve decimal places for rank/tie computations and canonicalizes true pair gaps before the one-percentage-point close-pair boundary. MAE, RMSE, signed error, tails and out-of-range metrics use raw predictions; all 80,840 such metric rows were verified bitwise unchanged. No prediction, tuning choice, split, observed item or fit changed. Original results/hashes are preserved in `analysis/results/study/original_raw_ties/`; the dated amendment and definitive scorer are in `analysis/notes/postscore_amendments/` and `analysis/diagnostics/score_canonical.py`. Independent bounded review accepted this correction; that assessment is separate from final-version approval. The dated plan-owner account is in `study/postscore_account.md`.

The corrected outputs pass all 1,420 main/robustness constant-shift comparisons and 22,360 independent rank-cell checks. Twelve decimals is a numerical convention, not proof that every near-equal learned prediction is mathematically identical; the true-score grid does not bound learned prediction gaps. Relative to twelve decimals, ten changes one rank vector and fourteen changes twenty-seven (six source-difference cells, twenty copies of one fixed MuSR anchor result, and one learned-imputer cell). Maximum per-cell Spearman differences are 0.000141 and 0.098376, respectively. The ten-decimal variant changes none of the 24 primary macro rank-contrast means or interval endpoints (four references, three budgets and two rank metrics). At fourteen decimals, maximum main macro contrast changes are 0.000382 for Spearman and 0.000371 for Kendall; maximum endpoint changes are 0.000856 and 0.000769. Neither precision changes the mean direction or paired-t interval zero-exclusion classification of these 24 contrasts; this does not remove per-cell sensitivity or adjust for multiple comparisons. Full affected cells and endpoints remain in `rank_precision_*` outputs. These measurements constrain sensitivity rather than asserting perfect numerical equivalence.

The original v1 authorization and pre-target macro-pairing amendment are retained. The original gate record is 14:04:42 UTC and the first target command launched at 14:04:50; formal feasibility-v2 publication extending GO to the exact amended scorer hash was 14:06:40. Its later evidence was appended without backdating. The predictor, tuning, data, split and draw bytes did not change; the plan owner explicitly judged a prediction rerun unnecessary. This chronology does not certify final-paper claims. Full matrices were physically available for the initial audit; the target budget is enforced controlled use through the observation API, not a cryptographic holdout.

E Reproduction, compute and asset boundaries

The delivery root README points to all standalone entry points. From `analysis/`, install the pinned `requirements.txt`, source `scripts/env.sh`, run `scripts/fetch_data.py` and `scripts/audit_data.py`, then `scripts/run_study.pydevelop`, `scripts/run_study.pypredict` and `diagnostics/score_canonical.py`. Use the committed configuration and its completed gate records; do not regenerate target membership. Input manifests pin upstream revision `55faff496947789f94e6b5908c77cbc3e84f81ad`, every file hash, source/target IDs, five development and twenty final draw namespaces, and seeds. The final artifact manifest connects raw predictions, scores, uncertainty, diagnostics and derived outputs. Rendering reads saved results without refitting. The presentation manifest gives each figure/table generator, command and exact inputs.

Measured environment: Python 3.11.2, NumPy 2.4.6, SciPy 1.17.1, pandas 3.0.6, scikit-learn 1.9.1 and CPU PyTorch 2.14.1; exact remaining versions are in `analysis/results/environment.json`. Commands request one BLAS/OpenMP thread; historical effective library pools were not instrumented; no GPU or new LLM calls are used. The sandbox has two CPU cores and 4 GiB memory; processes share host capacity. The two concurrent cache-reusing prediction commands consumed 67.468 process CPU seconds in total (37.752 and 29.825 seconds elapsed), maximum process peak RSS 473,992 KiB. Original scoring consumed 234.205 CPU seconds and corrected scoring 237.245 CPU seconds, the latter peaking at 1,250,604 KiB. These include loading/output, exclude source fitting and are not paid-API or new-model inference savings.

The 218 unaffected source-fit component records total 1,312.549 elapsed seconds; one original record is 165,044.541 seconds including a world suspension. It remains intact and cannot measure active computation. An identical-source timing-only replication took 173.822 elapsed/173.779 process CPU seconds and matched the objective without replacing parameters. These component measurements do not supply a complete original source-command CPU total, which was not instrumented. A fresh fit/tune run is budgeted at roughly 30–60 minutes on comparable hardware as an extrapolation, not an observed guarantee. Two isolated same-configuration source-only commands measured 1,460.129 process CPU seconds and a 1,155.498-second parallel wall span. Combined with original cached predictions and corrected scoring, this gives a 1,764.841-second (29.41 CPU-minute) reproduction-workload proxy. These phases were measured at different times; the total excludes acquisition, pilots, obsolete scoring, diagnostics, formula reporting, rendering and setup. It does not recover original source CPU or an original end-to-end wall time. Cached fits, shared anchor construction and reused p-/gp-inference must not be counted repeatedly as distinct executed work; the whole-command ledger and per-method-alone costs answer different questions.

The replay reproduced all 219 MIRT fits and 45 selected-parameter packages, but one GSM development correctness-anchor ridge loss grid differed by up to 0.451171 pp. A bounded source-only 1/2/1/2-thread check reproduced both grids, changing one of thirty development representatives while retaining selected $\alpha = 1$. This establishes thread sensitivity, not the original effective thread count. All 78 final correctness-/IRT-anchor selections and their same-budget mean predictions reproduced exactly. Preserved original selections/predictions remain the study record; fixed seeds alone do not ensure every intermediate representative is invariant (`analysis/notes/source_reproducibility.md`). The initial equivalence-audit assertion failure and repair are retained.

All initially capped attempts and allowed continuations are retained. One supplemental formula-diagnostic job failed when a file glob included timing files lacking target rows; the corrected glob was rerun without dropping primary data or methods (`analysis/notes/run_failures.json`). Preliminary runs, original/corrected scoring and supplemental diagnostics have their own records; the main prediction time is not a whole-project total.

The packaged binary score files and pinned upstream code carry the repository's MIT license, retained in `analysis/notes/sources/UPSTREAM_LICENSE`. They originate from Open LLM Leaderboard and HELM as credited in the supplied data specification; HELM software's Apache-2.0 license is not asserted to establish every upstream dataset right. No benchmark prompts, model weights or raw upstream task corpora are redistributed. Upstream task-specific rights and HELM identity alignment are not independently established; access to public packaged results is not a blanket license for those other assets. The delivered code/results document intended retrospective finite-matrix use and its limits rather than granting unverified third-party rights.

Permitted source URLs, versions, reading scope and hashes are in `manuscript/access-log.md`, `reading-manifest.json` and the analysis access log. No prohibited encounter was recognized; no withheld-work identity search or forward-citation search was attempted. Agents assisted context inspection, planning, methods, code, experiments, presentation, writing and independent review, as disclosed in Section 6; this was not solely language editing. Responsible future conference authorship, institutional approvals and external submission are not established by this research delivery.

F Complete empirical evidence

The main paper gives the all-method $k = 30$ summary and the thirteen-benchmark paired MAE comparison. The following displays retain all budgets, both score-error and rank metrics, heterogeneous outcomes, correction and selection controls, and bounded stress conditions. They use the same conditional item-draw uncertainty as Section 4; repeated fixed-anchor outputs do not measure fit uncertainty. Each caption identifies its numerical population and limitations. All displayed values are generated from committed measured results; the delivery's presentation manifest maps every native include to its hash-pinned inputs and generator.

F.1 Macro effects and benchmark budget curves

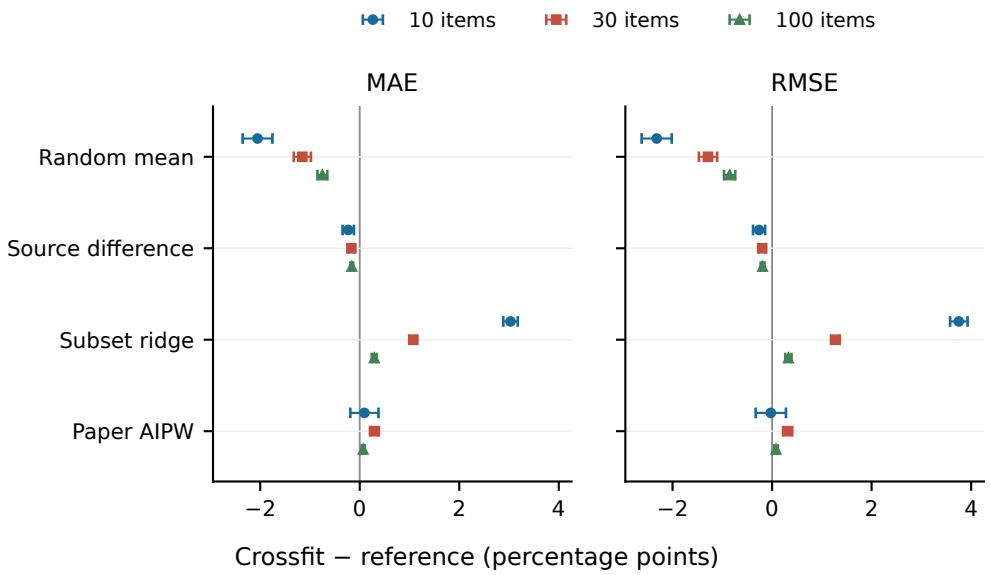


Figure 3: Equal-weight paired effects across all 13 benchmarks; dots show paired means. Each draw first averages within-benchmark crossfit-minus-reference differences; all 13 paired cells are defined. Negative error differences favor crossfit. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. Per-benchmark outcomes are retained in the appendix; a finite-set macro mean does not establish a universal winner.

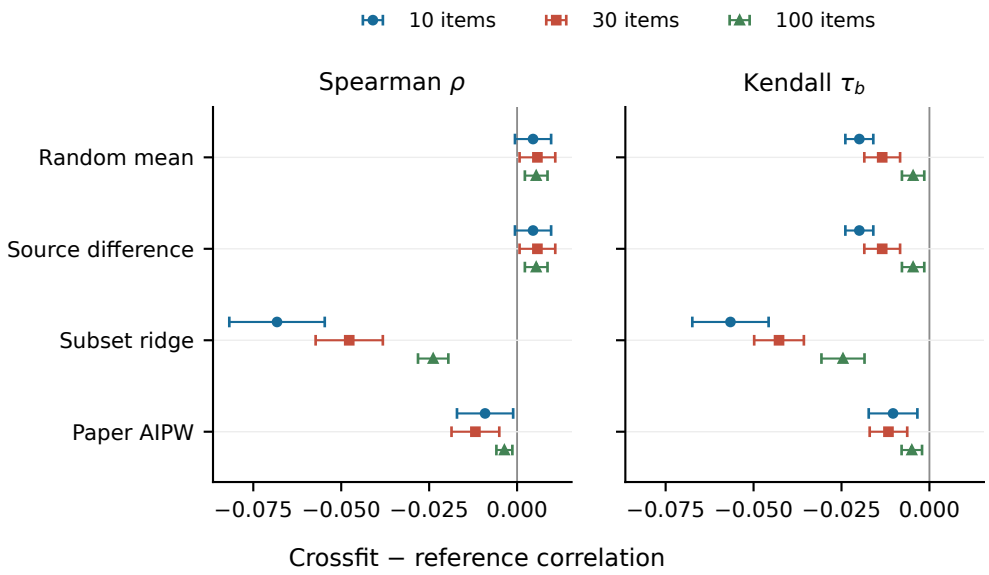


Figure 4: Equal-weight paired effects across all 13 benchmarks; dots show paired means. Each draw first averages within-benchmark crossfit-minus-reference differences; all 13 paired cells are defined. Positive rank differences favor crossfit. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. Per-benchmark outcomes are retained in the appendix; a finite-set macro mean does not establish a universal winner.

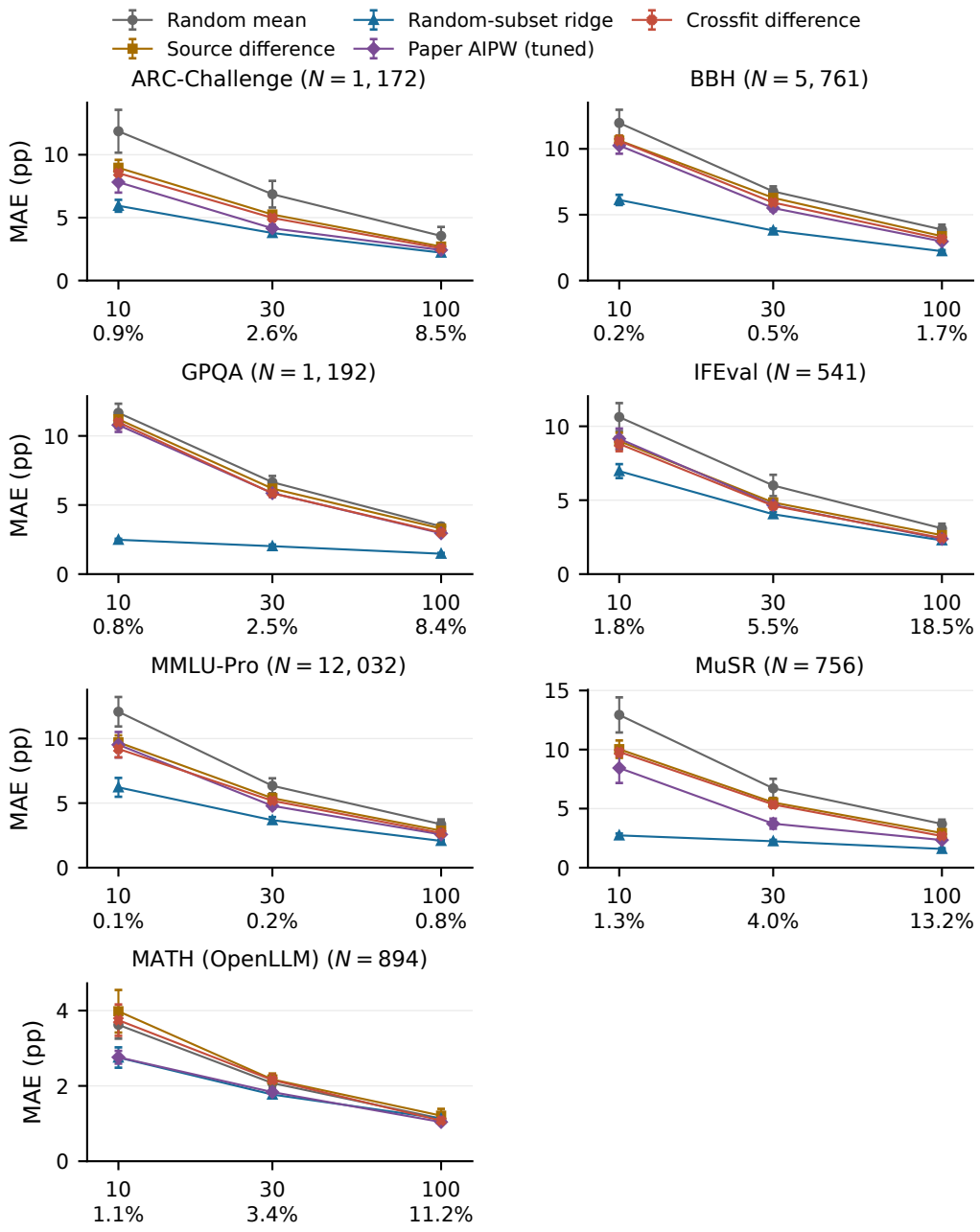


Figure 5: MAE versus observed item count for the seven OpenLLM matrices. Points are supplied means; tick labels give k and $100k/N$ for the full item count N. Linear error axes start at zero; each benchmark has its own vertical scale. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. All 18 methods and full numeric intervals are retained in the per-benchmark tables; the five curves show the prespecified core comparison.

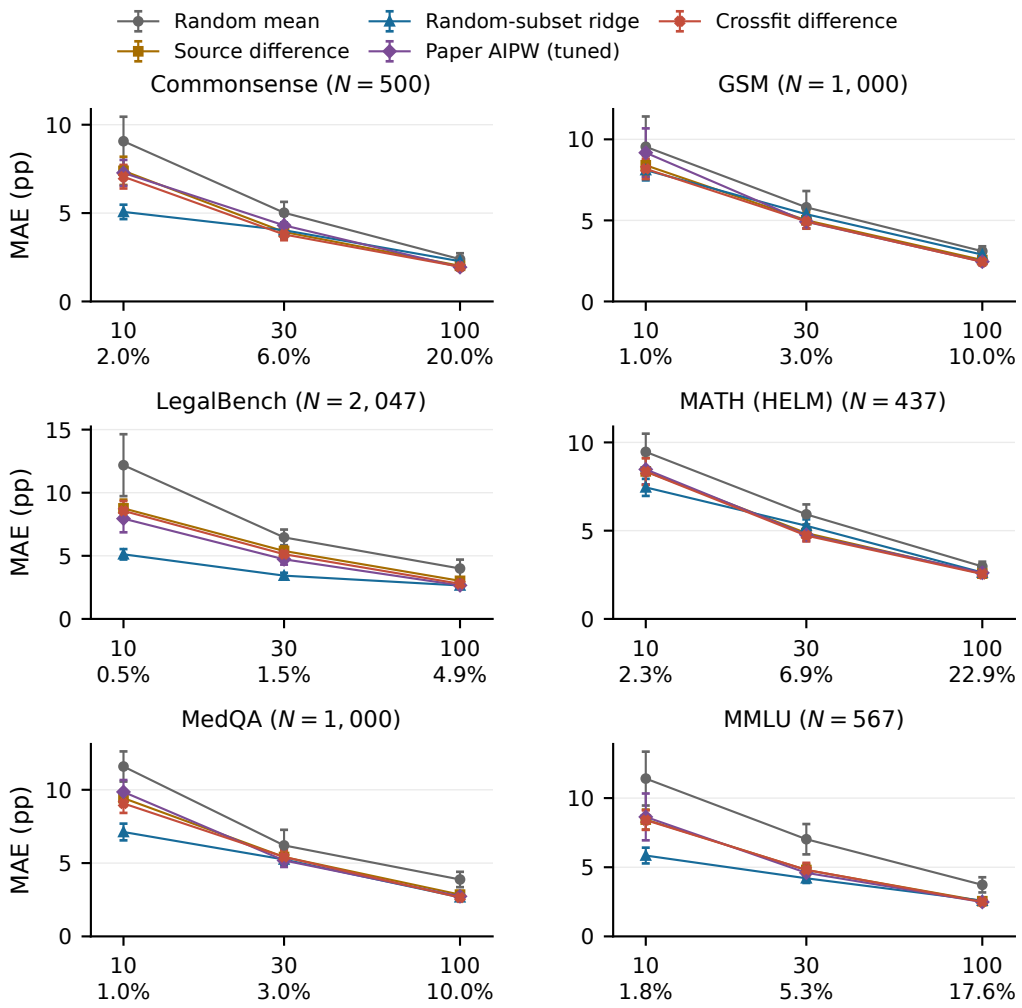


Figure 6: MAE versus observed item count for the six HELM matrices with opaque target row IDs. Points are supplied means; tick labels give k and $100k/N$ for the full item count N. Linear error axes start at zero; each benchmark has its own vertical scale. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. All 18 methods and full numeric intervals are retained in the per-benchmark tables; the five curves show the prespecified core comparison.

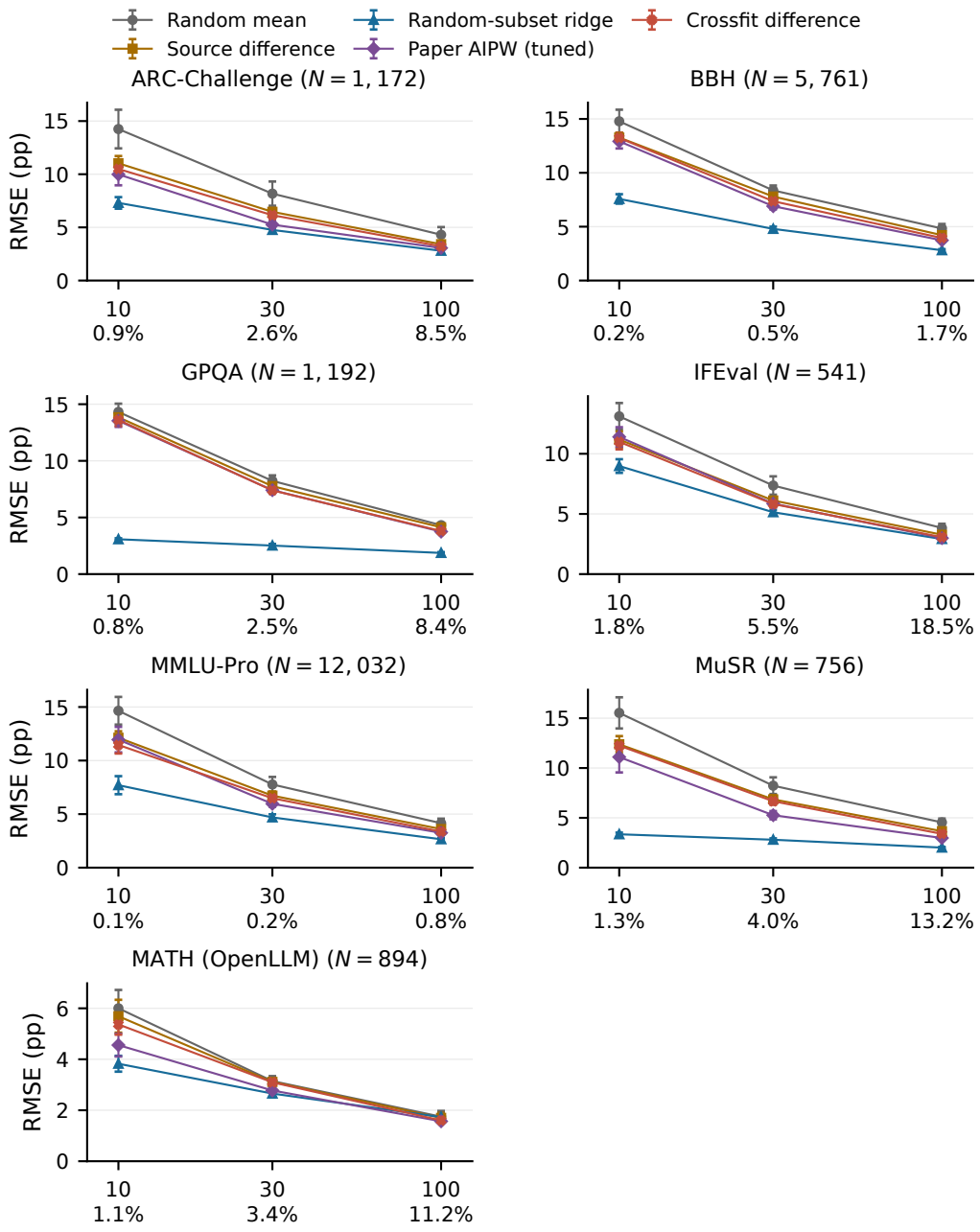


Figure 7: RMSE versus observed item count for the seven OpenLLM matrices. Points are supplied means; tick labels give k and $100k/N$ for the full item count N. Linear error axes start at zero; each benchmark has its own vertical scale. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. All 18 methods and full numeric intervals are retained in the per-benchmark tables; the five curves show the prespecified core comparison.

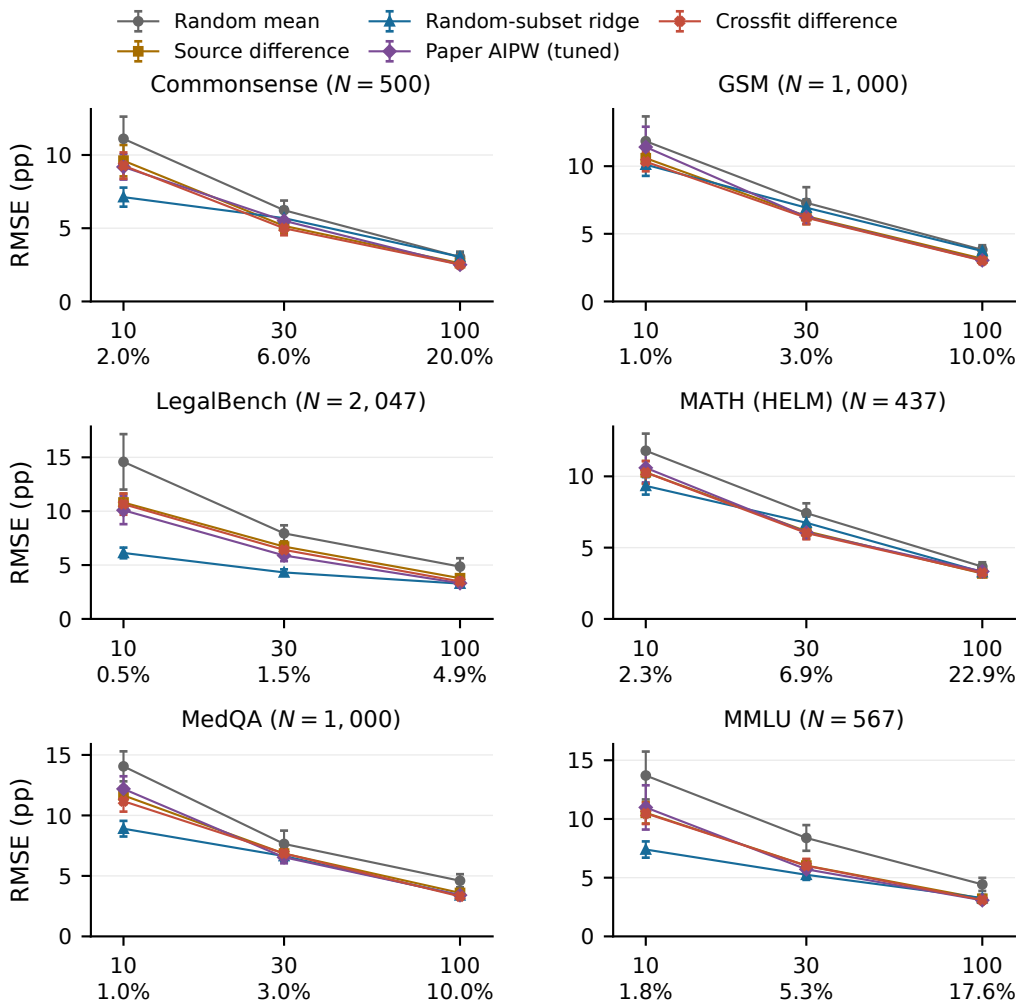


Figure 8: RMSE versus observed item count for the six HELM matrices with opaque target row IDs. Points are supplied means; tick labels give k and $100k/N$ for the full item count N. Linear error axes start at zero; each benchmark has its own vertical scale. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. All 18 methods and full numeric intervals are retained in the per-benchmark tables; the five curves show the prespecified core comparison.

F.2 Absolute all-method macro accuracy

Table 5: Equal-weight MAE across all 13 finite benchmarks, supplied mean [95% t interval] over 20 item orderings. Error is in percentage points; lower is better. Intervals quantify conditional Monte Carlo mean uncertainty, not future-model or future-draw coverage. The source prior uses zero items. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	11.13 [11.13, 11.13]	11.13 [11.13, 11.13]	11.13 [11.13, 11.13]
Random mean	10.62 [10.27, 10.96]	5.99 [5.81, 6.17]	3.25 [3.14, 3.36]
Source difference	8.79 [8.63, 8.95]	5.00 [4.93, 5.08]	2.66 [2.62, 2.70]
Random-subset ridge	5.53 [5.41, 5.66]	3.76 [3.68, 3.84]	2.21 [2.17, 2.24]
Paper AIPW (tuned)	8.47 [8.18, 8.76]	4.54 [4.47, 4.61]	2.43 [2.39, 2.47]
Code AIPW (alpha 10)	8.54 [8.28, 8.80]	4.56 [4.49, 4.63]	2.48 [2.45, 2.52]
Crossfit difference	8.56 [8.39, 8.73]	4.84 [4.76, 4.91]	2.50 [2.46, 2.54]
Crossfit clipped	8.45 [8.30, 8.60]	4.81 [4.74, 4.88]	2.50 [2.45, 2.54]
Half-fit impute (tuned)	5.86 [5.73, 5.98]	3.81 [3.74, 3.87]	2.31 [2.26, 2.35]
Half-fit impute (matched)	7.83 [7.72, 7.93]	4.97 [4.91, 5.02]	2.83 [2.80, 2.87]
Full-fit impute	5.53 [5.40, 5.66]	3.72 [3.64, 3.80]	2.16 [2.13, 2.19]
Correctness-anchor mean	6.76 [6.76, 6.76]	4.23 [4.23, 4.23]	3.14 [3.14, 3.14]
Correctness-anchor ridge	4.66 [4.66, 4.66]	3.24 [3.24, 3.24]	2.25 [2.25, 2.25]
Random MIRT p	7.97 [7.85, 8.08]	4.85 [4.79, 4.90]	2.51 [2.48, 2.55]
Random MIRT gp	6.56 [6.39, 6.73]	4.08 [4.00, 4.17]	2.32 [2.27, 2.36]
IRT-anchor mean	8.48 [8.48, 8.48]	4.76 [4.76, 4.76]	2.48 [2.48, 2.48]
IRT-anchor MIRT p	7.89 [7.89, 7.89]	4.55 [4.55, 4.55]	2.43 [2.43, 2.43]
IRT-anchor MIRT gp	6.00 [6.00, 6.00]	3.48 [3.48, 3.48]	2.03 [2.03, 2.03]

Table 6: Equal-weight RMSE across all 13 finite benchmarks, supplied mean [95% t interval] over 20 item orderings. Error is in percentage points; lower is better. Intervals quantify conditional Monte Carlo mean uncertainty, not future-model or future-draw coverage. The source prior uses zero items. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	13.78 [13.78, 13.78]	13.78 [13.78, 13.78]	13.78 [13.78, 13.78]
Random mean	13.06 [12.68, 13.43]	7.40 [7.22, 7.58]	4.01 [3.88, 4.14]
Source difference	11.00 [10.81, 11.18]	6.31 [6.22, 6.40]	3.35 [3.30, 3.40]
Random-subset ridge	6.98 [6.81, 7.15]	4.84 [4.74, 4.95]	2.83 [2.78, 2.88]
Paper AIPW (tuned)	10.76 [10.45, 11.08]	5.80 [5.72, 5.88]	3.08 [3.03, 3.13]
Code AIPW (alpha 10)	10.79 [10.48, 11.10]	5.78 [5.69, 5.86]	3.16 [3.12, 3.20]
Crossfit difference	10.74 [10.53, 10.95]	6.11 [6.02, 6.20]	3.16 [3.11, 3.20]
Crossfit clipped	10.63 [10.44, 10.83]	6.09 [6.01, 6.18]	3.16 [3.11, 3.20]
Half-fit impute (tuned)	7.28 [7.10, 7.45]	4.83 [4.73, 4.92]	2.94 [2.88, 3.00]
Half-fit impute (matched)	9.51 [9.37, 9.65]	6.26 [6.20, 6.33]	3.66 [3.61, 3.70]
Full-fit impute	6.98 [6.81, 7.15]	4.77 [4.67, 4.88]	2.76 [2.72, 2.80]
Correctness-anchor mean	8.93 [8.93, 8.93]	5.58 [5.58, 5.58]	4.13 [4.13, 4.13]
Correctness-anchor ridge	6.19 [6.19, 6.19]	4.25 [4.25, 4.25]	2.96 [2.96, 2.96]
Random MIRT p	9.31 [9.19, 9.43]	5.82 [5.77, 5.88]	3.12 [3.08, 3.17]
Random MIRT gp	8.13 [7.94, 8.33]	5.13 [5.04, 5.22]	2.92 [2.86, 2.97]
IRT-anchor mean	10.97 [10.97, 10.97]	5.88 [5.88, 5.88]	3.22 [3.22, 3.22]
IRT-anchor MIRT p	9.24 [9.24, 9.24]	5.50 [5.50, 5.50]	3.02 [3.02, 3.02]
IRT-anchor MIRT gp	7.71 [7.71, 7.71]	4.40 [4.40, 4.40]	2.60 [2.60, 2.60]

Table 7: Equal-weight Spearman ρ across all 13 finite benchmarks, supplied mean [95% t interval] over 20 item orderings. Correlations are within each benchmark; higher is better. The constant source prior is undefined on all 13 benchmarks and all 20 draws. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. All other rank cells are defined on all 13 benchmarks. Intervals quantify conditional Monte Carlo mean uncertainty, not future-model or future-draw coverage. The source prior uses zero items. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.644 [0.627, 0.661]	0.793 [0.780, 0.807]	0.900 [0.894, 0.906]
Source difference	0.644 [0.627, 0.661]	0.793 [0.780, 0.807]	0.900 [0.894, 0.906]
Random-subset ridge	0.717 [0.705, 0.729]	0.847 [0.840, 0.854]	0.930 [0.927, 0.933]
Paper AIPW (tuned)	0.658 [0.643, 0.673]	0.811 [0.804, 0.818]	0.909 [0.906, 0.913]
Code AIPW (alpha 10)	0.656 [0.642, 0.670]	0.810 [0.803, 0.816]	0.905 [0.902, 0.908]
Crossfit difference	0.649 [0.635, 0.662]	0.799 [0.789, 0.809]	0.906 [0.901, 0.910]
Crossfit clipped	0.649 [0.635, 0.662]	0.799 [0.789, 0.809]	0.906 [0.901, 0.910]
Half-fit impute (tuned)	0.718 [0.706, 0.730]	0.848 [0.840, 0.856]	0.929 [0.927, 0.932]
Half-fit impute (matched)	0.721 [0.710, 0.731]	0.856 [0.849, 0.862]	0.936 [0.933, 0.938]
Full-fit impute	0.717 [0.705, 0.729]	0.849 [0.841, 0.856]	0.930 [0.927, 0.933]
Correctness-anchor mean	0.755 [0.755, 0.755]	0.851 [0.851, 0.851]	0.890 [0.890, 0.890]
Correctness-anchor ridge	0.822 [0.822, 0.822]	0.899 [0.899, 0.899]	0.943 [0.943, 0.943]
Random MIRT p	0.697 [0.682, 0.712]	0.841 [0.834, 0.849]	0.928 [0.925, 0.931]
Random MIRT gp	0.698 [0.683, 0.712]	0.839 [0.832, 0.846]	0.929 [0.926, 0.932]
IRT-anchor mean	0.667 [0.667, 0.667]	0.818 [0.818, 0.818]	0.912 [0.912, 0.912]
IRT-anchor MIRT p	0.731 [0.731, 0.731]	0.862 [0.862, 0.862]	0.933 [0.933, 0.933]
IRT-anchor MIRT gp	0.727 [0.727, 0.727]	0.868 [0.868, 0.868]	0.935 [0.935, 0.935]

Table 8: Equal-weight Kendall τ_b across all 13 finite benchmarks, supplied mean [95% t interval] over 20 item orderings. Correlations are within each benchmark; higher is better. The constant source prior is undefined on all 13 benchmarks and all 20 draws. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. All other rank cells are defined on all 13 benchmarks. Intervals quantify conditional Monte Carlo mean uncertainty, not future-model or future-draw coverage. The source prior uses zero items. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.529 [0.515, 0.543]	0.665 [0.655, 0.676]	0.789 [0.784, 0.794]
Source difference	0.529 [0.515, 0.543]	0.665 [0.655, 0.676]	0.789 [0.784, 0.794]
Random-subset ridge	0.566 [0.555, 0.576]	0.695 [0.687, 0.702]	0.809 [0.804, 0.814]
Paper AIPW (tuned)	0.519 [0.507, 0.532]	0.664 [0.658, 0.669]	0.789 [0.785, 0.794]
Code AIPW (alpha 10)	0.517 [0.505, 0.529]	0.663 [0.657, 0.668]	0.784 [0.780, 0.787]
Crossfit difference	0.509 [0.497, 0.521]	0.652 [0.645, 0.659]	0.784 [0.780, 0.789]
Crossfit clipped	0.509 [0.497, 0.521]	0.652 [0.645, 0.659]	0.784 [0.780, 0.789]
Half-fit impute (tuned)	0.567 [0.556, 0.577]	0.695 [0.687, 0.704]	0.807 [0.803, 0.811]
Half-fit impute (matched)	0.570 [0.560, 0.580]	0.706 [0.699, 0.713]	0.819 [0.815, 0.822]
Full-fit impute	0.566 [0.555, 0.576]	0.697 [0.690, 0.705]	0.811 [0.807, 0.816]
Correctness-anchor mean	0.610 [0.610, 0.610]	0.725 [0.725, 0.725]	0.792 [0.792, 0.792]
Correctness-anchor ridge	0.663 [0.663, 0.663]	0.756 [0.756, 0.756]	0.828 [0.828, 0.828]
Random MIRT p	0.550 [0.538, 0.563]	0.694 [0.686, 0.703]	0.811 [0.807, 0.815]
Random MIRT gp	0.553 [0.541, 0.565]	0.693 [0.686, 0.701]	0.815 [0.811, 0.819]
IRT-anchor mean	0.528 [0.528, 0.528]	0.674 [0.674, 0.674]	0.789 [0.789, 0.789]
IRT-anchor MIRT p	0.574 [0.574, 0.574]	0.710 [0.710, 0.710]	0.811 [0.811, 0.811]
IRT-anchor MIRT gp	0.572 [0.572, 0.572]	0.721 [0.721, 0.721]	0.816 [0.816, 0.816]

F.3 Paired heterogeneous score errors

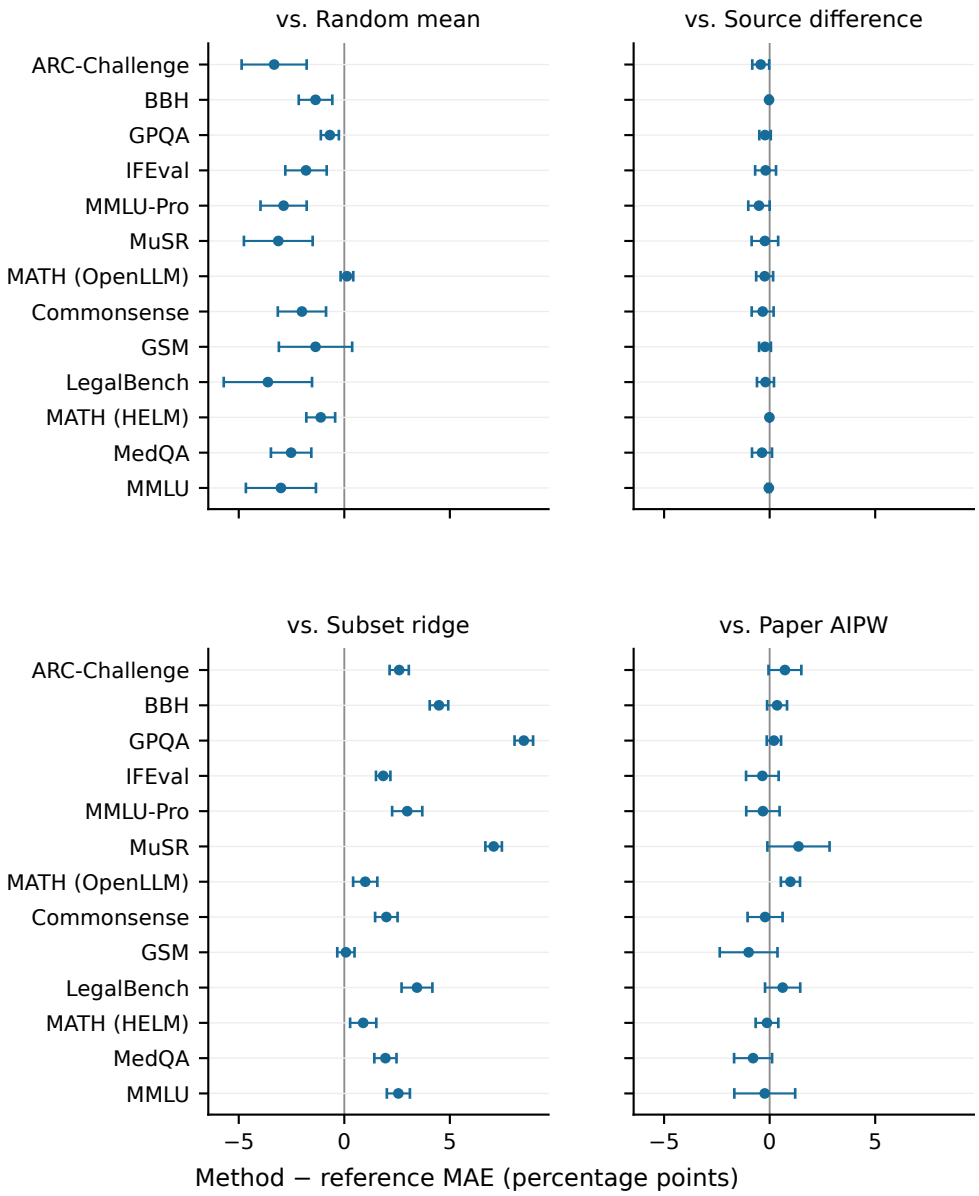


Figure 9: Per-benchmark crossfit-minus-reference MAE at $k = 10$. Negative differences favor crossfit; each reference shares the locked target pool and uniform item ordering. All 13 benchmarks and four core references are retained, including unfavorable outcomes. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. All four panels share one horizontal scale. Item fractions are in the frozen-pool table.

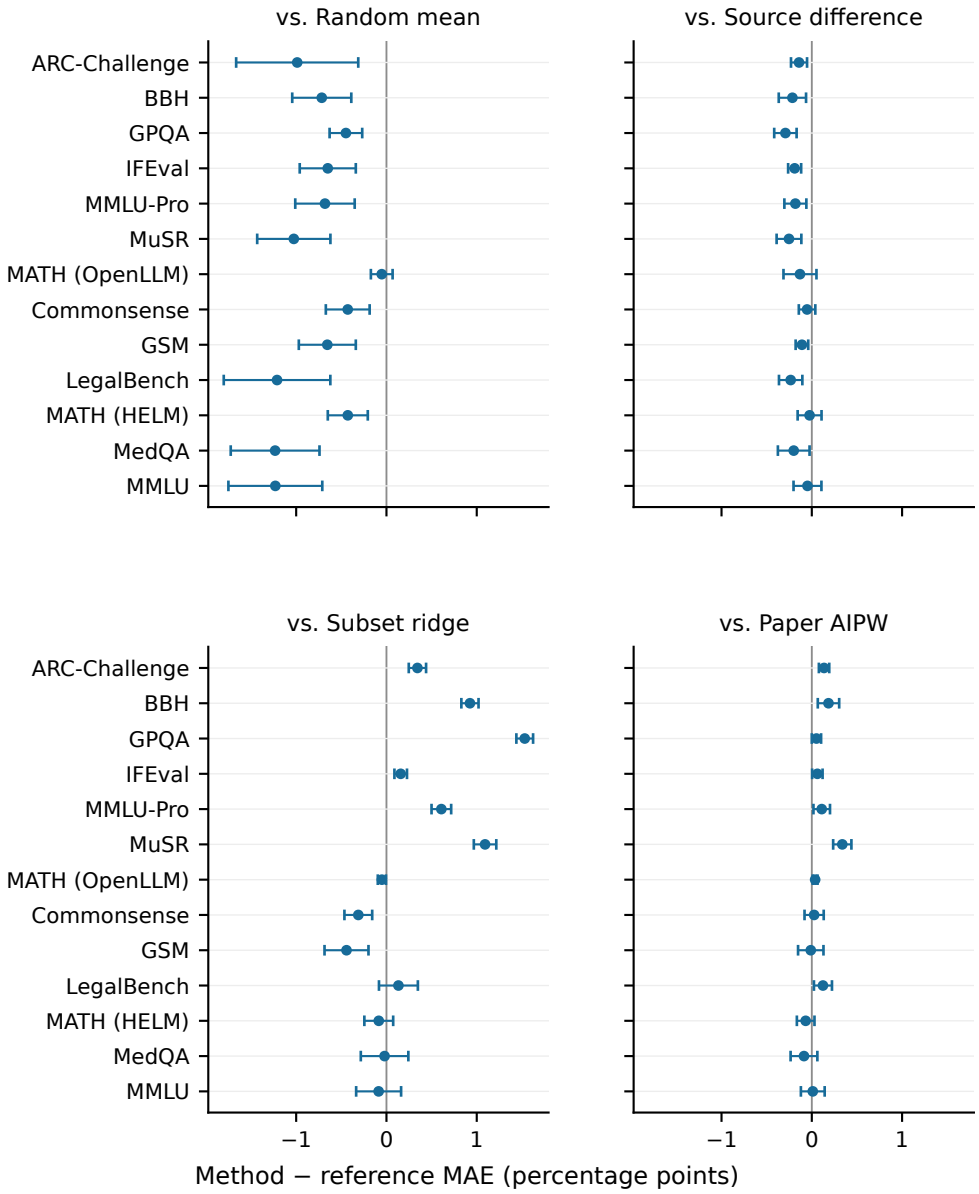


Figure 10: Per-benchmark crossfit-minus-reference MAE at $k = 100$. Negative differences favor crossfit; each reference shares the locked target pool and uniform item ordering. All 13 benchmarks and four core references are retained, including unfavorable outcomes. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. All four panels share one horizontal scale. Item fractions are in the frozen-pool table.

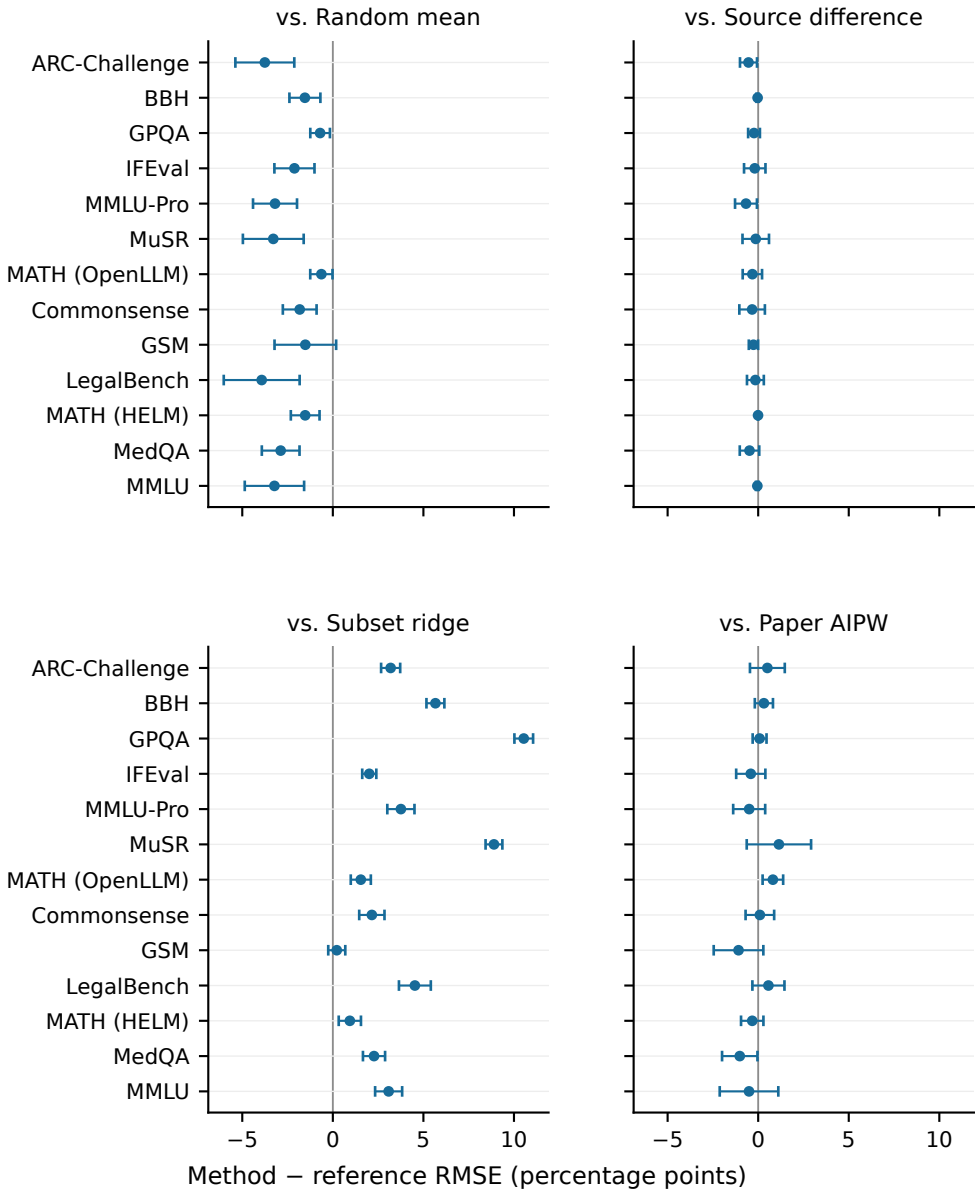


Figure 11: Per-benchmark crossfit-minus-reference RMSE at $k = 10$. Negative differences favor crossfit; each reference shares the locked target pool and uniform item ordering. All 13 benchmarks and four core references are retained, including unfavorable outcomes. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. All four panels share one horizontal scale. Item fractions are in the frozen-pool table.

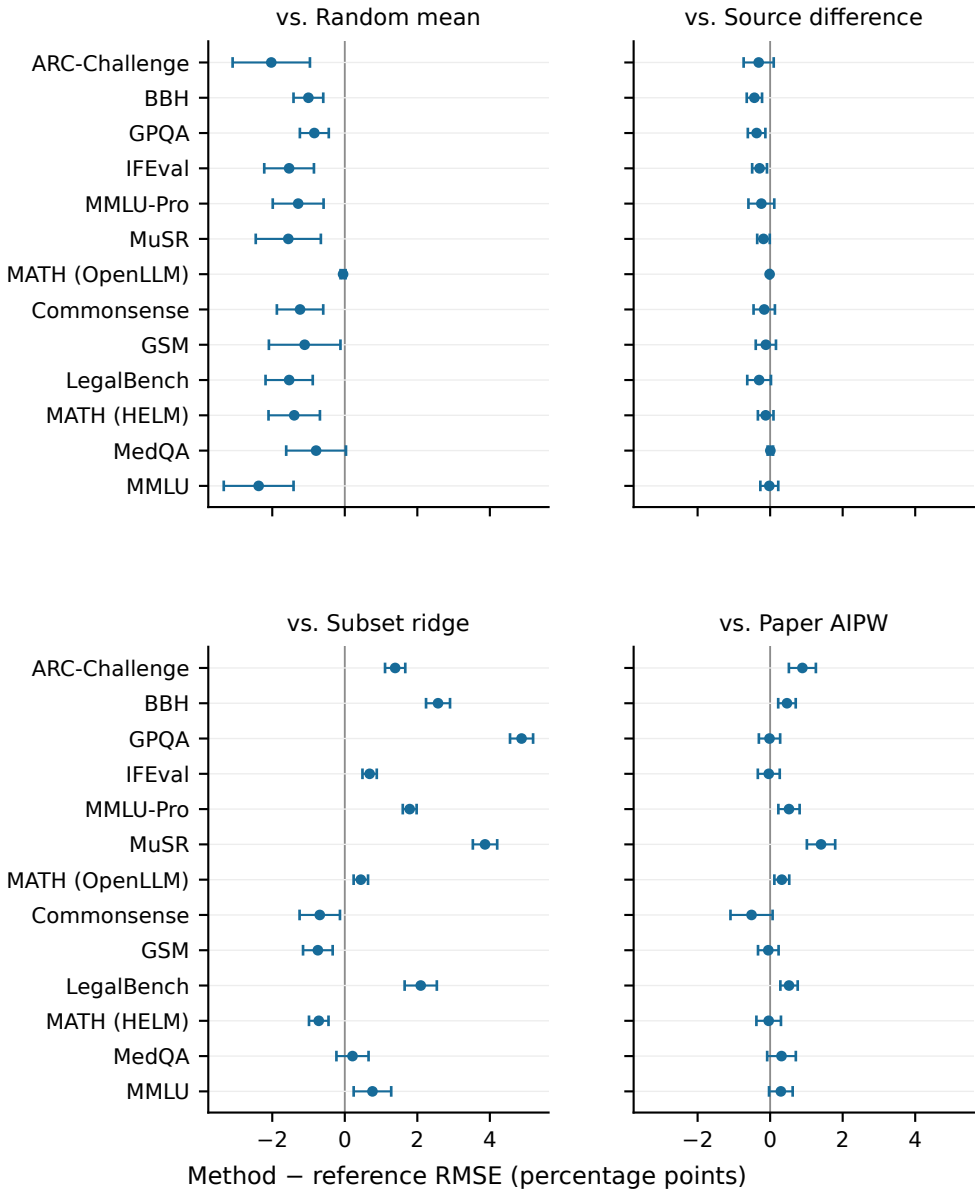


Figure 12: Per-benchmark crossfit-minus-reference RMSE at $k = 30$. Negative differences favor crossfit; each reference shares the locked target pool and uniform item ordering. All 13 benchmarks and four core references are retained, including unfavorable outcomes. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. All four panels share one horizontal scale. Item fractions are in the frozen-pool table.

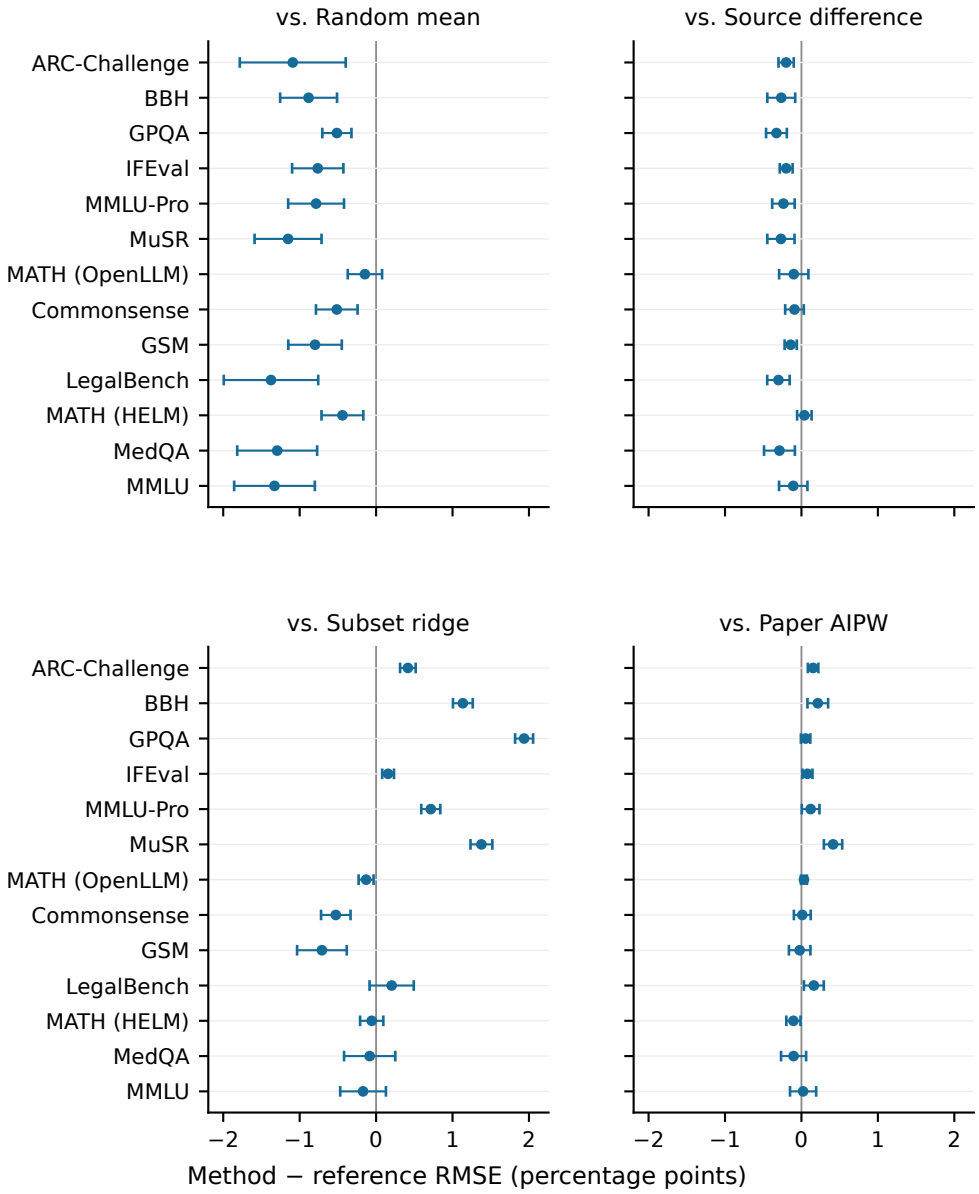


Figure 13: Per-benchmark crossfit-minus-reference RMSE at $k = 100$. Negative differences favor crossfit; each reference shares the locked target pool and uniform item ordering. All 13 benchmarks and four core references are retained, including unfavorable outcomes. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. All four panels share one horizontal scale. Item fractions are in the frozen-pool table.

F.4 Selection, estimation and correction controls

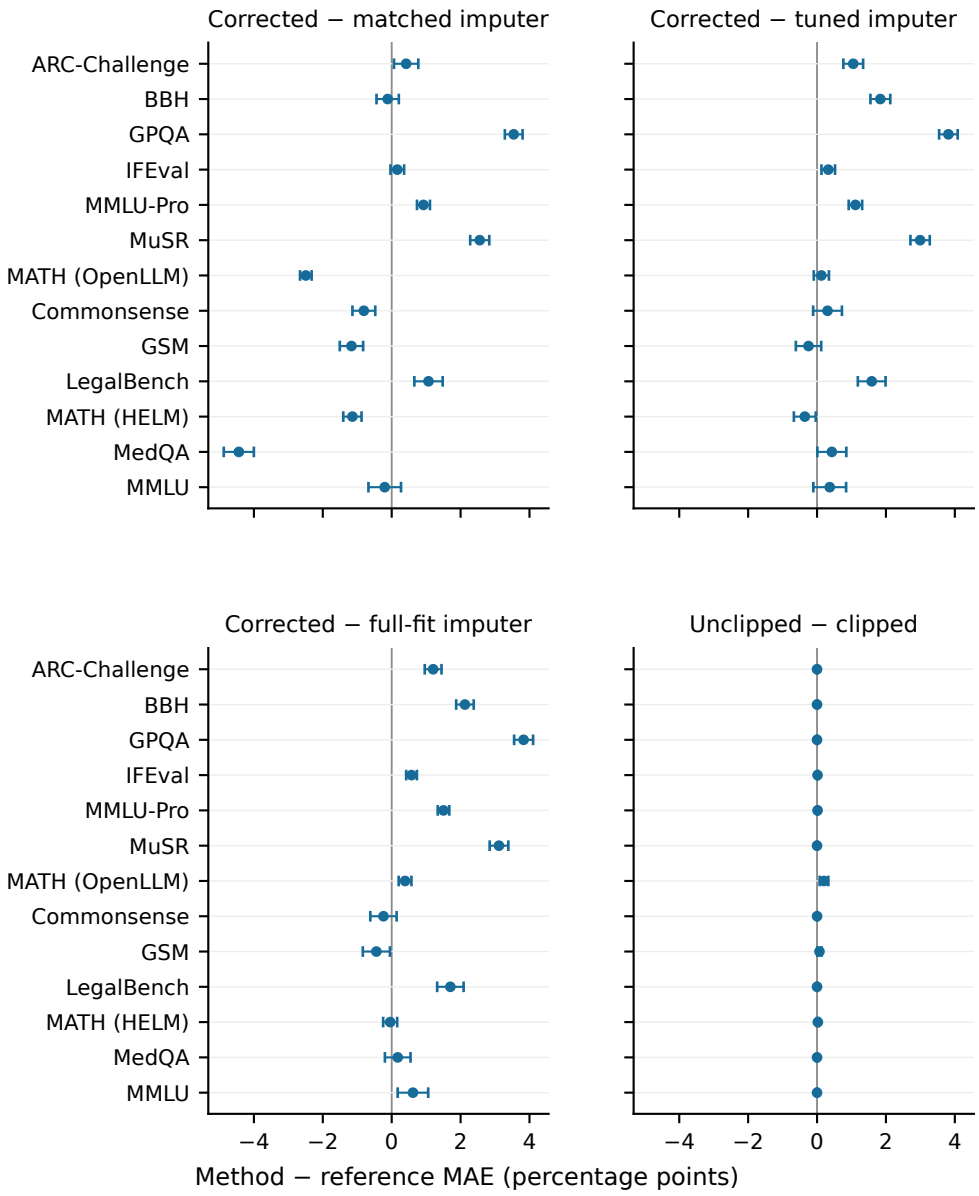


Figure 14: Correction and clipping controls at $k = 30$. Panels give the displayed method-minus-reference MAE; negative is lower error for the first method. The matched half-fit imputer uses the candidate alpha; the tuned imputer changes regularization as well as correction. Full-fit imputation uses all observed items. Clipping is a separately named biased ablation. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage.

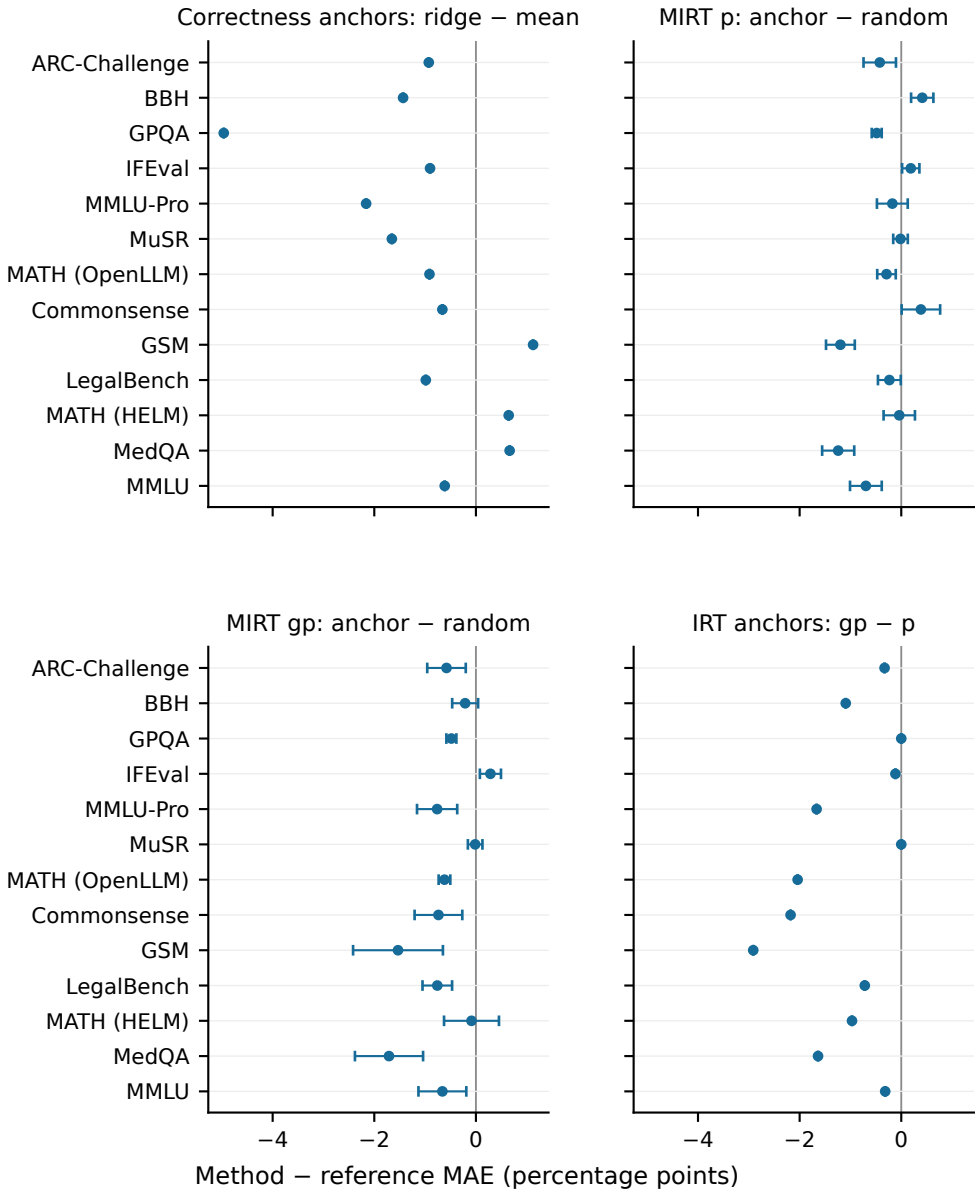


Figure 15: Selection and estimation controls at $k = 30$. Each title identifies the method-minus-reference MAE; negative favors the first method. Random and IRT-anchor MIRT use the same source-selected dimension; gp mixtures are development selected. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

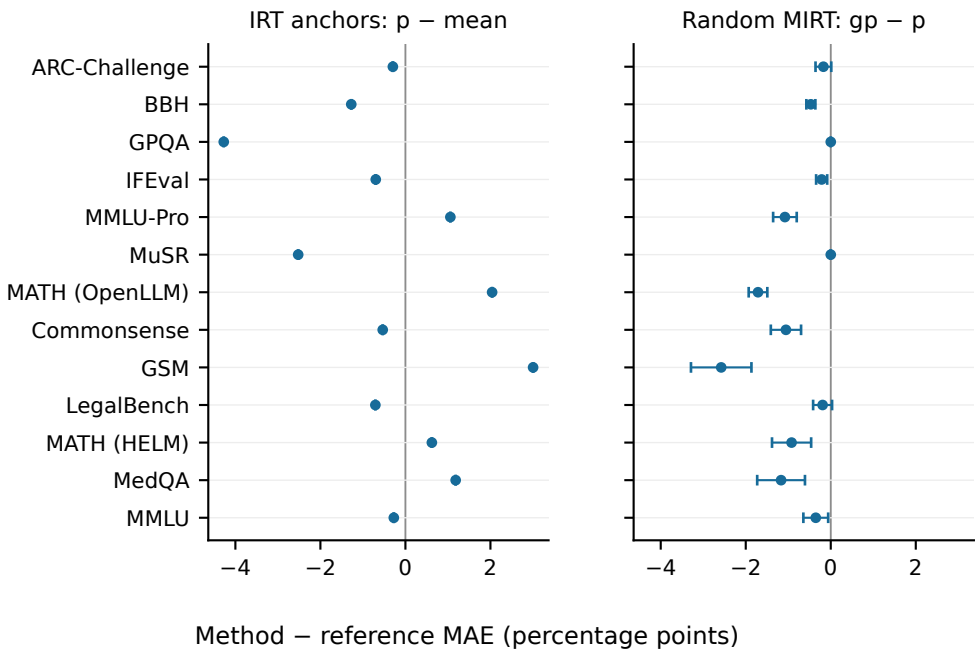


Figure 16: Selection and estimation controls at $k = 30$. Each title identifies the method-minus-reference MAE; negative favors the first method. Random and IRT-anchor MIRT use the same source-selected dimension; gp mixtures are development selected. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Table 9: Correctness anchors: ridge minus mean: per-benchmark MAE difference in percentage points, mean [95% paired t interval]. Negative favors the first method. All three nominal budgets observe exactly k items per target. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Benchmark	$k = 10$	$k = 30$	$k = 100$
ARC-Challenge	-1.03 [-1.03, -1.03]	-0.93 [-0.93, -0.93]	-0.35 [-0.35, -0.35]
BBH	-4.89 [-4.89, -4.89]	-1.43 [-1.43, -1.43]	-1.04 [-1.04, -1.04]
GPQA	-8.09 [-8.09, -8.09]	-4.96 [-4.96, -4.96]	-4.38 [-4.38, -4.38]
IFEval	-0.79 [-0.79, -0.79]	-0.90 [-0.90, -0.90]	-1.50 [-1.50, -1.50]
MMLU-Pro	-2.78 [-2.78, -2.78]	-2.16 [-2.16, -2.16]	-1.80 [-1.80, -1.80]
MuSR	-3.73 [-3.73, -3.73]	-1.65 [-1.65, -1.65]	-1.74 [-1.74, -1.74]
MATH (OpenLLM)	-0.35 [-0.35, -0.35]	-0.91 [-0.91, -0.91]	-0.63 [-0.63, -0.63]
Commonsense	-0.63 [-0.63, -0.63]	-0.66 [-0.66, -0.66]	-1.39 [-1.39, -1.39]
GSM	-1.39 [-1.39, -1.39]	1.13 [1.13, 1.13]	-0.07 [-0.07, -0.07]
LegalBench	-3.27 [-3.27, -3.27]	-0.99 [-0.99, -0.99]	0.37 [0.37, 0.37]
MATH (HELM)	-0.28 [-0.28, -0.28]	0.65 [0.65, 0.65]	0.01 [0.01, 0.01]
MedQA	0.26 [0.26, 0.26]	0.66 [0.66, 0.66]	0.00 [0.00, 0.00]
MMLU	-0.35 [-0.35, -0.35]	-0.61 [-0.61, -0.61]	1.01 [1.01, 1.01]

Table 10: MIRT p: anchor minus random: per-benchmark MAE difference in percentage points, mean [95% paired t interval]. Negative favors the first method. All three nominal budgets observe exactly k items per target. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Benchmark	$k = 10$	$k = 30$	$k = 100$
ARC-Challenge	0.83 [0.26, 1.40]	-0.42 [-0.74, -0.10]	0.14 [0.02, 0.26]
BBH	0.49 [0.12, 0.86]	0.41 [0.19, 0.63]	0.20 [0.10, 0.30]
GPQA	-0.26 [-0.34, -0.18]	-0.48 [-0.58, -0.38]	-0.15 [-0.23, -0.08]
IFEval	-2.43 [-3.10, -1.77]	0.19 [0.02, 0.36]	-0.00 [-0.09, 0.08]
MMLU-Pro	-0.37 [-0.88, 0.15]	-0.18 [-0.48, 0.13]	0.13 [-0.00, 0.27]
MuSR	-0.08 [-0.25, 0.09]	-0.02 [-0.16, 0.13]	0.20 [0.14, 0.26]
MATH (OpenLLM)	0.50 [0.02, 0.99]	-0.29 [-0.47, -0.11]	-0.24 [-0.30, -0.19]
Commonsense	2.74 [2.17, 3.31]	0.39 [0.01, 0.77]	0.14 [-0.08, 0.35]
GSM	-1.97 [-2.68, -1.26]	-1.20 [-1.48, -0.91]	0.06 [-0.16, 0.28]
LegalBench	-0.75 [-1.24, -0.26]	-0.23 [-0.46, -0.01]	-0.29 [-0.45, -0.13]
MATH (HELM)	-0.43 [-0.98, 0.12]	-0.04 [-0.35, 0.27]	-0.80 [-0.97, -0.62]
MedQA	0.18 [-0.28, 0.63]	-1.24 [-1.56, -0.93]	-0.11 [-0.33, 0.10]
MMLU	0.57 [0.10, 1.04]	-0.70 [-1.01, -0.38]	-0.31 [-0.49, -0.13]

Table 11: MIRT gp: anchor minus random: per-benchmark MAE difference in percentage points, mean [95% paired t interval]. Negative favors the first method. All three nominal budgets observe exactly k items per target. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Benchmark	$k = 10$	$k = 30$	$k = 100$
ARC-Challenge	0.60 [0.03, 1.17]	-0.58 [-0.96, -0.20]	0.04 [-0.08, 0.16]
BBH	-0.78 [-1.15, -0.41]	-0.21 [-0.47, 0.04]	0.08 [-0.05, 0.20]
GPQA	-0.26 [-0.34, -0.18]	-0.48 [-0.58, -0.38]	-0.15 [-0.24, -0.05]
IFEval	-1.64 [-2.30, -0.98]	0.29 [0.08, 0.49]	-0.15 [-0.25, -0.05]
MMLU-Pro	-1.12 [-1.70, -0.54]	-0.76 [-1.16, -0.37]	-0.21 [-0.36, -0.06]
MuSR	-0.08 [-0.25, 0.09]	-0.02 [-0.16, 0.13]	0.08 [0.02, 0.14]
MATH (OpenLLM)	-0.65 [-1.02, -0.29]	-0.62 [-0.73, -0.50]	-0.26 [-0.38, -0.15]
Commonsense	1.10 [-0.15, 2.34]	-0.74 [-1.21, -0.27]	-0.25 [-0.48, -0.01]
GSM	-3.14 [-4.88, -1.40]	-1.53 [-2.42, -0.65]	-0.05 [-0.29, 0.19]
LegalBench	-1.47 [-1.97, -0.98]	-0.76 [-1.05, -0.47]	-0.47 [-0.69, -0.25]
MATH (HELM)	0.07 [-0.78, 0.93]	-0.09 [-0.63, 0.45]	-1.07 [-1.29, -0.85]
MedQA	0.42 [-0.25, 1.10]	-1.71 [-2.38, -1.04]	-0.38 [-0.59, -0.17]
MMLU	-0.39 [-1.20, 0.41]	-0.66 [-1.13, -0.19]	-0.97 [-1.25, -0.69]

Table 12: IRT anchors: gp minus p: per-benchmark MAE difference in percentage points, mean [95% paired t interval]. Negative favors the first method. All three nominal budgets observe exactly k items per target. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Benchmark	$k = 10$	$k = 30$	$k = 100$
ARC-Challenge	-0.23 [-0.23, -0.23]	-0.33 [-0.33, -0.33]	-0.10 [-0.10, -0.10]
BBH	-1.27 [-1.27, -1.27]	-1.09 [-1.09, -1.09]	-0.30 [-0.30, -0.30]
GPQA	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
IFEval	-0.13 [-0.13, -0.13]	-0.12 [-0.12, -0.12]	-0.22 [-0.22, -0.22]
MMLU-Pro	-1.78 [-1.78, -1.78]	-1.67 [-1.67, -1.67]	-0.69 [-0.69, -0.69]
MuSR	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	-0.11 [-0.11, -0.11]
MATH (OpenLLM)	-5.05 [-5.05, -5.05]	-2.04 [-2.04, -2.04]	-0.51 [-0.51, -0.51]
Commonsense	-4.48 [-4.48, -4.48]	-2.18 [-2.18, -2.18]	-0.60 [-0.60, -0.60]
GSM	-5.99 [-5.99, -5.99]	-2.91 [-2.91, -2.91]	-0.77 [-0.77, -0.77]
LegalBench	-0.72 [-0.72, -0.72]	-0.72 [-0.72, -0.72]	-0.31 [-0.31, -0.31]
MATH (HELM)	-2.49 [-2.49, -2.49]	-0.97 [-0.97, -0.97]	-0.44 [-0.44, -0.44]
MedQA	-1.04 [-1.04, -1.04]	-1.64 [-1.64, -1.64]	-0.51 [-0.51, -0.51]
MMLU	-1.44 [-1.44, -1.44]	-0.32 [-0.32, -0.32]	-0.70 [-0.70, -0.70]

Table 13: IRT anchors: p minus mean: per-benchmark MAE difference in percentage points, mean [95% paired t interval]. Negative favors the first method. All three nominal budgets observe exactly k items per target. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Benchmark	$k = 10$	$k = 30$	$k = 100$
ARC-Challenge	-1.13 [-1.13, -1.13]	-0.30 [-0.30, -0.30]	-0.31 [-0.31, -0.31]
BBH	-8.05 [-8.05, -8.05]	-1.27 [-1.27, -1.27]	-0.37 [-0.37, -0.37]
GPQA	-9.44 [-9.44, -9.44]	-4.27 [-4.27, -4.27]	-1.75 [-1.75, -1.75]
IFEval	-2.23 [-2.23, -2.23]	-0.70 [-0.70, -0.70]	-0.04 [-0.04, -0.04]
MMLU-Pro	-1.53 [-1.53, -1.53]	1.06 [1.06, 1.06]	0.47 [0.47, 0.47]
MuSR	-6.50 [-6.50, -6.50]	-2.52 [-2.52, -2.52]	-0.80 [-0.80, -0.80]
MATH (OpenLLM)	5.05 [5.05, 5.05]	2.04 [2.04, 2.04]	0.51 [0.51, 0.51]
Commonsense	6.58 [6.58, 6.58]	-0.53 [-0.53, -0.53]	-0.39 [-0.39, -0.39]
GSM	4.10 [4.10, 4.10]	3.00 [3.00, 3.00]	0.62 [0.62, 0.62]
LegalBench	-0.18 [-0.18, -0.18]	-0.71 [-0.71, -0.71]	0.35 [0.35, 0.35]
MATH (HELM)	3.03 [3.03, 3.03]	0.62 [0.62, 0.62]	0.11 [0.11, 0.11]
MedQA	1.48 [1.48, 1.48]	1.18 [1.18, 1.18]	0.33 [0.33, 0.33]
MMLU	1.19 [1.19, 1.19]	-0.27 [-0.27, -0.27]	0.61 [0.61, 0.61]

Table 14: Random MIRT: gp minus p: per-benchmark MAE difference in percentage points, mean [95% paired t interval]. Negative favors the first method. All three nominal budgets observe exactly k items per target. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Benchmark	$k = 10$	$k = 30$	$k = 100$
ARC-Challenge	0.00 [0.00, 0.00]	-0.17 [-0.36, 0.01]	0.00 [0.00, 0.00]
BBH	0.00 [0.00, 0.00]	-0.47 [-0.57, -0.36]	-0.17 [-0.26, -0.09]
GPQA	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	-0.01 [-0.05, 0.03]
IFEval	-0.93 [-1.14, -0.71]	-0.21 [-0.34, -0.08]	-0.07 [-0.14, -0.00]
MMLU-Pro	-1.02 [-1.21, -0.83]	-1.08 [-1.36, -0.80]	-0.35 [-0.40, -0.29]
MuSR	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
MATH (OpenLLM)	-3.89 [-4.54, -3.25]	-1.71 [-1.93, -1.49]	-0.50 [-0.63, -0.36]
Commonsense	-2.84 [-3.88, -1.80]	-1.05 [-1.41, -0.70]	-0.21 [-0.41, -0.02]
GSM	-4.82 [-6.06, -3.58]	-2.58 [-3.29, -1.87]	-0.65 [-0.89, -0.41]
LegalBench	0.00 [0.00, 0.00]	-0.19 [-0.41, 0.03]	-0.14 [-0.26, -0.01]
MATH (HELM)	-3.00 [-3.91, -2.08]	-0.92 [-1.38, -0.46]	-0.17 [-0.36, 0.02]
MedQA	-1.29 [-1.65, -0.94]	-1.17 [-1.73, -0.61]	-0.25 [-0.45, -0.05]
MMLU	-0.47 [-0.92, -0.03]	-0.35 [-0.64, -0.06]	-0.04 [-0.30, 0.21]

Table 15: Crossfit-minus-reference MAE in percentage points across all 13 benchmarks, mean [95% paired t interval]. Negative favors crossfit. Every reference is retained, including tuned and matched-alpha imputation controls. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage.

Reference	$k = 10$	$k = 30$	$k = 100$
Random mean	-2.05 [-2.35, -1.75]	-1.15 [-1.33, -0.98]	-0.75 [-0.85, -0.65]
Source difference	-0.23 [-0.34, -0.12]	-0.17 [-0.22, -0.11]	-0.16 [-0.19, -0.13]
Random-subset ridge	3.03 [2.88, 3.18]	1.07 [1.00, 1.15]	0.29 [0.24, 0.34]
Paper AIPW (tuned)	0.09 [-0.19, 0.38]	0.30 [0.21, 0.38]	0.07 [0.04, 0.10]
Code AIPW (alpha 10)	0.02 [-0.20, 0.25]	0.27 [0.19, 0.36]	0.02 [-0.02, 0.05]
Half-fit impute (matched)	0.74 [0.58, 0.90]	-0.13 [-0.21, -0.05]	-0.34 [-0.39, -0.28]
Half-fit impute (tuned)	2.71 [2.56, 2.86]	1.03 [0.93, 1.12]	0.19 [0.14, 0.24]
Full-fit impute	3.03 [2.89, 3.18]	1.12 [1.04, 1.19]	0.34 [0.29, 0.39]
Crossfit clipped	0.11 [0.08, 0.15]	0.03 [0.01, 0.04]	0.00 [0.00, 0.00]

F.5 Less history and name-proxy exclusion

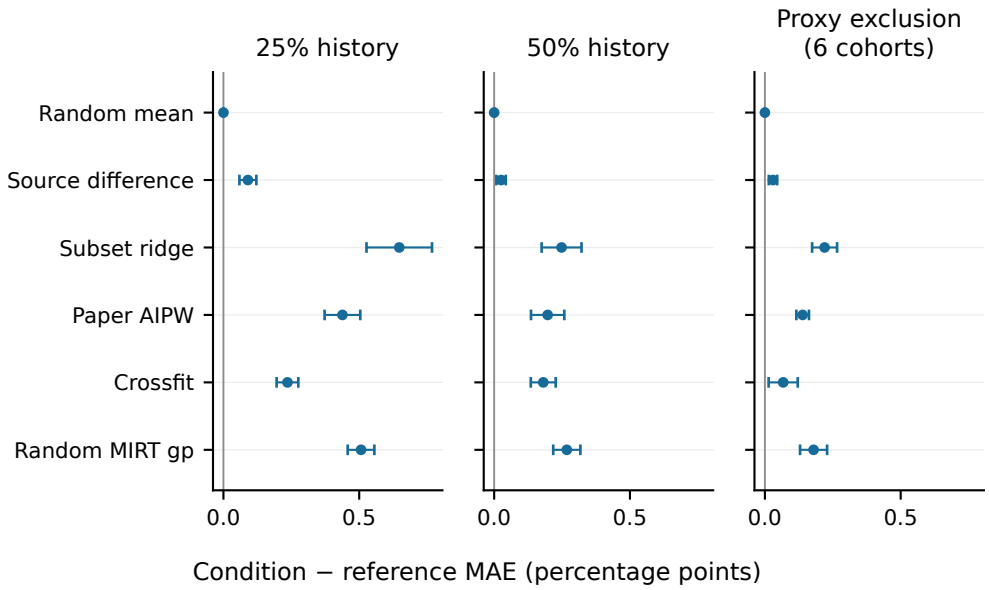


Figure 17: Sensitivity at $k = 30$: reduced-history minus full-history error across 13 benchmarks, and proxy-excluded minus ordinary-history error across six eligible cohorts. Each contrast uses the identical target cohort and item ordering. Positive values mean larger MAE under the stress condition. Name proxies do not establish ancestry or performance on a new target population; ARC is ineligible and HELM identity alignment is unresolved. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage.

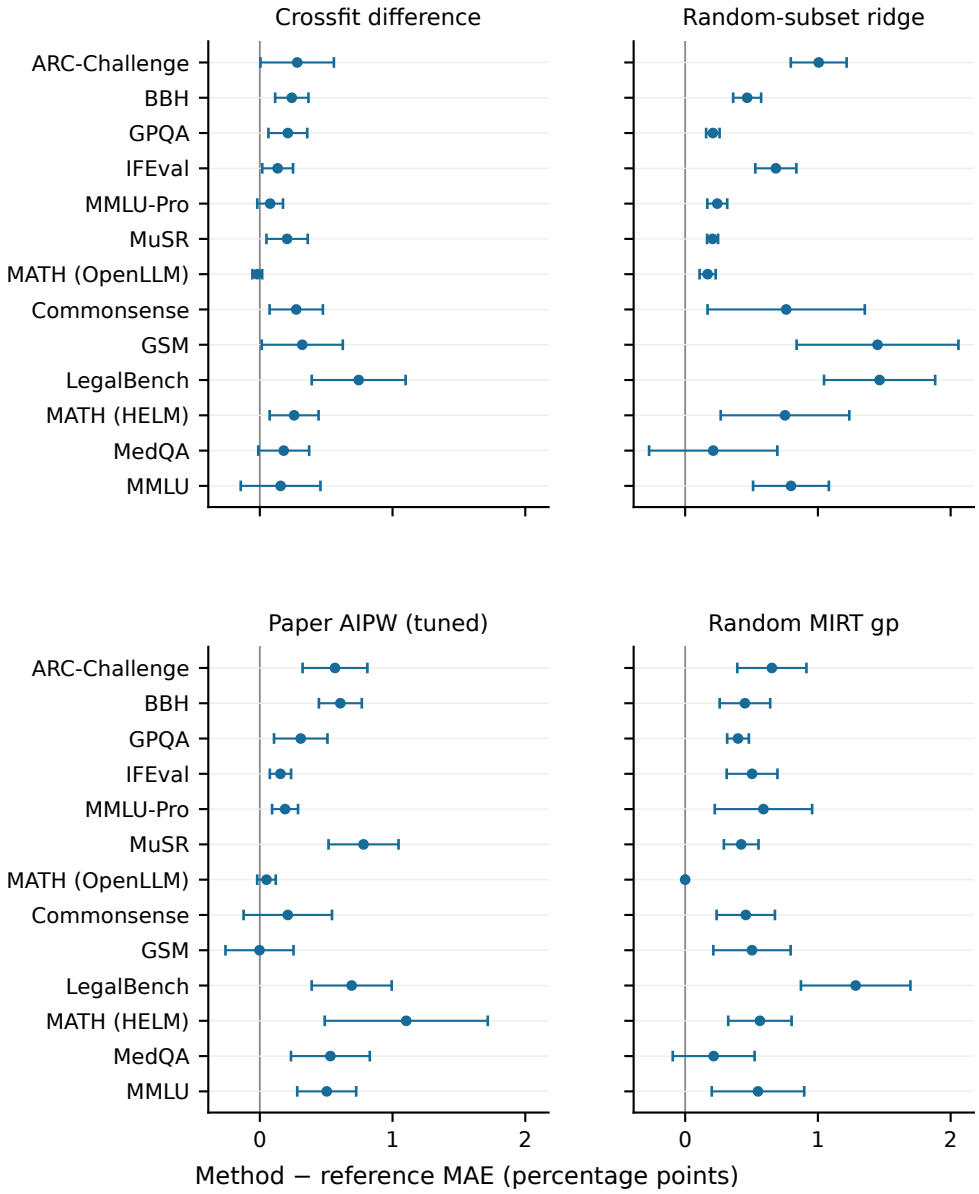


Figure 18: 25% history at $k = 30$: condition-minus-reference MAE on the same locked targets and item draws. Positive values mean worse error under the stress condition. All 13 matrices are retained. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage.

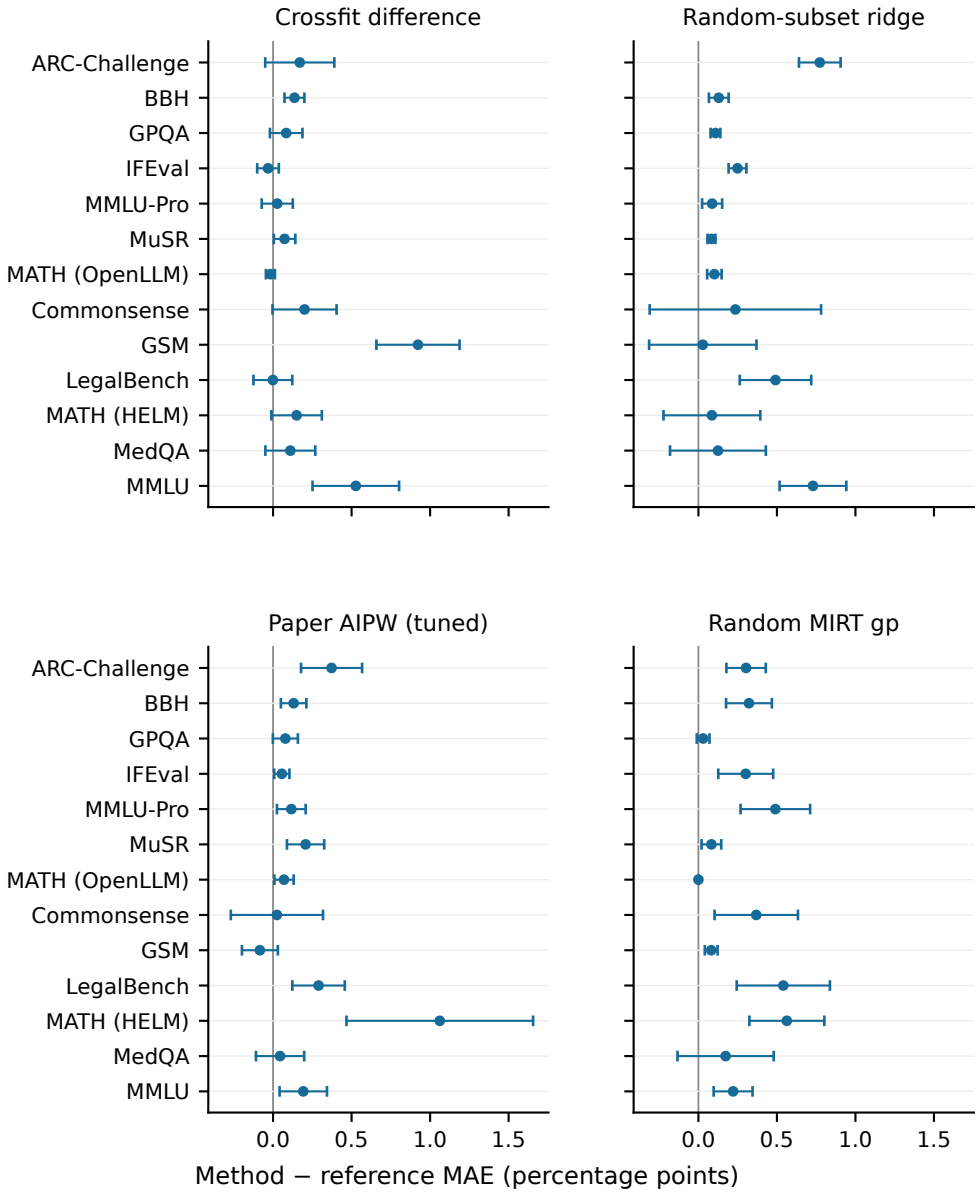


Figure 19: 50% history at $k = 30$: condition-minus-reference MAE on the same locked targets and item draws. Positive values mean worse error under the stress condition. All 13 matrices are retained. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage.

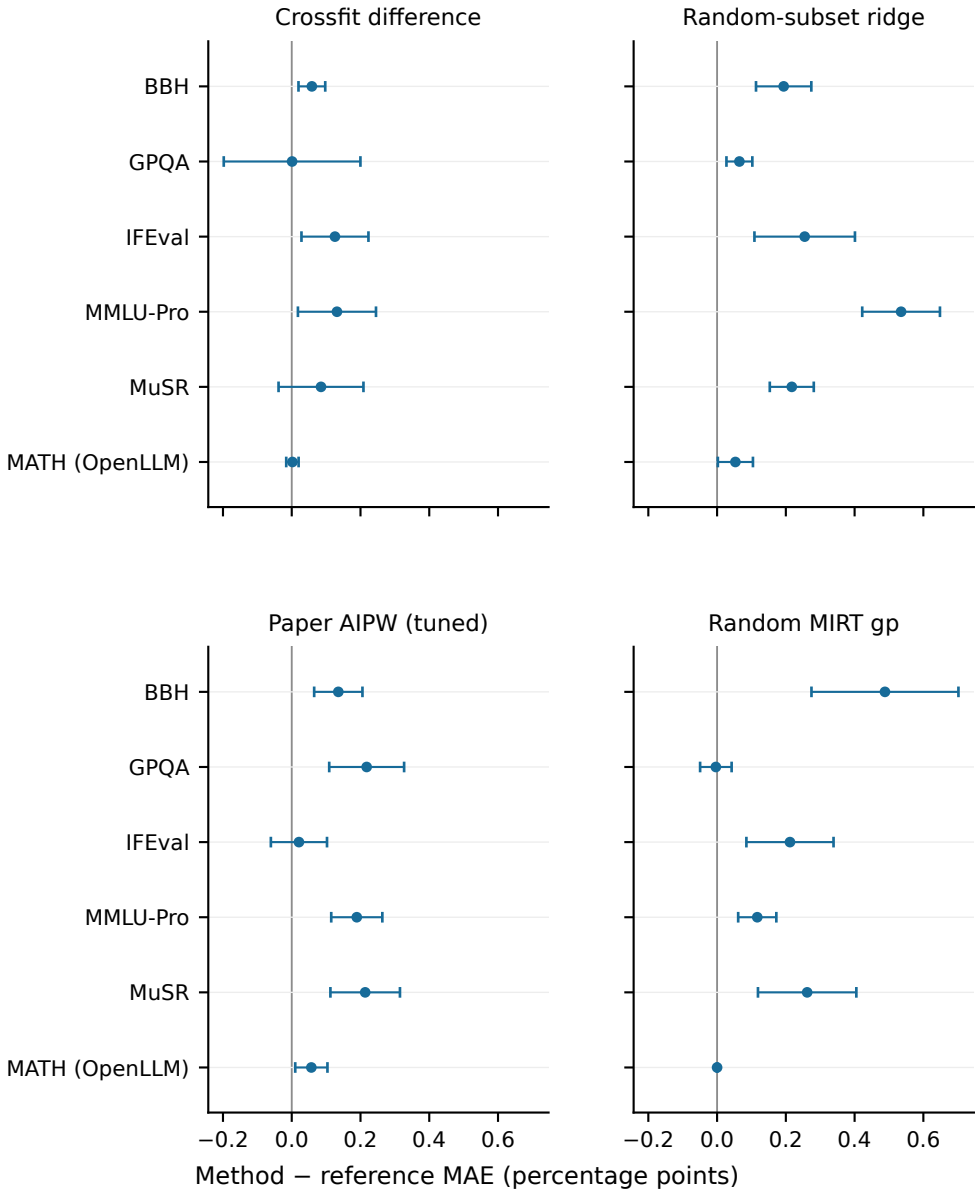


Figure 20: Proxy-excluded history at $k = 30$: condition-minus-reference MAE on the same locked targets and item draws. Positive values mean worse error under the stress condition. The six rows are eligible name-proxy cohorts, not verified ancestry; ARC and HELM are outside this comparison. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage.

Table 16: Condition-minus-reference MAE at $k = 30$, supplied paired mean [95% t interval]. Source-size means use 13 benchmarks; proxy exclusion uses six eligible same-target cohorts. Positive error differences mean worse performance. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage.

Method	25% history	50% history	Proxy exclusion
Source prior (0 items)	0.05 [0.05, 0.05]	0.06 [0.06, 0.06]	0.05 [0.05, 0.05]
Random mean	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	-0.00 [-0.00, 0.00]
Source difference	0.09 [0.06, 0.12]	0.03 [0.01, 0.04]	0.03 [0.01, 0.05]
Random-subset ridge	0.65 [0.53, 0.77]	0.25 [0.17, 0.32]	0.22 [0.17, 0.27]
Paper AIPW (tuned)	0.44 [0.37, 0.50]	0.20 [0.14, 0.26]	0.14 [0.12, 0.16]
Code AIPW (alpha 10)	0.29 [0.24, 0.33]	0.12 [0.09, 0.15]	0.12 [0.10, 0.15]
Crossfit difference	0.24 [0.20, 0.28]	0.18 [0.13, 0.23]	0.07 [0.01, 0.12]
Crossfit clipped	0.23 [0.19, 0.27]	0.18 [0.14, 0.23]	0.07 [0.02, 0.12]
Half-fit impute (tuned)	0.91 [0.79, 1.03]	0.38 [0.32, 0.44]	0.23 [0.19, 0.27]
Half-fit impute (matched)	4.46 [4.41, 4.50]	1.95 [1.89, 2.00]	0.65 [0.60, 0.70]
Full-fit impute	0.74 [0.63, 0.85]	0.27 [0.20, 0.35]	0.22 [0.17, 0.26]
Random MIRT p	1.75 [1.66, 1.84]	0.52 [0.47, 0.57]	0.32 [0.27, 0.37]
Random MIRT gp	0.51 [0.46, 0.56]	0.27 [0.22, 0.32]	0.18 [0.13, 0.23]

Table 17: Condition-minus-reference RMSE at $k = 30$, supplied paired mean [95% t interval]. Source-size means use 13 benchmarks; proxy exclusion uses six eligible same-target cohorts. Positive error differences mean worse performance. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage.

Method	25% history	50% history	Proxy exclusion
Source prior (0 items)	0.33 [0.33, 0.33]	0.18 [0.18, 0.18]	0.12 [0.12, 0.12]
Random mean	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	-0.00 [-0.00, 0.00]
Source difference	0.11 [0.08, 0.14]	0.04 [0.02, 0.06]	0.04 [0.02, 0.05]
Random-subset ridge	0.81 [0.68, 0.95]	0.32 [0.22, 0.41]	0.30 [0.23, 0.36]
Paper AIPW (tuned)	0.52 [0.44, 0.60]	0.25 [0.18, 0.32]	0.16 [0.13, 0.20]
Code AIPW (alpha 10)	0.36 [0.31, 0.42]	0.17 [0.14, 0.21]	0.16 [0.13, 0.19]
Crossfit difference	0.29 [0.24, 0.34]	0.23 [0.17, 0.29]	0.09 [0.02, 0.15]
Crossfit clipped	0.28 [0.23, 0.34]	0.23 [0.17, 0.29]	0.09 [0.02, 0.15]
Half-fit impute (tuned)	1.18 [1.03, 1.32]	0.51 [0.42, 0.60]	0.32 [0.26, 0.37]
Half-fit impute (matched)	5.82 [5.77, 5.87]	2.55 [2.47, 2.64]	0.90 [0.84, 0.96]
Full-fit impute	0.94 [0.81, 1.07]	0.36 [0.26, 0.45]	0.29 [0.23, 0.36]
Random MIRT p	1.91 [1.83, 2.00]	0.56 [0.51, 0.61]	0.43 [0.38, 0.48]
Random MIRT gp	0.62 [0.57, 0.68]	0.34 [0.30, 0.39]	0.23 [0.18, 0.29]

Table 18: Condition-minus-reference Spearman ρ at $k = 30$, supplied paired mean [95% t interval]. Source-size means use 13 benchmarks; proxy exclusion uses six eligible same-target cohorts. Positive correlation differences mean better ranking. Undefined means no valid paired rank draw; constant source-prior ranks are never zero filled. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage.

Method	25% history	50% history	Proxy exclusion
Source prior (0 items)	undefined	undefined	undefined
Random mean	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Source difference	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Random-subset ridge	-0.027 [-0.032, -0.021]	-0.010 [-0.014, -0.005]	-0.015 [-0.023, -0.007]
Paper AIPW (tuned)	-0.015 [-0.020, -0.010]	-0.005 [-0.008, -0.002]	-0.007 [-0.014, -0.000]
Code AIPW (alpha 10)	-0.009 [-0.013, -0.005]	-0.005 [-0.008, -0.002]	-0.003 [-0.008, 0.002]
Crossfit difference	-0.005 [-0.009, -0.002]	-0.004 [-0.006, -0.002]	0.003 [-0.006, 0.012]
Crossfit clipped	-0.005 [-0.009, -0.002]	-0.004 [-0.006, -0.002]	0.003 [-0.006, 0.012]
Half-fit impute (tuned)	-0.030 [-0.037, -0.024]	-0.011 [-0.016, -0.007]	-0.015 [-0.022, -0.008]
Half-fit impute (matched)	-0.021 [-0.028, -0.015]	-0.007 [-0.009, -0.005]	-0.011 [-0.019, -0.003]
Full-fit impute	-0.025 [-0.030, -0.020]	-0.011 [-0.015, -0.006]	-0.015 [-0.023, -0.007]
Random MIRT p	-0.023 [-0.027, -0.019]	-0.008 [-0.010, -0.006]	-0.011 [-0.019, -0.003]
Random MIRT gp	-0.025 [-0.030, -0.020]	-0.009 [-0.011, -0.007]	-0.010 [-0.017, -0.004]

Table 19: Condition-minus-reference Kendall τ_b at $k = 30$, supplied paired mean [95% t interval]. Source-size means use 13 benchmarks; proxy exclusion uses six eligible same-target cohorts. Positive correlation differences mean better ranking. Undefined means no valid paired rank draw; constant source-prior ranks are never zero filled. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage.

Method	25% history	50% history	Proxy exclusion
Source prior (0 items)	undefined	undefined	undefined
Random mean	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Source difference	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Random-subset ridge	-0.029 [-0.036, -0.022]	-0.009 [-0.014, -0.004]	-0.015 [-0.021, -0.008]
Paper AIPW (tuned)	-0.014 [-0.019, -0.009]	-0.004 [-0.007, -0.001]	-0.007 [-0.012, -0.002]
Code AIPW (alpha 10)	-0.008 [-0.012, -0.004]	-0.004 [-0.007, -0.001]	-0.004 [-0.009, 0.000]
Crossfit difference	-0.004 [-0.007, -0.001]	-0.004 [-0.007, -0.001]	0.003 [-0.005, 0.010]
Crossfit clipped	-0.004 [-0.007, -0.001]	-0.004 [-0.007, -0.002]	0.003 [-0.005, 0.010]
Half-fit impute (tuned)	-0.034 [-0.041, -0.026]	-0.011 [-0.016, -0.006]	-0.014 [-0.020, -0.009]
Half-fit impute (matched)	-0.020 [-0.026, -0.014]	-0.007 [-0.011, -0.004]	-0.010 [-0.017, -0.003]
Full-fit impute	-0.028 [-0.034, -0.021]	-0.011 [-0.016, -0.006]	-0.014 [-0.021, -0.007]
Random MIRT p	-0.023 [-0.028, -0.019]	-0.009 [-0.011, -0.007]	-0.010 [-0.020, -0.001]
Random MIRT gp	-0.022 [-0.026, -0.017]	-0.009 [-0.012, -0.007]	-0.007 [-0.014, -0.000]

F.6 Collisions, ties, tail errors and finite-draw decomposition

Table 20: Retrospective complete-response collisions and crossfit MAE at $k = 30$ (percentage points). Targets are distinct identities or opaque HELM rows; patterns are distinct complete target vectors; matches count target vectors occurring in history. Primary weights target records equally; pattern weights each distinct target vector equally. Match and nonmatch errors condition on complete responses only after prediction. Dashes mean an empty diagnostic subgroup, not zero error. Values are descriptive draw means; these outcome-conditioned populations do not define retraining or prove lineage independence.

Benchmark	Targets	Patterns	Matches	Primary	Pattern	Match	Nonmatch
ARC-Challenge	64	64	1	4.98	4.98	4.67	4.99
BBH	129	129	1	5.92	5.92	4.24	5.94
GPQA	129	127	3	5.85	5.89	3.38	5.91
IFEval	129	128	0	4.62	4.63	–	4.62
MMLU-Pro	129	127	2	5.18	5.21	2.52	5.22
MuSR	129	128	0	5.35	5.35	–	5.35
MATH (OpenLLM)	129	119	13	2.16	2.28	0.76	2.31
Commonsense	25	25	1	3.79	3.79	2.84	3.83
GSM	25	25	0	4.94	4.94	–	4.94
LegalBench	24	24	0	5.13	5.13	–	5.13
MATH (HELM)	25	25	0	4.72	4.72	–	4.72
MedQA	25	24	0	5.42	5.47	–	5.42
MMLU	25	25	0	4.81	4.81	–	4.81

Table 21: Unordered within-benchmark pair counts at $k = 30$. Truth ties have exactly equal canonicalized full scores. Close pairs have a nonzero true gap at most one percentage point; wide pairs have a larger gap. The final three columns give mean predicted-tie counts across 20 draws. Truth categories partition the target pairs; predicted ties are reported separately from inversions. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged.

Benchmark	Truth ties	Close pairs	Wide pairs	Random ties	Crossfit ties	Ridge ties
ARC-Challenge	3	112	1901	141.7	2.2	2.2
BBH	5	447	7804	563.8	0.3	0.3
GPQA	77	1581	6598	961.5	20.4	20.4
IFEval	29	303	7924	421.4	8.8	8.8
MMLU-Pro	5	567	7684	559.0	10.4	10.4
MuSR	83	1060	7113	999.0	1.9	1.9
MATH (OpenLLM)	187	2173	5896	2868.2	2496.4	2496.4
Commonsense	2	17	281	32.9	9.1	9.1
GSM	1	13	286	14.1	1.4	1.4
LegalBench	0	23	253	26.1	0.1	0.1
MATH (HELM)	1	10	289	13.3	0.3	0.4
MedQA	2	10	288	17.1	1.2	1.2
MMLU	1	10	289	21.8	0.3	0.3

Table 22: Mean within-bin inversion and predicted-tie percentages at $k = 30$, equal weighting of the 13 benchmarks and 20 draws. Every benchmark has nonempty close and wide bins. A predicted tie is not an inversion, so the two rates must be read together. The zero-item source prior ties every pair and therefore has no defined Spearman or Kendall correlation, even though its inversion rate is zero. These are descriptive fixed-pool means. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged.

Method	Close inv. (%)	Close ties (%)	Wide inv. (%)	Wide ties (%)
Source prior (0 items)	0.00	100.00	0.00	100.00
Random mean	37.60	20.70	11.85	8.25
Source difference	37.60	20.70	11.85	8.25
Random-subset ridge	44.66	5.90	12.23	1.64
Paper AIPW (tuned)	44.57	5.86	14.06	1.64
Code AIPW (alpha 10)	44.82	5.86	14.10	1.64
Crossfit difference	44.60	5.86	14.69	1.64
Crossfit clipped	44.53	5.96	14.69	1.65
Half-fit impute (tuned)	45.33	5.86	12.17	1.64
Half-fit impute (matched)	44.33	5.86	11.64	1.64
Full-fit impute	44.68	5.86	12.09	1.64
Correctness-anchor mean	46.54	3.19	11.27	0.43
Correctness-anchor ridge	47.43	2.51	9.25	0.18
Random MIRT p	44.83	5.86	12.35	1.64
Random MIRT gp	44.33	6.19	12.33	2.00
IRT-anchor mean	42.22	4.44	13.76	0.93
IRT-anchor MIRT p	41.74	4.35	11.69	0.79
IRT-anchor MIRT gp	41.65	4.36	11.15	0.80

Table 23: Score diagnostics at $k = 10$: equal-weight means over the 13 benchmarks and 20 draws. Bias is signed predicted-minus-true error in percentage points. Tail is the mean of per-benchmark/draw 90th-percentile absolute errors, not a pooled-target quantile. Out is the percentage of raw predictions outside $[0, 1]$. The clipped candidate is separately named; the source prior observes zero items. These are descriptive fixed-pool diagnostics, not inferential intervals.

Method	Bias (pp)	Tail (pp)	Out (%)
Source prior (0 items)	-0.93	20.97	0.00
Random mean	-0.49	20.75	0.00
Source difference	+0.12	17.62	5.14
Random-subset ridge	-0.09	10.91	0.02
Paper AIPW (tuned)	-0.20	17.10	0.25
Code AIPW (alpha 10)	-0.12	17.18	0.29
Crossfit difference	+0.13	17.29	4.99
Crossfit clipped	+0.17	17.13	0.00
Half-fit impute (tuned)	-0.16	11.11	0.00
Half-fit impute (matched)	-0.83	14.41	0.00
Full-fit impute	-0.08	10.91	0.02
Correctness-anchor mean	-0.31	13.81	0.00
Correctness-anchor ridge	+0.16	9.58	0.00
Random MIRT p	-1.30	14.01	0.00
Random MIRT gp	-0.89	12.82	0.00
IRT-anchor mean	-0.74	16.61	0.00
IRT-anchor MIRT p	-1.65	13.91	0.00
IRT-anchor MIRT gp	-1.38	11.83	0.00

Table 24: Score diagnostics at $k = 30$: equal-weight means over the 13 benchmarks and 20 draws. Bias is signed predicted-minus-true error in percentage points. Tail is the mean of per-benchmark/draw 90th-percentile absolute errors, not a pooled-target quantile. Out is the percentage of raw predictions outside $[0, 1]$. The clipped candidate is separately named; the source prior observes zero items. These are descriptive fixed-pool diagnostics, not inferential intervals.

Method	Bias (pp)	Tail (pp)	Out (%)
Source prior (0 items)	-0.93	20.97	0.00
Random mean	-0.29	11.76	0.00
Source difference	-0.06	10.12	3.01
Random-subset ridge	-0.04	7.85	0.02
Paper AIPW (tuned)	-0.04	9.17	0.05
Code AIPW (alpha 10)	-0.03	9.24	0.03
Crossfit difference	-0.04	9.85	2.96
Crossfit clipped	-0.03	9.82	0.00
Half-fit impute (tuned)	-0.09	7.70	0.00
Half-fit impute (matched)	-0.19	9.53	0.00
Full-fit impute	-0.08	7.68	0.02
Correctness-anchor mean	-0.77	8.90	0.00
Correctness-anchor ridge	-0.10	6.52	0.00
Random MIRT p	-0.75	9.01	0.00
Random MIRT gp	-0.53	8.20	0.00
IRT-anchor mean	+0.01	9.25	0.00
IRT-anchor MIRT p	-0.82	8.59	0.00
IRT-anchor MIRT gp	-0.41	7.32	0.00

Table 25: Score diagnostics at $k = 100$: equal-weight means over the 13 benchmarks and 20 draws. Bias is signed predicted-minus-true error in percentage points. Tail is the mean of per-benchmark/draw 90th-percentile absolute errors, not a pooled-target quantile. Out is the percentage of raw predictions outside $[0, 1]$. The clipped candidate is separately named; the source prior observes zero items. These are descriptive fixed-pool diagnostics, not inferential intervals.

Method	Bias (pp)	Tail (pp)	Out (%)
Source prior (0 items)	-0.93	20.97	0.00
Random mean	-0.03	6.34	0.02
Source difference	-0.04	5.35	0.77
Random-subset ridge	-0.02	4.52	0.00
Paper AIPW (tuned)	-0.03	4.89	0.00
Code AIPW (alpha 10)	-0.06	5.05	0.03
Crossfit difference	-0.06	5.05	0.94
Crossfit clipped	-0.06	5.05	0.00
Half-fit impute (tuned)	-0.03	4.71	0.00
Half-fit impute (matched)	-0.06	5.74	0.00
Full-fit impute	-0.03	4.41	0.00
Correctness-anchor mean	-0.52	6.49	0.00
Correctness-anchor ridge	+0.07	4.47	0.31
Random MIRT p	-0.17	4.92	0.00
Random MIRT gp	-0.04	4.65	0.00
IRT-anchor mean	+0.21	5.05	0.00
IRT-anchor MIRT p	-0.34	4.89	0.00
IRT-anchor MIRT gp	+0.01	3.97	0.00

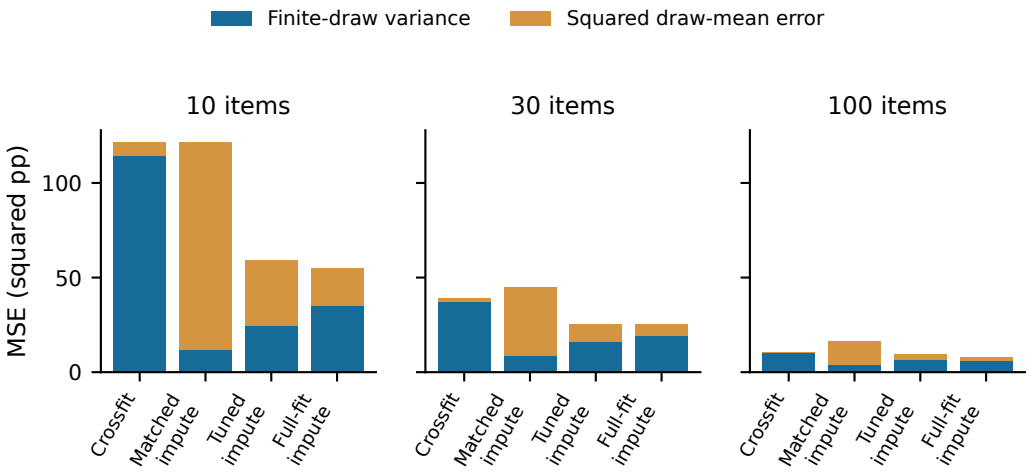


Figure 21: Descriptive decomposition of mean squared error over the actual 20 draws, then averaged equally over benchmarks. At each target the stored identity uses variance with divisor 20 plus squared error of the draw-mean prediction. The orange component is not an unbiased estimate of squared population design bias. Matched imputation uses candidate alpha, tuned imputation selects its own alpha, and full-fit imputation uses all k responses. No interval or generalization claim is inferred from this finite-draw identity.

F.7 Formula, numerical, optimization and compute diagnostics

Table 26: Supplementary matched-alpha formula diagnostic: code10 minus paper10, mean [95% paired t interval] across 13 benchmarks. Both estimates use the same alpha-10 ridge fit and the same observed items; $\widehat{\mu}_{\mathrm{paper10}} = (1 - k/N)\widehat{\mu}_{\mathrm{code10}} + (k/N)\bar{y}_S$. This isolates the return formula; the primary tuned-paper versus fixed-code comparison additionally changes alpha. Negative MAE/RMSE differences favor code10. The supplied 95% Student t intervals describe the mean across 20 independent item orderings, conditional on the fixed historical and target pools. They quantify Monte Carlo mean uncertainty, not future-model or future-draw coverage. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged.

Metric	$k = 10$	$k = 30$	$k = 100$
MAE (pp)	-0.0062 [-0.0083, -0.0042]	-0.0022 [-0.0061, 0.0016]	0.0498 [0.0392, 0.0604]
RMSE (pp)	-0.0062 [-0.0079, -0.0045]	-0.0027 [-0.0064, 0.0010]	0.0662 [0.0549, 0.0775]
Spearman ρ	0.0000 [-0.0002, 0.0002]	-0.0002 [-0.0007, 0.0002]	-0.0022 [-0.0028, -0.0016]
Kendall τ_b	0.0001 [-0.0002, 0.0003]	-0.0002 [-0.0007, 0.0003]	-0.0035 [-0.0043, -0.0026]
Bias (pp)	0.0065 [-0.0012, 0.0141]	0.0117 [-0.0062, 0.0296]	-0.0131 [-0.0413, 0.0151]

Table 27: Rank-only numerical sensitivity relative to the final 12-decimal convention. Changed counts compare 20,940 defined pairs of method-cell correlations out of 22,360 cells; 1,420 constant-prior cells are undefined. Cell max is the largest individual correlation change; macro max and endpoint max bound changes in the supplied primary macro contrasts and their endpoints. Fourteen decimals still fragment mathematical ties; near-equality of learned predictions is not universally a mathematical tie. The post-scoring repair changes no predictions or raw error fields, and the initial raw-rank outputs are archived.

Decimals	Metric	Changed	Cell max	Macro max	Endpoint max
10	ρ	1	0.000141	0.000000	0.000000
10	τ_b	1	0.000160	0.000000	0.000000
14	ρ	27	0.098376	0.000382	0.000856
14	τ_b	27	0.092516	0.000371	0.000769

Table 28: Audited cached MAP-MIRT source fits across all scenarios: development fits, selected historical refits and fits needing the frozen continuation. All 219 final records met the relative-objective stopping rule; this does not establish global optimality or a small gradient. Grad. is the largest saved absolute gradient component. Elapsed sums each distinct unaffected saved fit once, in seconds, and is not CPU time. † excludes one MMLU-Pro fit whose original 165044.54 seconds includes world suspension; the separate timing replication is reported with command costs. Source caches serve multiple budgets and methods.

Benchmark	Dev.	Refit	Continued	Grad.	Elapsed (s)
ARC-Challenge	12	4	0	0.111	21.22
BBH	16	5	2	0.423	282.71
GPQA	16	4	0	0.306	24.65
IFEval	16	5	0	0.053	18.94
MMLU-Pro	16	5	6	1.250	855.29†
MuSR	16	4	0	0.068	19.41
MATH (OpenLLM)	16	4	5	0.077	53.93
Commonsense	10	3	0	0.021	3.21
GSM	10	3	0	0.082	7.58
LegalBench	10	3	0	0.030	11.97
MATH (HELM)	10	3	0	0.017	2.84
MedQA	10	3	0	0.018	6.14
MMLU	10	5	0	0.022	4.65

Table 29: Measured command costs, including cache reads, loading and output. CPU is user plus system time; elapsed is wall time; RSS is process peak MiB (1024 KiB). The reproduction proxy sums source-replay, original cached-prediction and corrected-scoring CPU from separate runs; original source CPU and original end-to-end wall time are unknown. Concurrent source replay spans 1155.498 wall seconds. Do not sum concurrent elapsed times or process memory peaks. The proxy excludes acquisition, pilots, initial scoring, diagnostic audits, formula reporting, rendering and setup. The single-fit replay only times the suspension-affected source fit. A differing development loss grid limits exact replay equivalence (see source-reproduction checks). No LLM evaluation or API cost was measured.

Command	CPU (s)	Elapsed (s)	RSS (MiB)
Source replay A	1084.819	1155.498	422.1
Source replay B	375.310	446.082	307.2
Cached prediction A	37.743	37.752	462.9
Cached prediction B	29.725	29.825	416.4
Corrected scoring	237.245	237.369	1221.3
Reproduction proxy	1764.841	—	—
Initial scoring (excluded)	234.205	234.290	1223.4
Single-fit replay (excluded)	173.779	173.822	409.3

Table 30: Assigned method-alone elapsed costs in seconds for the $k = 30$ primary grid (13 benchmarks, 20 draws), excluding cached MIRT fitting. Selection includes source-dependent statistic construction; prediction covers the target batch using available responses. Fixed anchors are selected once per benchmark and budget. Mean/ridge anchor costs share construction and must not be summed as separate physical executions; gp variants reuse their corresponding p inference. These allocations answer an amortization question and do not sum to measured command CPU or include running a language model.

Method	Selection (s)	Target batch (s)
Source prior (0 items)	0.1117	0.0033
Random mean	0.0037	0.0028
Source difference	0.1175	0.0033
Random-subset ridge	0.1767	0.0031
Paper AIPW (tuned)	0.1859	0.0030
Code AIPW (alpha 10)	0.1769	0.0030
Crossfit difference	0.3147	0.0028
Crossfit clipped	0.3064	0.0069
Half-fit impute (tuned)	0.2893	0.0028
Half-fit impute (matched)	0.2859	0.0027
Full-fit impute	0.1605	0.0027
Correctness-anchor mean	4.4905	0.0038
Correctness-anchor ridge	4.5010	0.0029
Random MIRT p	0.0000	1.8733
Random MIRT gp	0.0000	shared with p
IRT-anchor mean	0.5020	0.0027
IRT-anchor MIRT p	0.0000	1.8603
IRT-anchor MIRT gp	0.0000	shared with p

Table 31: Retrospective source replay checks; the original scientific artifacts are preserved. Fit/parameter agreement uses the analysis audit tolerances. One GSM correctness-anchor ridge development grid differs; a bounded repeated one-/two-thread diagnostic reproduces the replay/original grids, with one of 30 development representatives changing. This does not identify the unrecorded historical thread count. All 78 final anchor selections and mean predictions reproduce exactly. The source-equivalence assertion failed and is disclosed rather than treating every development loss as equal. A separate supplemental formula glob failure was repaired by excluding timing files; neither failure removes primary observations. Rank repair and MIRT continuation diagnostics are reported separately.

Check	Result
MIRT fit parameters/objectives/attempts	219 / 219
Selected-parameter packages	45 / 45
Final anchor selections/mean predictions	78 / 78 exact
Differing development loss grids	1 (GSM/main/$k = 30$)
Largest development-loss change	0.451171 pp
Selected correctness-anchor ridge alpha	1 in both runs
Historical effective thread count	unrecorded

F.8 Complete per-benchmark, all-method results

All eighteen primary methods and three budgets are retained for each benchmark and metric. A dash with an explicit undefined count is not a zero correlation. Benchmark-specific intervals retain their fixed historical and target pools.

Table 32: ARC-Challenge: MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 0.85%, 2.56%, 8.53% of $N = 1,172$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	11.21 [11.21, 11.21]	11.21 [11.21, 11.21]	11.21 [11.21, 11.21]
Random mean	11.87 [10.16, 13.57]	6.86 [5.80, 7.92]	3.55 [2.84, 4.26]
Source difference	8.97 [8.34, 9.59]	5.25 [4.98, 5.51]	2.70 [2.55, 2.85]
Random-subset ridge	5.94 [5.47, 6.41]	3.78 [3.51, 4.04]	2.22 [2.11, 2.32]
Paper AIPW (tuned)	7.82 [6.99, 8.65]	4.15 [3.87, 4.43]	2.42 [2.31, 2.54]
Code AIPW (alpha 10)	8.11 [7.23, 8.99]	4.24 [3.97, 4.50]	2.37 [2.25, 2.49]
Crossfit difference	8.54 [7.97, 9.12]	4.98 [4.72, 5.24]	2.56 [2.43, 2.69]
Crossfit clipped	8.54 [7.96, 9.11]	4.98 [4.72, 5.24]	2.56 [2.43, 2.69]
Half-fit impute (tuned)	6.58 [6.08, 7.07]	3.93 [3.64, 4.22]	2.25 [2.16, 2.34]
Half-fit impute (matched)	7.49 [6.98, 8.01]	4.56 [4.21, 4.91]	4.35 [4.23, 4.47]
Full-fit impute	5.94 [5.47, 6.41]	3.78 [3.51, 4.04]	2.20 [2.09, 2.31]
Correctness-anchor mean	5.28 [5.28, 5.28]	3.83 [3.83, 3.83]	2.49 [2.49, 2.49]
Correctness-anchor ridge	4.24 [4.24, 4.24]	2.90 [2.90, 2.90]	2.14 [2.14, 2.14]
Random MIRT p	7.27 [6.70, 7.84]	4.52 [4.20, 4.84]	2.37 [2.25, 2.49]
Random MIRT gp	7.27 [6.70, 7.84]	4.35 [3.97, 4.72]	2.37 [2.25, 2.49]
IRT-anchor mean	9.23 [9.23, 9.23]	4.39 [4.39, 4.39]	2.82 [2.82, 2.82]
IRT-anchor MIRT p	8.10 [8.10, 8.10]	4.09 [4.09, 4.09]	2.51 [2.51, 2.51]
IRT-anchor MIRT gp	7.87 [7.87, 7.87]	3.77 [3.77, 3.77]	2.41 [2.41, 2.41]

Table 33: ARC-Challenge: RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 0.85%, 2.56%, 8.53% of $N = 1,172$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	13.66 [13.66, 13.66]	13.66 [13.66, 13.66]	13.66 [13.66, 13.66]
Random mean	14.25 [12.43, 16.07]	8.17 [7.03, 9.32]	4.31 [3.58, 5.03]
Source difference	11.03 [10.35, 11.71]	6.46 [6.12, 6.81]	3.42 [3.27, 3.56]
Random-subset ridge	7.30 [6.76, 7.85]	4.76 [4.43, 5.09]	2.80 [2.69, 2.91]
Paper AIPW (tuned)	9.99 [8.96, 11.02]	5.26 [4.89, 5.63]	3.06 [2.95, 3.18]
Code AIPW (alpha 10)	10.23 [9.15, 11.32]	5.29 [4.96, 5.62]	3.00 [2.87, 3.13]
Crossfit difference	10.49 [9.85, 11.14]	6.15 [5.81, 6.49]	3.22 [3.09, 3.34]
Crossfit clipped	10.48 [9.84, 11.12]	6.15 [5.81, 6.49]	3.22 [3.09, 3.34]
Half-fit impute (tuned)	8.05 [7.49, 8.60]	4.87 [4.53, 5.22]	2.85 [2.73, 2.98]
Half-fit impute (matched)	9.12 [8.55, 9.68]	5.67 [5.26, 6.08]	5.56 [5.42, 5.69]
Full-fit impute	7.30 [6.76, 7.85]	4.76 [4.43, 5.09]	2.78 [2.67, 2.89]
Correctness-anchor mean	6.55 [6.55, 6.55]	5.28 [5.28, 5.28]	3.30 [3.30, 3.30]
Correctness-anchor ridge	5.53 [5.53, 5.53]	3.45 [3.45, 3.45]	2.72 [2.72, 2.72]
Random MIRT p	8.77 [8.15, 9.40]	5.61 [5.23, 5.99]	2.98 [2.85, 3.12]
Random MIRT gp	8.77 [8.15, 9.40]	5.32 [4.88, 5.75]	2.98 [2.85, 3.12]
IRT-anchor mean	11.48 [11.48, 11.48]	5.52 [5.52, 5.52]	3.39 [3.39, 3.39]
IRT-anchor MIRT p	9.71 [9.71, 9.71]	5.03 [5.03, 5.03]	3.09 [3.09, 3.09]
IRT-anchor MIRT gp	9.47 [9.47, 9.47]	4.54 [4.54, 4.54]	2.93 [2.93, 2.93]

Table 34: ARC-Challenge: Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 0.85%, 2.56%, 8.53% of $N = 1,172$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.695 [0.617, 0.774]	0.875 [0.859, 0.891]	0.949 [0.942, 0.955]
Source difference	0.695 [0.617, 0.774]	0.875 [0.859, 0.891]	0.949 [0.942, 0.955]
Random-subset ridge	0.752 [0.701, 0.803]	0.897 [0.885, 0.909]	0.957 [0.953, 0.960]
Paper AIPW (tuned)	0.733 [0.682, 0.785]	0.892 [0.879, 0.905]	0.953 [0.948, 0.957]
Code AIPW (alpha 10)	0.720 [0.661, 0.779]	0.890 [0.877, 0.902]	0.954 [0.948, 0.959]
Crossfit difference	0.713 [0.646, 0.780]	0.876 [0.861, 0.891]	0.951 [0.945, 0.957]
Crossfit clipped	0.713 [0.646, 0.780]	0.876 [0.861, 0.891]	0.951 [0.945, 0.957]
Half-fit impute (tuned)	0.754 [0.706, 0.802]	0.892 [0.878, 0.906]	0.956 [0.951, 0.960]
Half-fit impute (matched)	0.753 [0.707, 0.800]	0.894 [0.881, 0.907]	0.953 [0.949, 0.957]
Full-fit impute	0.752 [0.701, 0.802]	0.897 [0.886, 0.909]	0.957 [0.953, 0.960]
Correctness-anchor mean	0.883 [0.883, 0.883]	0.914 [0.914, 0.914]	0.977 [0.977, 0.977]
Correctness-anchor ridge	0.875 [0.875, 0.875]	0.925 [0.925, 0.925]	0.970 [0.970, 0.970]
Random MIRT p	0.756 [0.710, 0.802]	0.897 [0.883, 0.912]	0.960 [0.956, 0.963]
Random MIRT gp	0.756 [0.710, 0.802]	0.897 [0.884, 0.911]	0.960 [0.956, 0.963]
IRT-anchor mean	0.604 [0.604, 0.604]	0.877 [0.877, 0.877]	0.943 [0.943, 0.943]
IRT-anchor MIRT p	0.641 [0.641, 0.641]	0.902 [0.902, 0.902]	0.948 [0.948, 0.948]
IRT-anchor MIRT gp	0.634 [0.634, 0.634]	0.896 [0.896, 0.896]	0.952 [0.952, 0.952]

Table 35: ARC-Challenge: Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 0.85%, 2.56%, 8.53% of $N = 1,172$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.559 [0.490, 0.627]	0.726 [0.704, 0.747]	0.834 [0.823, 0.845]
Source difference	0.559 [0.490, 0.627]	0.726 [0.704, 0.747]	0.834 [0.823, 0.845]
Random-subset ridge	0.583 [0.535, 0.631]	0.739 [0.721, 0.756]	0.839 [0.831, 0.847]
Paper AIPW (tuned)	0.566 [0.521, 0.610]	0.730 [0.711, 0.748]	0.833 [0.825, 0.841]
Code AIPW (alpha 10)	0.556 [0.506, 0.607]	0.726 [0.707, 0.744]	0.833 [0.823, 0.843]
Crossfit difference	0.547 [0.489, 0.605]	0.705 [0.684, 0.726]	0.829 [0.818, 0.840]
Crossfit clipped	0.547 [0.489, 0.605]	0.705 [0.684, 0.726]	0.829 [0.818, 0.840]
Half-fit impute (tuned)	0.583 [0.540, 0.627]	0.731 [0.710, 0.752]	0.836 [0.827, 0.845]
Half-fit impute (matched)	0.585 [0.542, 0.627]	0.734 [0.714, 0.754]	0.835 [0.827, 0.843]
Full-fit impute	0.583 [0.535, 0.630]	0.738 [0.721, 0.756]	0.840 [0.832, 0.848]
Correctness-anchor mean	0.717 [0.717, 0.717]	0.766 [0.766, 0.766]	0.878 [0.878, 0.878]
Correctness-anchor ridge	0.707 [0.707, 0.707]	0.788 [0.788, 0.788]	0.873 [0.873, 0.873]
Random MIRT p	0.587 [0.545, 0.629]	0.739 [0.718, 0.760]	0.847 [0.840, 0.854]
Random MIRT gp	0.587 [0.545, 0.629]	0.738 [0.718, 0.758]	0.847 [0.840, 0.854]
IRT-anchor mean	0.431 [0.431, 0.431]	0.706 [0.706, 0.706]	0.804 [0.804, 0.804]
IRT-anchor MIRT p	0.465 [0.465, 0.465]	0.742 [0.742, 0.742]	0.822 [0.822, 0.822]
IRT-anchor MIRT gp	0.459 [0.459, 0.459]	0.735 [0.735, 0.735]	0.833 [0.833, 0.833]

Table 36: BBH: MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 0.17%, 0.52%, 1.74% of $N = 5,761$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	10.27 [10.27, 10.27]	10.27 [10.27, 10.27]	10.27 [10.27, 10.27]
Random mean	11.97 [10.97, 12.97]	6.77 [6.39, 7.15]	3.87 [3.50, 4.24]
Source difference	10.63 [10.26, 11.01]	6.28 [5.96, 6.60]	3.37 [3.20, 3.53]
Random-subset ridge	6.12 [5.73, 6.51]	3.80 [3.65, 3.95]	2.23 [2.14, 2.31]
Paper AIPW (tuned)	10.25 [9.63, 10.88]	5.51 [5.26, 5.75]	2.97 [2.86, 3.08]
Code AIPW (alpha 10)	10.25 [9.63, 10.88]	5.51 [5.27, 5.75]	2.98 [2.86, 3.09]
Crossfit difference	10.60 [10.24, 10.97]	5.92 [5.65, 6.20]	3.15 [3.01, 3.30]
Crossfit clipped	10.60 [10.24, 10.97]	5.92 [5.65, 6.20]	3.15 [3.01, 3.30]
Half-fit impute (tuned)	6.68 [6.28, 7.08]	4.09 [3.90, 4.27]	2.35 [2.26, 2.44]
Half-fit impute (matched)	9.92 [9.86, 9.99]	6.04 [5.81, 6.27]	2.35 [2.26, 2.44]
Full-fit impute	6.12 [5.73, 6.51]	3.80 [3.65, 3.95]	2.23 [2.14, 2.31]
Correctness-anchor mean	8.47 [8.47, 8.47]	4.57 [4.57, 4.57]	2.92 [2.92, 2.92]
Correctness-anchor ridge	3.58 [3.58, 3.58]	3.13 [3.13, 3.13]	1.88 [1.88, 1.88]
Random MIRT p	7.65 [7.28, 8.02]	5.07 [4.85, 5.29]	2.78 [2.68, 2.88]
Random MIRT gp	7.65 [7.28, 8.02]	4.60 [4.34, 4.85]	2.60 [2.48, 2.73]
IRT-anchor mean	16.18 [16.18, 16.18]	6.75 [6.75, 6.75]	3.35 [3.35, 3.35]
IRT-anchor MIRT p	8.14 [8.14, 8.14]	5.48 [5.48, 5.48]	2.98 [2.98, 2.98]
IRT-anchor MIRT gp	6.87 [6.87, 6.87]	4.39 [4.39, 4.39]	2.68 [2.68, 2.68]

Table 37: BBH: RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 0.17%, 0.52%, 1.74% of $N = 5,761$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	11.97 [11.97, 11.97]	11.97 [11.97, 11.97]	11.97 [11.97, 11.97]
Random mean	14.78 [13.70, 15.86]	8.37 [7.92, 8.82]	4.83 [4.41, 5.25]
Source difference	13.27 [12.81, 13.73]	7.80 [7.40, 8.19]	4.21 [4.01, 4.41]
Random-subset ridge	7.58 [7.14, 8.01]	4.79 [4.59, 5.00]	2.81 [2.71, 2.91]
Paper AIPW (tuned)	12.92 [12.26, 13.59]	6.90 [6.60, 7.20]	3.73 [3.59, 3.88]
Code AIPW (alpha 10)	12.92 [12.26, 13.59]	6.90 [6.61, 7.20]	3.74 [3.60, 3.89]
Crossfit difference	13.24 [12.79, 13.69]	7.36 [7.02, 7.71]	3.95 [3.76, 4.14]
Crossfit clipped	13.23 [12.79, 13.68]	7.36 [7.02, 7.71]	3.95 [3.76, 4.14]
Half-fit impute (tuned)	8.09 [7.67, 8.51]	5.14 [4.90, 5.39]	2.98 [2.88, 3.09]
Half-fit impute (matched)	11.58 [11.52, 11.64]	7.28 [7.03, 7.54]	2.98 [2.88, 3.09]
Full-fit impute	7.57 [7.14, 8.01]	4.79 [4.59, 5.00]	2.81 [2.71, 2.91]
Correctness-anchor mean	11.06 [11.06, 11.06]	5.63 [5.63, 5.63]	3.77 [3.77, 3.77]
Correctness-anchor ridge	4.67 [4.67, 4.67]	3.89 [3.89, 3.89]	2.37 [2.37, 2.37]
Random MIRT p	9.16 [8.79, 9.53]	6.20 [5.94, 6.46]	3.50 [3.39, 3.62]
Random MIRT gp	9.16 [8.79, 9.53]	5.68 [5.36, 5.99]	3.27 [3.10, 3.43]
IRT-anchor mean	19.77 [19.77, 19.77]	8.36 [8.36, 8.36]	4.19 [4.19, 4.19]
IRT-anchor MIRT p	9.87 [9.87, 9.87]	6.82 [6.82, 6.82]	3.77 [3.77, 3.77]
IRT-anchor MIRT gp	9.10 [9.10, 9.10]	5.70 [5.70, 5.70]	3.41 [3.41, 3.41]

Table 38: BBH: Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 0.17%, 0.52%, 1.74% of $N = 5,761$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.624 [0.560, 0.688]	0.822 [0.788, 0.855]	0.939 [0.932, 0.947]
Source difference	0.624 [0.560, 0.688]	0.822 [0.788, 0.855]	0.939 [0.932, 0.947]
Random-subset ridge	0.778 [0.746, 0.811]	0.917 [0.910, 0.925]	0.967 [0.964, 0.970]
Paper AIPW (tuned)	0.670 [0.634, 0.706]	0.866 [0.857, 0.875]	0.949 [0.945, 0.953]
Code AIPW (alpha 10)	0.670 [0.634, 0.706]	0.866 [0.857, 0.875]	0.949 [0.945, 0.953]
Crossfit difference	0.634 [0.582, 0.686]	0.844 [0.824, 0.863]	0.945 [0.940, 0.949]
Crossfit clipped	0.634 [0.582, 0.686]	0.844 [0.824, 0.863]	0.945 [0.940, 0.949]
Half-fit impute (tuned)	0.777 [0.744, 0.810]	0.915 [0.907, 0.922]	0.967 [0.964, 0.970]
Half-fit impute (matched)	0.775 [0.744, 0.805]	0.914 [0.906, 0.922]	0.967 [0.964, 0.970]
Full-fit impute	0.778 [0.746, 0.811]	0.917 [0.910, 0.925]	0.967 [0.964, 0.970]
Correctness-anchor mean	0.909 [0.909, 0.909]	0.929 [0.929, 0.929]	0.958 [0.958, 0.958]
Correctness-anchor ridge	0.917 [0.917, 0.917]	0.948 [0.948, 0.948]	0.973 [0.973, 0.973]
Random MIRT p	0.782 [0.751, 0.814]	0.909 [0.901, 0.917]	0.963 [0.959, 0.966]
Random MIRT gp	0.782 [0.751, 0.814]	0.901 [0.892, 0.909]	0.962 [0.959, 0.966]
IRT-anchor mean	0.609 [0.609, 0.609]	0.852 [0.852, 0.852]	0.931 [0.931, 0.931]
IRT-anchor MIRT p	0.728 [0.728, 0.728]	0.900 [0.900, 0.900]	0.964 [0.964, 0.964]
IRT-anchor MIRT gp	0.675 [0.675, 0.675]	0.883 [0.883, 0.883]	0.962 [0.962, 0.962]

Table 39: BBH: Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 0.17%, 0.52%, 1.74% of $N = 5,761$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.482 [0.429, 0.534]	0.649 [0.616, 0.682]	0.797 [0.785, 0.809]
Source difference	0.482 [0.429, 0.534]	0.649 [0.616, 0.682]	0.797 [0.785, 0.809]
Random-subset ridge	0.586 [0.557, 0.616]	0.747 [0.736, 0.757]	0.845 [0.838, 0.852]
Paper AIPW (tuned)	0.488 [0.458, 0.518]	0.677 [0.667, 0.687]	0.805 [0.797, 0.813]
Code AIPW (alpha 10)	0.488 [0.458, 0.518]	0.677 [0.667, 0.687]	0.804 [0.796, 0.812]
Crossfit difference	0.457 [0.416, 0.498]	0.650 [0.630, 0.671]	0.796 [0.788, 0.805]
Crossfit clipped	0.457 [0.416, 0.498]	0.650 [0.630, 0.671]	0.796 [0.788, 0.805]
Half-fit impute (tuned)	0.585 [0.556, 0.615]	0.742 [0.731, 0.753]	0.845 [0.838, 0.852]
Half-fit impute (matched)	0.583 [0.554, 0.611]	0.741 [0.730, 0.752]	0.845 [0.838, 0.852]
Full-fit impute	0.586 [0.557, 0.616]	0.747 [0.736, 0.757]	0.845 [0.838, 0.852]
Correctness-anchor mean	0.739 [0.739, 0.739]	0.760 [0.760, 0.760]	0.824 [0.824, 0.824]
Correctness-anchor ridge	0.750 [0.750, 0.750]	0.793 [0.793, 0.793]	0.860 [0.860, 0.860]
Random MIRT p	0.590 [0.560, 0.620]	0.734 [0.723, 0.746]	0.837 [0.829, 0.845]
Random MIRT gp	0.590 [0.560, 0.620]	0.722 [0.711, 0.734]	0.836 [0.828, 0.844]
IRT-anchor mean	0.447 [0.447, 0.447]	0.656 [0.656, 0.656]	0.774 [0.774, 0.774]
IRT-anchor MIRT p	0.522 [0.522, 0.522]	0.721 [0.721, 0.721]	0.843 [0.843, 0.843]
IRT-anchor MIRT gp	0.492 [0.492, 0.492]	0.696 [0.696, 0.696]	0.839 [0.839, 0.839]

Table 40: GPQA: MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 0.84%, 2.52%, 8.39% of $N = 1,192$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	2.79 [2.79, 2.79]	2.79 [2.79, 2.79]	2.79 [2.79, 2.79]
Random mean	11.67 [11.01, 12.33]	6.65 [6.20, 7.10]	3.45 [3.25, 3.66]
Source difference	11.21 [10.69, 11.72]	6.19 [5.85, 6.53]	3.30 [3.12, 3.47]
Random-subset ridge	2.48 [2.41, 2.56]	2.02 [1.90, 2.14]	1.47 [1.41, 1.53]
Paper AIPW (tuned)	10.79 [10.29, 11.28]	5.86 [5.59, 6.13]	2.95 [2.88, 3.03]
Code AIPW (alpha 10)	10.78 [10.27, 11.29]	5.87 [5.59, 6.14]	3.19 [3.11, 3.27]
Crossfit difference	10.99 [10.53, 11.44]	5.85 [5.61, 6.09]	3.01 [2.90, 3.11]
Crossfit clipped	10.95 [10.50, 11.40]	5.85 [5.61, 6.09]	3.01 [2.90, 3.11]
Half-fit impute (tuned)	2.51 [2.44, 2.57]	2.04 [1.92, 2.16]	1.43 [1.36, 1.49]
Half-fit impute (matched)	2.63 [2.60, 2.67]	2.31 [2.22, 2.40]	1.67 [1.60, 1.74]
Full-fit impute	2.48 [2.41, 2.56]	2.02 [1.90, 2.14]	1.48 [1.42, 1.54]
Correctness-anchor mean	10.18 [10.18, 10.18]	6.58 [6.58, 6.58]	5.56 [5.56, 5.56]
Correctness-anchor ridge	2.09 [2.09, 2.09]	1.61 [1.61, 1.61]	1.18 [1.18, 1.18]
Random MIRT p	2.41 [2.33, 2.49]	2.22 [2.12, 2.31]	1.81 [1.74, 1.89]
Random MIRT gp	2.41 [2.33, 2.49]	2.22 [2.12, 2.31]	1.81 [1.71, 1.90]
IRT-anchor mean	11.59 [11.59, 11.59]	6.01 [6.01, 6.01]	3.41 [3.41, 3.41]
IRT-anchor MIRT p	2.15 [2.15, 2.15]	1.73 [1.73, 1.73]	1.66 [1.66, 1.66]
IRT-anchor MIRT gp	2.15 [2.15, 2.15]	1.73 [1.73, 1.73]	1.66 [1.66, 1.66]

Table 41: GPQA: RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 0.84%, 2.52%, 8.39% of $N = 1,192$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	3.40 [3.40, 3.40]	3.40 [3.40, 3.40]	3.40 [3.40, 3.40]
Random mean	14.32 [13.59, 15.05]	8.23 [7.75, 8.72]	4.32 [4.10, 4.53]
Source difference	13.84 [13.27, 14.42]	7.77 [7.38, 8.15]	4.13 [3.94, 4.33]
Random-subset ridge	3.08 [2.99, 3.16]	2.52 [2.37, 2.66]	1.87 [1.80, 1.94]
Paper AIPW (tuned)	13.54 [13.00, 14.08]	7.41 [7.10, 7.72]	3.75 [3.64, 3.86]
Code AIPW (alpha 10)	13.51 [12.95, 14.06]	7.42 [7.11, 7.73]	4.07 [3.97, 4.18]
Crossfit difference	13.62 [13.08, 14.15]	7.39 [7.11, 7.68]	3.80 [3.68, 3.93]
Crossfit clipped	13.54 [13.01, 14.07]	7.39 [7.11, 7.68]	3.80 [3.68, 3.93]
Half-fit impute (tuned)	3.06 [2.98, 3.14]	2.53 [2.40, 2.67]	1.80 [1.72, 1.88]
Half-fit impute (matched)	3.20 [3.16, 3.25]	2.83 [2.72, 2.93]	2.09 [2.01, 2.18]
Full-fit impute	3.08 [2.99, 3.16]	2.52 [2.38, 2.66]	1.88 [1.81, 1.95]
Correctness-anchor mean	12.53 [12.53, 12.53]	8.19 [8.19, 8.19]	6.96 [6.96, 6.96]
Correctness-anchor ridge	2.77 [2.77, 2.77]	1.99 [1.99, 1.99]	1.51 [1.51, 1.51]
Random MIRT p	3.21 [3.11, 3.30]	2.86 [2.72, 2.99]	2.30 [2.20, 2.40]
Random MIRT gp	3.21 [3.11, 3.30]	2.86 [2.72, 2.99]	2.29 [2.17, 2.42]
IRT-anchor mean	15.00 [15.00, 15.00]	7.62 [7.62, 7.62]	4.35 [4.35, 4.35]
IRT-anchor MIRT p	2.96 [2.96, 2.96]	2.37 [2.37, 2.37]	2.05 [2.05, 2.05]
IRT-anchor MIRT gp	2.96 [2.96, 2.96]	2.37 [2.37, 2.37]	2.05 [2.05, 2.05]

Table 42: GPQA: Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 0.84%, 2.52%, 8.39% of $N = 1,192$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.208 [0.147, 0.269]	0.325 [0.229, 0.422]	0.538 [0.474, 0.602]
Source difference	0.208 [0.147, 0.269]	0.325 [0.229, 0.422]	0.538 [0.474, 0.602]
Random-subset ridge	0.392 [0.357, 0.427]	0.577 [0.526, 0.628]	0.766 [0.750, 0.783]
Paper AIPW (tuned)	0.183 [0.129, 0.236]	0.337 [0.282, 0.393]	0.587 [0.551, 0.624]
Code AIPW (alpha 10)	0.182 [0.129, 0.236]	0.336 [0.281, 0.390]	0.558 [0.530, 0.587]
Crossfit difference	0.208 [0.149, 0.266]	0.336 [0.255, 0.416]	0.570 [0.525, 0.615]
Crossfit clipped	0.208 [0.149, 0.266]	0.336 [0.255, 0.416]	0.570 [0.525, 0.615]
Half-fit impute (tuned)	0.394 [0.358, 0.429]	0.580 [0.528, 0.633]	0.786 [0.770, 0.802]
Half-fit impute (matched)	0.387 [0.349, 0.425]	0.580 [0.525, 0.636]	0.792 [0.774, 0.810]
Full-fit impute	0.392 [0.357, 0.427]	0.576 [0.526, 0.627]	0.765 [0.748, 0.781]
Correctness-anchor mean	0.149 [0.149, 0.149]	0.244 [0.244, 0.244]	0.212 [0.212, 0.212]
Correctness-anchor ridge	0.584 [0.584, 0.584]	0.730 [0.730, 0.730]	0.828 [0.828, 0.828]
Random MIRT p	0.297 [0.231, 0.362]	0.475 [0.417, 0.533]	0.679 [0.657, 0.701]
Random MIRT gp	0.297 [0.231, 0.362]	0.475 [0.417, 0.533]	0.695 [0.664, 0.727]
IRT-anchor mean	0.179 [0.179, 0.179]	0.338 [0.338, 0.338]	0.576 [0.576, 0.576]
IRT-anchor MIRT p	0.489 [0.489, 0.489]	0.742 [0.742, 0.742]	0.766 [0.766, 0.766]
IRT-anchor MIRT gp	0.489 [0.489, 0.489]	0.742 [0.742, 0.742]	0.766 [0.766, 0.766]

Table 43: GPQA: Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 0.84%, 2.52%, 8.39% of $N = 1,192$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.156 [0.110, 0.201]	0.237 [0.164, 0.310]	0.395 [0.345, 0.445]
Source difference	0.156 [0.110, 0.201]	0.237 [0.164, 0.310]	0.395 [0.345, 0.445]
Random-subset ridge	0.276 [0.250, 0.302]	0.415 [0.375, 0.456]	0.579 [0.563, 0.595]
Paper AIPW (tuned)	0.128 [0.091, 0.164]	0.233 [0.195, 0.272]	0.423 [0.394, 0.453]
Code AIPW (alpha 10)	0.128 [0.092, 0.164]	0.232 [0.194, 0.270]	0.400 [0.378, 0.423]
Crossfit difference	0.144 [0.104, 0.185]	0.232 [0.174, 0.289]	0.410 [0.374, 0.446]
Crossfit clipped	0.144 [0.104, 0.185]	0.232 [0.174, 0.289]	0.410 [0.374, 0.446]
Half-fit impute (tuned)	0.277 [0.251, 0.303]	0.418 [0.378, 0.459]	0.597 [0.580, 0.615]
Half-fit impute (matched)	0.272 [0.244, 0.301]	0.415 [0.372, 0.458]	0.600 [0.581, 0.620]
Full-fit impute	0.276 [0.250, 0.302]	0.415 [0.375, 0.455]	0.578 [0.561, 0.594]
Correctness-anchor mean	0.115 [0.115, 0.115]	0.166 [0.166, 0.166]	0.155 [0.155, 0.155]
Correctness-anchor ridge	0.407 [0.407, 0.407]	0.538 [0.538, 0.538]	0.648 [0.648, 0.648]
Random MIRT p	0.206 [0.160, 0.253]	0.335 [0.292, 0.377]	0.494 [0.476, 0.512]
Random MIRT gp	0.206 [0.160, 0.253]	0.335 [0.292, 0.377]	0.512 [0.484, 0.541]
IRT-anchor mean	0.124 [0.124, 0.124]	0.231 [0.231, 0.231]	0.419 [0.419, 0.419]
IRT-anchor MIRT p	0.323 [0.323, 0.323]	0.535 [0.535, 0.535]	0.567 [0.567, 0.567]
IRT-anchor MIRT gp	0.323 [0.323, 0.323]	0.535 [0.535, 0.535]	0.567 [0.567, 0.567]

Table 44: IFEval: MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.85%, 5.55%, 18.48% of $N = 541$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	17.22 [17.22, 17.22]	17.22 [17.22, 17.22]	17.22 [17.22, 17.22]
Random mean	10.62 [9.68, 11.57]	6.00 [5.28, 6.72]	3.09 [2.77, 3.41]
Source difference	9.00 [8.45, 9.56]	4.85 [4.66, 5.03]	2.63 [2.53, 2.74]
Random-subset ridge	6.97 [6.50, 7.44]	4.05 [3.91, 4.19]	2.28 [2.21, 2.35]
Paper AIPW (tuned)	9.16 [8.47, 9.85]	4.67 [4.43, 4.92]	2.38 [2.31, 2.46]
Code AIPW (alpha 10)	9.11 [8.41, 9.80]	4.68 [4.43, 4.93]	2.55 [2.45, 2.65]
Crossfit difference	8.81 [8.32, 9.30]	4.62 [4.40, 4.84]	2.44 [2.37, 2.52]
Crossfit clipped	8.67 [8.21, 9.13]	4.61 [4.40, 4.81]	2.44 [2.37, 2.52]
Half-fit impute (tuned)	7.66 [7.07, 8.25]	4.30 [4.17, 4.42]	2.32 [2.26, 2.38]
Half-fit impute (matched)	8.15 [7.53, 8.77]	4.46 [4.33, 4.59]	3.09 [3.03, 3.16]
Full-fit impute	6.97 [6.50, 7.44]	4.04 [3.90, 4.19]	2.28 [2.21, 2.35]
Correctness-anchor mean	6.46 [6.46, 6.46]	4.82 [4.82, 4.82]	3.76 [3.76, 3.76]
Correctness-anchor ridge	5.67 [5.67, 5.67]	3.91 [3.91, 3.91]	2.26 [2.26, 2.26]
Random MIRT p	8.98 [8.32, 9.64]	4.56 [4.39, 4.73]	2.47 [2.39, 2.56]
Random MIRT gp	8.05 [7.39, 8.71]	4.35 [4.14, 4.56]	2.40 [2.30, 2.50]
IRT-anchor mean	8.78 [8.78, 8.78]	5.45 [5.45, 5.45]	2.51 [2.51, 2.51]
IRT-anchor MIRT p	6.55 [6.55, 6.55]	4.75 [4.75, 4.75]	2.47 [2.47, 2.47]
IRT-anchor MIRT gp	6.41 [6.41, 6.41]	4.64 [4.64, 4.64]	2.25 [2.25, 2.25]

Table 45: IFEval: RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.85%, 5.55%, 18.48% of $N = 541$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	20.49 [20.49, 20.49]	20.49 [20.49, 20.49]	20.49 [20.49, 20.49]
Random mean	13.11 [12.01, 14.22]	7.36 [6.60, 8.12]	3.84 [3.49, 4.18]
Source difference	11.18 [10.51, 11.85]	6.12 [5.88, 6.35]	3.27 [3.15, 3.39]
Random-subset ridge	8.98 [8.41, 9.55]	5.14 [4.96, 5.32]	2.91 [2.83, 3.00]
Paper AIPW (tuned)	11.40 [10.60, 12.21]	5.86 [5.56, 6.17]	2.99 [2.91, 3.08]
Code AIPW (alpha 10)	11.33 [10.52, 12.15]	5.87 [5.56, 6.18]	3.25 [3.13, 3.38]
Crossfit difference	10.99 [10.37, 11.62]	5.83 [5.56, 6.09]	3.07 [2.98, 3.16]
Crossfit clipped	10.81 [10.23, 11.39]	5.81 [5.55, 6.07]	3.07 [2.98, 3.16]
Half-fit impute (tuned)	9.59 [8.95, 10.23]	5.46 [5.31, 5.62]	2.95 [2.88, 3.01]
Half-fit impute (matched)	10.12 [9.47, 10.77]	5.65 [5.49, 5.81]	3.96 [3.87, 4.05]
Full-fit impute	8.98 [8.41, 9.55]	5.14 [4.96, 5.31]	2.91 [2.83, 2.99]
Correctness-anchor mean	8.40 [8.40, 8.40]	6.12 [6.12, 6.12]	4.59 [4.59, 4.59]
Correctness-anchor ridge	7.51 [7.51, 7.51]	5.15 [5.15, 5.15]	2.88 [2.88, 2.88]
Random MIRT p	10.64 [9.89, 11.39]	5.70 [5.51, 5.88]	3.13 [3.02, 3.24]
Random MIRT gp	9.88 [9.13, 10.63]	5.44 [5.22, 5.66]	3.02 [2.90, 3.14]
IRT-anchor mean	10.44 [10.44, 10.44]	6.69 [6.69, 6.69]	3.28 [3.28, 3.28]
IRT-anchor MIRT p	7.94 [7.94, 7.94]	6.13 [6.13, 6.13]	3.03 [3.03, 3.03]
IRT-anchor MIRT gp	8.01 [8.01, 8.01]	5.85 [5.85, 5.85]	2.75 [2.75, 2.75]

Table 46: IFEval: Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.85%, 5.55%, 18.48% of $N = 541$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.859 [0.831, 0.886]	0.945 [0.938, 0.951]	0.977 [0.975, 0.979]
Source difference	0.859 [0.831, 0.886]	0.945 [0.938, 0.951]	0.977 [0.975, 0.979]
Random-subset ridge	0.874 [0.855, 0.892]	0.951 [0.947, 0.955]	0.979 [0.978, 0.981]
Paper AIPW (tuned)	0.864 [0.844, 0.884]	0.946 [0.941, 0.952]	0.979 [0.977, 0.980]
Code AIPW (alpha 10)	0.865 [0.846, 0.885]	0.946 [0.941, 0.952]	0.976 [0.974, 0.978]
Crossfit difference	0.859 [0.836, 0.882]	0.945 [0.940, 0.951]	0.978 [0.976, 0.980]
Crossfit clipped	0.859 [0.836, 0.882]	0.945 [0.940, 0.951]	0.978 [0.976, 0.980]
Half-fit impute (tuned)	0.874 [0.855, 0.892]	0.948 [0.944, 0.952]	0.980 [0.979, 0.981]
Half-fit impute (matched)	0.875 [0.856, 0.893]	0.949 [0.945, 0.953]	0.980 [0.978, 0.981]
Full-fit impute	0.874 [0.855, 0.892]	0.951 [0.947, 0.955]	0.979 [0.978, 0.981]
Correctness-anchor mean	0.902 [0.902, 0.902]	0.958 [0.958, 0.958]	0.978 [0.978, 0.978]
Correctness-anchor ridge	0.908 [0.908, 0.908]	0.960 [0.960, 0.960]	0.987 [0.987, 0.987]
Random MIRT p	0.869 [0.848, 0.889]	0.953 [0.949, 0.956]	0.980 [0.979, 0.981]
Random MIRT gp	0.870 [0.849, 0.891]	0.953 [0.949, 0.957]	0.981 [0.980, 0.982]
IRT-anchor mean	0.893 [0.893, 0.893]	0.912 [0.912, 0.912]	0.975 [0.975, 0.975]
IRT-anchor MIRT p	0.916 [0.916, 0.916]	0.934 [0.934, 0.934]	0.979 [0.979, 0.979]
IRT-anchor MIRT gp	0.901 [0.901, 0.901]	0.925 [0.925, 0.925]	0.982 [0.982, 0.982]

Table 47: IFEval: Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.85%, 5.55%, 18.48% of $N = 541$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.711 [0.682, 0.741]	0.823 [0.813, 0.833]	0.890 [0.885, 0.895]
Source difference	0.711 [0.682, 0.741]	0.823 [0.813, 0.833]	0.890 [0.885, 0.895]
Random-subset ridge	0.703 [0.681, 0.724]	0.818 [0.810, 0.825]	0.891 [0.887, 0.895]
Paper AIPW (tuned)	0.691 [0.667, 0.715]	0.809 [0.800, 0.818]	0.888 [0.885, 0.892]
Code AIPW (alpha 10)	0.692 [0.669, 0.715]	0.809 [0.800, 0.818]	0.880 [0.875, 0.885]
Crossfit difference	0.684 [0.659, 0.709]	0.806 [0.797, 0.816]	0.886 [0.882, 0.891]
Crossfit clipped	0.684 [0.659, 0.709]	0.806 [0.797, 0.816]	0.886 [0.882, 0.891]
Half-fit impute (tuned)	0.703 [0.681, 0.724]	0.813 [0.806, 0.821]	0.892 [0.889, 0.895]
Half-fit impute (matched)	0.704 [0.681, 0.726]	0.815 [0.807, 0.822]	0.891 [0.888, 0.894]
Full-fit impute	0.703 [0.681, 0.724]	0.818 [0.810, 0.825]	0.890 [0.886, 0.894]
Correctness-anchor mean	0.744 [0.744, 0.744]	0.830 [0.830, 0.830]	0.882 [0.882, 0.882]
Correctness-anchor ridge	0.744 [0.744, 0.744]	0.840 [0.840, 0.840]	0.913 [0.913, 0.913]
Random MIRT p	0.698 [0.674, 0.721]	0.820 [0.813, 0.827]	0.892 [0.888, 0.895]
Random MIRT gp	0.699 [0.675, 0.723]	0.823 [0.815, 0.830]	0.894 [0.890, 0.897]
IRT-anchor mean	0.728 [0.728, 0.728]	0.757 [0.757, 0.757]	0.877 [0.877, 0.877]
IRT-anchor MIRT p	0.758 [0.758, 0.758]	0.791 [0.791, 0.791]	0.887 [0.887, 0.887]
IRT-anchor MIRT gp	0.737 [0.737, 0.737]	0.777 [0.777, 0.777]	0.896 [0.896, 0.896]

Table 48: MMLU-Pro: MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 0.08%, 0.25%, 0.83% of $N = 12,032$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	10.81 [10.81, 10.81]	10.81 [10.81, 10.81]	10.81 [10.81, 10.81]
Random mean	12.08 [10.94, 13.22]	6.34 [5.77, 6.92]	3.36 [2.99, 3.74]
Source difference	9.71 [9.17, 10.24]	5.38 [5.05, 5.72]	2.86 [2.71, 3.01]
Random-subset ridge	6.22 [5.49, 6.95]	3.68 [3.44, 3.92]	2.07 [1.99, 2.16]
Paper AIPW (tuned)	9.52 [8.53, 10.51]	4.79 [4.57, 5.00]	2.57 [2.45, 2.69]
Code AIPW (alpha 10)	9.55 [8.58, 10.53]	4.79 [4.57, 5.00]	2.57 [2.45, 2.69]
Crossfit difference	9.20 [8.55, 9.86]	5.18 [4.87, 5.50]	2.68 [2.56, 2.80]
Crossfit clipped	9.04 [8.44, 9.63]	5.17 [4.86, 5.47]	2.68 [2.56, 2.80]
Half-fit impute (tuned)	6.92 [6.21, 7.64]	4.07 [3.78, 4.36]	2.18 [2.09, 2.27]
Half-fit impute (matched)	7.22 [6.53, 7.91]	4.26 [3.96, 4.56]	2.18 [2.09, 2.27]
Full-fit impute	6.22 [5.49, 6.95]	3.68 [3.44, 3.92]	2.07 [1.99, 2.16]
Correctness-anchor mean	6.93 [6.93, 6.93]	4.95 [4.95, 4.95]	3.33 [3.33, 3.33]
Correctness-anchor ridge	4.15 [4.15, 4.15]	2.79 [2.79, 2.79]	1.53 [1.53, 1.53]
Random MIRT p	8.88 [8.37, 9.39]	5.96 [5.66, 6.26]	3.14 [3.00, 3.28]
Random MIRT gp	7.86 [7.28, 8.44]	4.88 [4.49, 5.28]	2.79 [2.64, 2.95]
IRT-anchor mean	10.04 [10.04, 10.04]	4.73 [4.73, 4.73]	2.80 [2.80, 2.80]
IRT-anchor MIRT p	8.52 [8.52, 8.52]	5.78 [5.78, 5.78]	3.27 [3.27, 3.27]
IRT-anchor MIRT gp	6.74 [6.74, 6.74]	4.12 [4.12, 4.12]	2.58 [2.58, 2.58]

Table 49: MMLU-Pro: RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 0.08%, 0.25%, 0.83% of $N = 12,032$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	12.91 [12.91, 12.91]	12.91 [12.91, 12.91]	12.91 [12.91, 12.91]
Random mean	14.65 [13.36, 15.94]	7.77 [7.07, 8.47]	4.15 [3.73, 4.57]
Source difference	12.13 [11.52, 12.74]	6.72 [6.35, 7.10]	3.60 [3.42, 3.79]
Random-subset ridge	7.70 [6.86, 8.54]	4.69 [4.40, 4.98]	2.65 [2.54, 2.76]
Paper AIPW (tuned)	11.96 [10.76, 13.15]	5.96 [5.69, 6.24]	3.25 [3.12, 3.38]
Code AIPW (alpha 10)	11.89 [10.68, 13.10]	5.96 [5.69, 6.24]	3.25 [3.12, 3.38]
Crossfit difference	11.46 [10.66, 12.26]	6.48 [6.13, 6.84]	3.37 [3.24, 3.50]
Crossfit clipped	11.24 [10.48, 11.99]	6.45 [6.10, 6.80]	3.37 [3.24, 3.50]
Half-fit impute (tuned)	8.37 [7.55, 9.19]	5.15 [4.82, 5.48]	2.78 [2.66, 2.89]
Half-fit impute (matched)	8.66 [7.85, 9.47]	5.33 [4.98, 5.67]	2.78 [2.66, 2.89]
Full-fit impute	7.70 [6.86, 8.54]	4.69 [4.40, 4.98]	2.65 [2.54, 2.76]
Correctness-anchor mean	9.20 [9.20, 9.20]	6.01 [6.01, 6.01]	4.09 [4.09, 4.09]
Correctness-anchor ridge	5.41 [5.41, 5.41]	3.65 [3.65, 3.65]	1.90 [1.90, 1.90]
Random MIRT p	10.56 [9.95, 11.17]	7.27 [6.90, 7.63]	3.97 [3.82, 4.12]
Random MIRT gp	9.50 [8.82, 10.18]	6.12 [5.63, 6.61]	3.52 [3.36, 3.68]
IRT-anchor mean	12.49 [12.49, 12.49]	6.01 [6.01, 6.01]	3.58 [3.58, 3.58]
IRT-anchor MIRT p	10.20 [10.20, 10.20]	7.04 [7.04, 7.04]	4.12 [4.12, 4.12]
IRT-anchor MIRT gp	8.81 [8.81, 8.81]	5.25 [5.25, 5.25]	3.29 [3.29, 3.29]

Table 50: MMLU-Pro: Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 0.08%, 0.25%, 0.83% of $N = 12,032$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.676 [0.598, 0.754]	0.878 [0.851, 0.904]	0.960 [0.956, 0.964]
Source difference	0.676 [0.598, 0.754]	0.878 [0.851, 0.904]	0.960 [0.956, 0.964]
Random-subset ridge	0.772 [0.711, 0.832]	0.926 [0.918, 0.935]	0.971 [0.969, 0.973]
Paper AIPW (tuned)	0.718 [0.662, 0.773]	0.903 [0.894, 0.913]	0.965 [0.963, 0.967]
Code AIPW (alpha 10)	0.718 [0.660, 0.776]	0.903 [0.894, 0.913]	0.965 [0.963, 0.967]
Crossfit difference	0.711 [0.649, 0.774]	0.891 [0.875, 0.907]	0.962 [0.960, 0.965]
Crossfit clipped	0.711 [0.649, 0.774]	0.891 [0.875, 0.907]	0.962 [0.960, 0.965]
Half-fit impute (tuned)	0.765 [0.706, 0.824]	0.920 [0.910, 0.930]	0.970 [0.968, 0.972]
Half-fit impute (matched)	0.768 [0.708, 0.827]	0.921 [0.911, 0.931]	0.970 [0.968, 0.972]
Full-fit impute	0.772 [0.711, 0.832]	0.926 [0.918, 0.935]	0.971 [0.969, 0.973]
Correctness-anchor mean	0.863 [0.863, 0.863]	0.936 [0.936, 0.936]	0.968 [0.968, 0.968]
Correctness-anchor ridge	0.915 [0.915, 0.915]	0.956 [0.956, 0.956]	0.986 [0.986, 0.986]
Random MIRT p	0.682 [0.607, 0.758]	0.889 [0.876, 0.902]	0.957 [0.953, 0.961]
Random MIRT gp	0.712 [0.642, 0.782]	0.895 [0.877, 0.913]	0.962 [0.959, 0.966]
IRT-anchor mean	0.714 [0.714, 0.714]	0.895 [0.895, 0.895]	0.962 [0.962, 0.962]
IRT-anchor MIRT p	0.706 [0.706, 0.706]	0.889 [0.889, 0.889]	0.957 [0.957, 0.957]
IRT-anchor MIRT gp	0.740 [0.740, 0.740]	0.913 [0.913, 0.913]	0.963 [0.963, 0.963]

Table 51: MMLU-Pro: Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 0.08%, 0.25%, 0.83% of $N = 12,032$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.537 [0.469, 0.606]	0.719 [0.690, 0.747]	0.837 [0.827, 0.846]
Source difference	0.537 [0.469, 0.606]	0.719 [0.690, 0.747]	0.837 [0.827, 0.846]
Random-subset ridge	0.596 [0.539, 0.652]	0.762 [0.748, 0.776]	0.855 [0.850, 0.860]
Paper AIPW (tuned)	0.545 [0.496, 0.594]	0.727 [0.714, 0.741]	0.838 [0.833, 0.844]
Code AIPW (alpha 10)	0.544 [0.493, 0.594]	0.727 [0.714, 0.741]	0.838 [0.833, 0.844]
Crossfit difference	0.537 [0.482, 0.591]	0.713 [0.694, 0.733]	0.832 [0.826, 0.839]
Crossfit clipped	0.537 [0.482, 0.592]	0.713 [0.694, 0.733]	0.832 [0.826, 0.839]
Half-fit impute (tuned)	0.588 [0.533, 0.643]	0.751 [0.735, 0.767]	0.853 [0.847, 0.858]
Half-fit impute (matched)	0.591 [0.534, 0.647]	0.754 [0.738, 0.770]	0.853 [0.847, 0.858]
Full-fit impute	0.596 [0.539, 0.652]	0.762 [0.748, 0.776]	0.855 [0.850, 0.860]
Correctness-anchor mean	0.697 [0.697, 0.697]	0.802 [0.802, 0.802]	0.852 [0.852, 0.852]
Correctness-anchor ridge	0.761 [0.761, 0.761]	0.825 [0.825, 0.825]	0.902 [0.902, 0.902]
Random MIRT p	0.515 [0.449, 0.581]	0.709 [0.692, 0.725]	0.828 [0.821, 0.835]
Random MIRT gp	0.544 [0.483, 0.606]	0.720 [0.698, 0.743]	0.837 [0.829, 0.844]
IRT-anchor mean	0.519 [0.519, 0.519]	0.721 [0.721, 0.721]	0.826 [0.826, 0.826]
IRT-anchor MIRT p	0.522 [0.522, 0.522]	0.709 [0.709, 0.709]	0.821 [0.821, 0.821]
IRT-anchor MIRT gp	0.548 [0.548, 0.548]	0.746 [0.746, 0.746]	0.830 [0.830, 0.830]

Table 52: MuSR: MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.32%, 3.97%, 13.23% of $N = 756$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	3.51 [3.51, 3.51]	3.51 [3.51, 3.51]	3.51 [3.51, 3.51]
Random mean	12.93 [11.45, 14.42]	6.72 [5.91, 7.52]	3.70 [3.34, 4.07]
Source difference	10.03 [9.29, 10.77]	5.52 [5.16, 5.87]	2.93 [2.77, 3.09]
Random-subset ridge	2.73 [2.57, 2.89]	2.24 [2.12, 2.36]	1.58 [1.52, 1.65]
Paper AIPW (tuned)	8.44 [7.17, 9.71]	3.72 [3.30, 4.15]	2.34 [2.27, 2.41]
Code AIPW (alpha 10)	8.92 [7.84, 9.99]	4.16 [3.83, 4.48]	2.41 [2.33, 2.49]
Crossfit difference	9.81 [9.41, 10.20]	5.35 [5.07, 5.63]	2.68 [2.57, 2.79]
Crossfit clipped	9.80 [9.41, 10.19]	5.35 [5.07, 5.63]	2.68 [2.57, 2.79]
Half-fit impute (tuned)	2.88 [2.74, 3.01]	2.36 [2.25, 2.47]	1.63 [1.56, 1.69]
Half-fit impute (matched)	2.88 [2.74, 3.01]	2.79 [2.69, 2.89]	1.64 [1.58, 1.69]
Full-fit impute	2.73 [2.57, 2.89]	2.23 [2.11, 2.35]	1.58 [1.51, 1.65]
Correctness-anchor mean	6.08 [6.08, 6.08]	3.63 [3.63, 3.63]	3.34 [3.34, 3.34]
Correctness-anchor ridge	2.35 [2.35, 2.35]	1.98 [1.98, 1.98]	1.60 [1.60, 1.60]
Random MIRT p	3.22 [3.05, 3.38]	2.45 [2.31, 2.59]	1.58 [1.52, 1.64]
Random MIRT gp	3.22 [3.05, 3.38]	2.45 [2.31, 2.59]	1.58 [1.52, 1.64]
IRT-anchor mean	9.64 [9.64, 9.64]	4.96 [4.96, 4.96]	2.57 [2.57, 2.57]
IRT-anchor MIRT p	3.14 [3.14, 3.14]	2.44 [2.44, 2.44]	1.78 [1.78, 1.78]
IRT-anchor MIRT gp	3.14 [3.14, 3.14]	2.44 [2.44, 2.44]	1.66 [1.66, 1.66]

Table 53: MuSR: RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.32%, 3.97%, 13.23% of $N = 756$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	4.11 [4.11, 4.11]	4.11 [4.11, 4.11]	4.11 [4.11, 4.11]
Random mean	15.53 [13.97, 17.10]	8.23 [7.39, 9.07]	4.54 [4.16, 4.93]
Source difference	12.38 [11.56, 13.20]	6.85 [6.43, 7.28]	3.66 [3.47, 3.85]
Random-subset ridge	3.35 [3.18, 3.52]	2.80 [2.67, 2.94]	2.01 [1.94, 2.08]
Paper AIPW (tuned)	11.10 [9.56, 12.65]	5.27 [4.86, 5.68]	2.98 [2.87, 3.08]
Code AIPW (alpha 10)	11.47 [10.12, 12.83]	5.29 [4.92, 5.67]	3.06 [2.94, 3.18]
Crossfit difference	12.25 [11.78, 12.71]	6.67 [6.31, 7.03]	3.39 [3.25, 3.53]
Crossfit clipped	12.23 [11.77, 12.69]	6.67 [6.31, 7.03]	3.39 [3.25, 3.53]
Half-fit impute (tuned)	3.45 [3.31, 3.59]	2.91 [2.80, 3.03]	2.07 [1.99, 2.14]
Half-fit impute (matched)	3.45 [3.31, 3.59]	3.34 [3.24, 3.44]	2.09 [2.02, 2.15]
Full-fit impute	3.35 [3.18, 3.52]	2.80 [2.66, 2.94]	2.00 [1.93, 2.08]
Correctness-anchor mean	7.69 [7.69, 7.69]	4.78 [4.78, 4.78]	4.29 [4.29, 4.29]
Correctness-anchor ridge	2.95 [2.95, 2.95]	2.55 [2.55, 2.55]	2.02 [2.02, 2.02]
Random MIRT p	3.92 [3.75, 4.08]	3.05 [2.90, 3.21]	2.02 [1.95, 2.08]
Random MIRT gp	3.92 [3.75, 4.08]	3.05 [2.90, 3.21]	2.02 [1.95, 2.08]
IRT-anchor mean	12.04 [12.04, 12.04]	6.17 [6.17, 6.17]	3.21 [3.21, 3.21]
IRT-anchor MIRT p	3.80 [3.80, 3.80]	3.31 [3.31, 3.31]	2.21 [2.21, 2.21]
IRT-anchor MIRT gp	3.80 [3.80, 3.80]	3.31 [3.31, 3.31]	2.09 [2.09, 2.09]

Table 54: MuSR: Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.32%, 3.97%, 13.23% of $N = 756$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.278 [0.144, 0.411]	0.469 [0.360, 0.579]	0.752 [0.719, 0.785]
Source difference	0.278 [0.144, 0.411]	0.469 [0.360, 0.579]	0.752 [0.719, 0.785]
Random-subset ridge	0.563 [0.501, 0.625]	0.726 [0.692, 0.761]	0.872 [0.862, 0.881]
Paper AIPW (tuned)	0.310 [0.227, 0.392]	0.576 [0.519, 0.633]	0.790 [0.776, 0.804]
Code AIPW (alpha 10)	0.291 [0.203, 0.380]	0.545 [0.498, 0.593]	0.782 [0.766, 0.798]
Crossfit difference	0.255 [0.151, 0.358]	0.494 [0.413, 0.574]	0.774 [0.758, 0.790]
Crossfit clipped	0.255 [0.151, 0.358]	0.494 [0.413, 0.574]	0.774 [0.758, 0.790]
Half-fit impute (tuned)	0.571 [0.514, 0.627]	0.722 [0.689, 0.755]	0.865 [0.855, 0.875]
Half-fit impute (matched)	0.571 [0.514, 0.627]	0.723 [0.692, 0.754]	0.870 [0.862, 0.879]
Full-fit impute	0.563 [0.501, 0.625]	0.727 [0.692, 0.761]	0.872 [0.862, 0.881]
Correctness-anchor mean	0.469 [0.469, 0.469]	0.625 [0.625, 0.625]	0.745 [0.745, 0.745]
Correctness-anchor ridge	0.700 [0.700, 0.700]	0.797 [0.797, 0.797]	0.870 [0.870, 0.870]
Random MIRT p	0.446 [0.345, 0.547]	0.671 [0.627, 0.716]	0.870 [0.862, 0.878]
Random MIRT gp	0.446 [0.345, 0.547]	0.671 [0.627, 0.716]	0.870 [0.862, 0.878]
IRT-anchor mean	0.420 [0.420, 0.420]	0.521 [0.521, 0.521]	0.791 [0.791, 0.791]
IRT-anchor MIRT p	0.575 [0.575, 0.575]	0.621 [0.621, 0.621]	0.852 [0.852, 0.852]
IRT-anchor MIRT gp	0.575 [0.575, 0.575]	0.621 [0.621, 0.621]	0.870 [0.870, 0.870]

Table 55: MuSR: Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.32%, 3.97%, 13.23% of $N = 756$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.210 [0.109, 0.311]	0.348 [0.266, 0.430]	0.571 [0.539, 0.603]
Source difference	0.210 [0.109, 0.311]	0.348 [0.266, 0.430]	0.571 [0.539, 0.603]
Random-subset ridge	0.401 [0.353, 0.448]	0.536 [0.506, 0.567]	0.689 [0.676, 0.701]
Paper AIPW (tuned)	0.216 [0.157, 0.275]	0.410 [0.365, 0.455]	0.596 [0.582, 0.611]
Code AIPW (alpha 10)	0.203 [0.140, 0.266]	0.384 [0.349, 0.420]	0.588 [0.572, 0.605]
Crossfit difference	0.176 [0.104, 0.248]	0.346 [0.288, 0.405]	0.577 [0.562, 0.593]
Crossfit clipped	0.176 [0.104, 0.248]	0.346 [0.288, 0.405]	0.577 [0.562, 0.593]
Half-fit impute (tuned)	0.405 [0.361, 0.450]	0.531 [0.502, 0.559]	0.681 [0.669, 0.694]
Half-fit impute (matched)	0.405 [0.361, 0.450]	0.530 [0.503, 0.557]	0.688 [0.677, 0.699]
Full-fit impute	0.401 [0.353, 0.448]	0.537 [0.506, 0.567]	0.689 [0.676, 0.702]
Correctness-anchor mean	0.323 [0.323, 0.323]	0.452 [0.452, 0.452]	0.564 [0.564, 0.564]
Correctness-anchor ridge	0.517 [0.517, 0.517]	0.600 [0.600, 0.600]	0.677 [0.677, 0.677]
Random MIRT p	0.314 [0.242, 0.387]	0.490 [0.453, 0.527]	0.688 [0.677, 0.699]
Random MIRT gp	0.314 [0.242, 0.387]	0.490 [0.453, 0.527]	0.688 [0.677, 0.699]
IRT-anchor mean	0.292 [0.292, 0.292]	0.364 [0.364, 0.364]	0.603 [0.603, 0.603]
IRT-anchor MIRT p	0.404 [0.404, 0.404]	0.452 [0.452, 0.452]	0.657 [0.657, 0.657]
IRT-anchor MIRT gp	0.404 [0.404, 0.404]	0.452 [0.452, 0.452]	0.682 [0.682, 0.682]

Table 56: MATH (OpenLLM): MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.12%, 3.36%, 11.19% of $N = 894$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	4.98 [4.98, 4.98]	4.98 [4.98, 4.98]	4.98 [4.98, 4.98]
Random mean	3.62 [3.25, 3.99]	2.07 [1.96, 2.19]	1.13 [1.01, 1.25]
Source difference	3.98 [3.42, 4.55]	2.17 [2.01, 2.33]	1.21 [1.03, 1.39]
Random-subset ridge	2.75 [2.48, 3.02]	1.77 [1.67, 1.87]	1.13 [1.09, 1.16]
Paper AIPW (tuned)	2.76 [2.59, 2.93]	1.84 [1.74, 1.93]	1.04 [1.01, 1.07]
Code AIPW (alpha 10)	2.91 [2.77, 3.04]	1.83 [1.74, 1.91]	1.05 [1.02, 1.09]
Crossfit difference	3.75 [3.33, 4.16]	2.16 [2.01, 2.30]	1.08 [1.04, 1.12]
Crossfit clipped	3.22 [3.05, 3.38]	1.95 [1.88, 2.03]	1.06 [1.03, 1.10]
Half-fit impute (tuned)	3.22 [2.98, 3.45]	2.03 [1.91, 2.15]	1.17 [1.12, 1.22]
Half-fit impute (matched)	3.27 [3.02, 3.51]	4.65 [4.61, 4.68]	1.28 [1.23, 1.33]
Full-fit impute	2.75 [2.48, 3.02]	1.77 [1.67, 1.87]	1.11 [1.07, 1.14]
Correctness-anchor mean	2.24 [2.24, 2.24]	2.07 [2.07, 2.07]	1.58 [1.58, 1.58]
Correctness-anchor ridge	1.89 [1.89, 1.89]	1.16 [1.16, 1.16]	0.95 [0.95, 0.95]
Random MIRT p	7.52 [7.03, 8.00]	3.79 [3.60, 3.97]	1.63 [1.57, 1.68]
Random MIRT gp	3.62 [3.25, 3.99]	2.07 [1.96, 2.19]	1.13 [1.01, 1.25]
IRT-anchor mean	2.97 [2.97, 2.97]	1.46 [1.46, 1.46]	0.87 [0.87, 0.87]
IRT-anchor MIRT p	8.02 [8.02, 8.02]	3.50 [3.50, 3.50]	1.38 [1.38, 1.38]
IRT-anchor MIRT gp	2.97 [2.97, 2.97]	1.46 [1.46, 1.46]	0.87 [0.87, 0.87]

Table 57: MATH (OpenLLM): RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.12%, 3.36%, 11.19% of $N = 894$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	6.73 [6.73, 6.73]	6.73 [6.73, 6.73]	6.73 [6.73, 6.73]
Random mean	6.00 [5.28, 6.72]	3.14 [2.95, 3.34]	1.75 [1.52, 1.97]
Source difference	5.69 [5.04, 6.33]	3.11 [2.95, 3.27]	1.70 [1.50, 1.90]
Random-subset ridge	3.82 [3.51, 4.13]	2.65 [2.51, 2.79]	1.73 [1.66, 1.80]
Paper AIPW (tuned)	4.56 [4.11, 5.00]	2.78 [2.64, 2.91]	1.57 [1.50, 1.63]
Code AIPW (alpha 10)	4.79 [4.48, 5.09]	2.73 [2.62, 2.85]	1.59 [1.52, 1.67]
Crossfit difference	5.37 [4.97, 5.77]	3.10 [2.95, 3.24]	1.60 [1.53, 1.67]
Crossfit clipped	5.14 [4.83, 5.45]	3.01 [2.88, 3.13]	1.59 [1.53, 1.66]
Half-fit impute (tuned)	4.26 [3.94, 4.57]	2.84 [2.70, 2.99]	1.73 [1.64, 1.81]
Half-fit impute (matched)	4.34 [4.02, 4.65]	6.31 [6.28, 6.35]	1.92 [1.85, 1.99]
Full-fit impute	3.82 [3.51, 4.13]	2.65 [2.51, 2.79]	1.68 [1.61, 1.76]
Correctness-anchor mean	4.53 [4.53, 4.53]	4.33 [4.33, 4.33]	3.70 [3.70, 3.70]
Correctness-anchor ridge	3.12 [3.12, 3.12]	2.13 [2.13, 2.13]	2.12 [2.12, 2.12]
Random MIRT p	8.07 [7.60, 8.54]	4.19 [4.00, 4.38]	1.91 [1.84, 1.98]
Random MIRT gp	6.00 [5.28, 6.72]	3.14 [2.95, 3.34]	1.75 [1.52, 1.97]
IRT-anchor mean	4.73 [4.73, 4.73]	2.14 [2.14, 2.14]	1.26 [1.26, 1.26]
IRT-anchor MIRT p	8.65 [8.65, 8.65]	3.75 [3.75, 3.75]	1.57 [1.57, 1.57]
IRT-anchor MIRT gp	4.73 [4.73, 4.73]	2.14 [2.14, 2.14]	1.26 [1.26, 1.26]

Table 58: MATH (OpenLLM): Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.12%, 3.36%, 11.19% of $N = 894$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.585 [0.526, 0.645]	0.761 [0.737, 0.785]	0.885 [0.874, 0.897]
Source difference	0.585 [0.526, 0.645]	0.761 [0.737, 0.785]	0.885 [0.874, 0.897]
Random-subset ridge	0.587 [0.530, 0.643]	0.765 [0.743, 0.787]	0.871 [0.859, 0.882]
Paper AIPW (tuned)	0.590 [0.533, 0.648]	0.765 [0.743, 0.788]	0.883 [0.873, 0.893]
Code AIPW (alpha 10)	0.586 [0.530, 0.643]	0.764 [0.742, 0.786]	0.880 [0.870, 0.891]
Crossfit difference	0.586 [0.528, 0.645]	0.762 [0.739, 0.785]	0.882 [0.871, 0.893]
Crossfit clipped	0.586 [0.528, 0.645]	0.762 [0.739, 0.785]	0.882 [0.871, 0.893]
Half-fit impute (tuned)	0.587 [0.529, 0.645]	0.759 [0.733, 0.784]	0.869 [0.857, 0.880]
Half-fit impute (matched)	0.587 [0.529, 0.645]	0.763 [0.740, 0.787]	0.874 [0.862, 0.886]
Full-fit impute	0.587 [0.530, 0.643]	0.765 [0.743, 0.787]	0.874 [0.864, 0.885]
Correctness-anchor mean	0.789 [0.789, 0.789]	0.931 [0.931, 0.931]	0.962 [0.962, 0.962]
Correctness-anchor ridge	0.791 [0.791, 0.791]	0.926 [0.926, 0.926]	0.948 [0.948, 0.948]
Random MIRT p	0.589 [0.532, 0.647]	0.767 [0.745, 0.789]	0.886 [0.875, 0.896]
Random MIRT gp	0.585 [0.526, 0.645]	0.761 [0.737, 0.785]	0.885 [0.874, 0.897]
IRT-anchor mean	0.622 [0.622, 0.622]	0.842 [0.842, 0.842]	0.896 [0.896, 0.896]
IRT-anchor MIRT p	0.624 [0.624, 0.624]	0.848 [0.848, 0.848]	0.901 [0.901, 0.901]
IRT-anchor MIRT gp	0.622 [0.622, 0.622]	0.842 [0.842, 0.842]	0.896 [0.896, 0.896]

Table 59: MATH (OpenLLM): Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.12%, 3.36%, 11.19% of $N = 894$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.490 [0.439, 0.541]	0.642 [0.620, 0.664]	0.768 [0.755, 0.781]
Source difference	0.490 [0.439, 0.541]	0.642 [0.620, 0.664]	0.768 [0.755, 0.781]
Random-subset ridge	0.484 [0.437, 0.532]	0.630 [0.609, 0.650]	0.732 [0.719, 0.746]
Paper AIPW (tuned)	0.487 [0.439, 0.534]	0.630 [0.609, 0.651]	0.746 [0.734, 0.757]
Code AIPW (alpha 10)	0.481 [0.435, 0.527]	0.629 [0.608, 0.649]	0.743 [0.731, 0.755]
Crossfit difference	0.482 [0.434, 0.530]	0.627 [0.605, 0.649]	0.744 [0.731, 0.756]
Crossfit clipped	0.482 [0.434, 0.530]	0.627 [0.605, 0.649]	0.744 [0.731, 0.756]
Half-fit impute (tuned)	0.484 [0.435, 0.533]	0.623 [0.598, 0.648]	0.727 [0.714, 0.740]
Half-fit impute (matched)	0.485 [0.436, 0.533]	0.629 [0.607, 0.651]	0.735 [0.721, 0.749]
Full-fit impute	0.484 [0.437, 0.532]	0.630 [0.609, 0.651]	0.736 [0.725, 0.748]
Correctness-anchor mean	0.655 [0.655, 0.655]	0.809 [0.809, 0.809]	0.870 [0.870, 0.870]
Correctness-anchor ridge	0.659 [0.659, 0.659]	0.804 [0.804, 0.804]	0.837 [0.837, 0.837]
Random MIRT p	0.486 [0.438, 0.534]	0.634 [0.613, 0.654]	0.751 [0.739, 0.764]
Random MIRT gp	0.490 [0.439, 0.541]	0.642 [0.620, 0.664]	0.768 [0.755, 0.781]
IRT-anchor mean	0.509 [0.509, 0.509]	0.702 [0.702, 0.702]	0.760 [0.760, 0.760]
IRT-anchor MIRT p	0.512 [0.512, 0.512]	0.714 [0.714, 0.714]	0.768 [0.768, 0.768]
IRT-anchor MIRT gp	0.509 [0.509, 0.509]	0.702 [0.702, 0.702]	0.760 [0.760, 0.760]

Table 60: Commonsense: MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 2.00%, 6.00%, 20.00% of $N = 500$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	11.70 [11.70, 11.70]	11.70 [11.70, 11.70]	11.70 [11.70, 11.70]
Random mean	9.07 [7.68, 10.45]	5.02 [4.41, 5.63]	2.39 [2.06, 2.73]
Source difference	7.39 [6.59, 8.19]	3.93 [3.59, 4.28]	2.02 [1.81, 2.22]
Random-subset ridge	5.07 [4.66, 5.47]	4.03 [3.59, 4.47]	2.28 [2.06, 2.49]
Paper AIPW (tuned)	7.28 [6.54, 8.01]	4.31 [3.75, 4.86]	1.94 [1.76, 2.12]
Code AIPW (alpha 10)	7.27 [6.53, 8.00]	4.06 [3.61, 4.51]	1.99 [1.82, 2.16]
Crossfit difference	7.06 [6.39, 7.73]	3.79 [3.47, 4.11]	1.96 [1.79, 2.13]
Crossfit clipped	6.88 [6.25, 7.51]	3.79 [3.47, 4.11]	1.96 [1.79, 2.13]
Half-fit impute (tuned)	5.00 [4.58, 5.42]	3.49 [3.11, 3.86]	2.07 [1.86, 2.28]
Half-fit impute (matched)	5.29 [4.87, 5.70]	4.60 [4.23, 4.96]	2.35 [2.16, 2.54]
Full-fit impute	5.06 [4.66, 5.47]	4.03 [3.59, 4.47]	2.23 [2.02, 2.44]
Correctness-anchor mean	6.47 [6.47, 6.47]	5.21 [5.21, 5.21]	4.26 [4.26, 4.26]
Correctness-anchor ridge	5.84 [5.84, 5.84]	4.55 [4.55, 4.55]	2.86 [2.86, 2.86]
Random MIRT p	10.18 [9.61, 10.75]	4.90 [4.53, 5.28]	2.19 [1.98, 2.40]
Random MIRT gp	7.34 [6.09, 8.59]	3.85 [3.38, 4.32]	1.97 [1.74, 2.21]
IRT-anchor mean	6.34 [6.34, 6.34]	5.82 [5.82, 5.82]	2.71 [2.71, 2.71]
IRT-anchor MIRT p	12.92 [12.92, 12.92]	5.29 [5.29, 5.29]	2.33 [2.33, 2.33]
IRT-anchor MIRT gp	8.44 [8.44, 8.44]	3.11 [3.11, 3.11]	1.73 [1.73, 1.73]

Table 61: Commonsense: RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 2.00%, 6.00%, 20.00% of $N = 500$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	18.32 [18.32, 18.32]	18.32 [18.32, 18.32]	18.32 [18.32, 18.32]
Random mean	11.11 [9.59, 12.62]	6.23 [5.57, 6.89]	3.03 [2.66, 3.40]
Source difference	9.61 [8.54, 10.68]	5.16 [4.69, 5.63]	2.61 [2.34, 2.88]
Random-subset ridge	7.12 [6.47, 7.77]	5.69 [4.94, 6.44]	3.04 [2.79, 3.30]
Paper AIPW (tuned)	9.18 [8.32, 10.04]	5.51 [4.84, 6.18]	2.51 [2.28, 2.73]
Code AIPW (alpha 10)	9.17 [8.32, 10.03]	5.17 [4.61, 5.74]	2.56 [2.34, 2.78]
Crossfit difference	9.28 [8.38, 10.17]	5.00 [4.52, 5.47]	2.52 [2.30, 2.73]
Crossfit clipped	9.06 [8.20, 9.93]	5.00 [4.52, 5.47]	2.52 [2.30, 2.73]
Half-fit impute (tuned)	7.22 [6.44, 7.99]	4.76 [4.06, 5.47]	2.81 [2.50, 3.12]
Half-fit impute (matched)	7.57 [6.86, 8.28]	6.94 [6.36, 7.51]	3.58 [3.30, 3.86]
Full-fit impute	7.12 [6.47, 7.77]	5.68 [4.94, 6.43]	2.96 [2.71, 3.21]
Correctness-anchor mean	8.73 [8.73, 8.73]	6.72 [6.72, 6.72]	6.15 [6.15, 6.15]
Correctness-anchor ridge	8.27 [8.27, 8.27]	6.38 [6.38, 6.38]	3.87 [3.87, 3.87]
Random MIRT p	11.18 [10.62, 11.75]	5.71 [5.28, 6.14]	2.68 [2.44, 2.92]
Random MIRT gp	8.94 [7.60, 10.28]	4.90 [4.35, 5.45]	2.53 [2.26, 2.80]
IRT-anchor mean	10.32 [10.32, 10.32]	6.61 [6.61, 6.61]	3.72 [3.72, 3.72]
IRT-anchor MIRT p	13.86 [13.86, 13.86]	6.14 [6.14, 6.14]	3.06 [3.06, 3.06]
IRT-anchor MIRT gp	10.39 [10.39, 10.39]	4.24 [4.24, 4.24]	2.56 [2.56, 2.56]

Table 62: Commonsense: Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 2.00%, 6.00%, 20.00% of $N = 500$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.705 [0.612, 0.799]	0.865 [0.838, 0.893]	0.957 [0.949, 0.966]
Source difference	0.705 [0.612, 0.799]	0.865 [0.838, 0.893]	0.957 [0.949, 0.966]
Random-subset ridge	0.753 [0.687, 0.819]	0.827 [0.788, 0.866]	0.947 [0.936, 0.957]
Paper AIPW (tuned)	0.727 [0.651, 0.803]	0.848 [0.817, 0.879]	0.957 [0.951, 0.964]
Code AIPW (alpha 10)	0.727 [0.651, 0.803]	0.865 [0.836, 0.894]	0.954 [0.948, 0.960]
Crossfit difference	0.707 [0.626, 0.788]	0.864 [0.837, 0.890]	0.954 [0.946, 0.961]
Crossfit clipped	0.708 [0.627, 0.789]	0.864 [0.837, 0.890]	0.954 [0.946, 0.961]
Half-fit impute (tuned)	0.746 [0.691, 0.801]	0.855 [0.827, 0.884]	0.951 [0.941, 0.961]
Half-fit impute (matched)	0.755 [0.700, 0.810]	0.867 [0.840, 0.894]	0.960 [0.952, 0.968]
Full-fit impute	0.753 [0.687, 0.818]	0.827 [0.788, 0.866]	0.947 [0.937, 0.957]
Correctness-anchor mean	0.809 [0.809, 0.809]	0.961 [0.961, 0.961]	0.956 [0.956, 0.956]
Correctness-anchor ridge	0.806 [0.806, 0.806]	0.885 [0.885, 0.885]	0.938 [0.938, 0.938]
Random MIRT p	0.744 [0.677, 0.812]	0.875 [0.848, 0.901]	0.960 [0.955, 0.966]
Random MIRT gp	0.732 [0.644, 0.821]	0.869 [0.841, 0.898]	0.962 [0.955, 0.968]
IRT-anchor mean	0.548 [0.548, 0.548]	0.918 [0.918, 0.918]	0.960 [0.960, 0.960]
IRT-anchor MIRT p	0.708 [0.708, 0.708]	0.874 [0.874, 0.874]	0.973 [0.973, 0.973]
IRT-anchor MIRT gp	0.649 [0.649, 0.649]	0.918 [0.918, 0.918]	0.973 [0.973, 0.973]

Table 63: Commonsense: Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 2.00%, 6.00%, 20.00% of $N = 500$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.600 [0.515, 0.685]	0.754 [0.719, 0.790]	0.874 [0.857, 0.890]
Source difference	0.600 [0.515, 0.685]	0.754 [0.719, 0.790]	0.874 [0.857, 0.890]
Random-subset ridge	0.630 [0.570, 0.690]	0.688 [0.642, 0.734]	0.841 [0.824, 0.858]
Paper AIPW (tuned)	0.606 [0.537, 0.675]	0.712 [0.672, 0.752]	0.861 [0.849, 0.873]
Code AIPW (alpha 10)	0.606 [0.537, 0.675]	0.732 [0.693, 0.771]	0.854 [0.843, 0.865]
Crossfit difference	0.584 [0.513, 0.655]	0.729 [0.693, 0.765]	0.856 [0.843, 0.870]
Crossfit clipped	0.585 [0.514, 0.656]	0.729 [0.693, 0.765]	0.856 [0.843, 0.870]
Half-fit impute (tuned)	0.617 [0.566, 0.668]	0.720 [0.683, 0.756]	0.853 [0.835, 0.871]
Half-fit impute (matched)	0.633 [0.580, 0.685]	0.736 [0.701, 0.772]	0.874 [0.859, 0.890]
Full-fit impute	0.630 [0.570, 0.690]	0.688 [0.642, 0.734]	0.842 [0.825, 0.858]
Correctness-anchor mean	0.672 [0.672, 0.672]	0.866 [0.866, 0.866]	0.884 [0.884, 0.884]
Correctness-anchor ridge	0.655 [0.655, 0.655]	0.714 [0.714, 0.714]	0.829 [0.829, 0.829]
Random MIRT p	0.622 [0.557, 0.688]	0.748 [0.712, 0.785]	0.873 [0.863, 0.884]
Random MIRT gp	0.612 [0.531, 0.694]	0.739 [0.700, 0.778]	0.873 [0.860, 0.886]
IRT-anchor mean	0.454 [0.454, 0.454]	0.787 [0.787, 0.787]	0.868 [0.868, 0.868]
IRT-anchor MIRT p	0.543 [0.543, 0.543]	0.749 [0.749, 0.749]	0.890 [0.890, 0.890]
IRT-anchor MIRT gp	0.502 [0.502, 0.502]	0.803 [0.803, 0.803]	0.890 [0.890, 0.890]

Table 64: GSM: MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.00%, 3.00%, 10.00% of $N = 1,000$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	19.61 [19.61, 19.61]	19.61 [19.61, 19.61]	19.61 [19.61, 19.61]
Random mean	9.54 [7.67, 11.41]	5.81 [4.80, 6.81]	3.10 [2.80, 3.40]
Source difference	8.40 [7.81, 8.99]	5.01 [4.49, 5.53]	2.55 [2.33, 2.78]
Random-subset ridge	8.10 [7.47, 8.74]	5.38 [4.93, 5.84]	2.89 [2.67, 3.11]
Paper AIPW (tuned)	9.18 [7.67, 10.68]	4.92 [4.51, 5.32]	2.46 [2.23, 2.69]
Code AIPW (alpha 10)	8.61 [7.88, 9.33]	4.93 [4.53, 5.33]	2.54 [2.35, 2.73]
Crossfit difference	8.18 [7.55, 8.81]	4.94 [4.51, 5.37]	2.44 [2.24, 2.64]
Crossfit clipped	8.05 [7.43, 8.67]	4.87 [4.47, 5.26]	2.44 [2.24, 2.63]
Half-fit impute (tuned)	8.60 [7.99, 9.22]	5.18 [4.86, 5.51]	2.99 [2.73, 3.24]
Half-fit impute (matched)	10.84 [10.25, 11.44]	6.10 [5.84, 6.37]	6.89 [6.71, 7.08]
Full-fit impute	8.10 [7.47, 8.74]	5.38 [4.93, 5.84]	2.53 [2.34, 2.72]
Correctness-anchor mean	8.14 [8.14, 8.14]	3.78 [3.78, 3.78]	3.12 [3.12, 3.12]
Correctness-anchor ridge	6.75 [6.75, 6.75]	4.91 [4.91, 4.91]	3.05 [3.05, 3.05]
Random MIRT p	13.53 [12.82, 14.24]	7.88 [7.60, 8.16]	3.44 [3.22, 3.66]
Random MIRT gp	8.71 [6.97, 10.45]	5.30 [4.42, 6.19]	2.79 [2.55, 3.03]
IRT-anchor mean	7.46 [7.46, 7.46]	3.68 [3.68, 3.68]	2.89 [2.89, 2.89]
IRT-anchor MIRT p	11.56 [11.56, 11.56]	6.68 [6.68, 6.68]	3.51 [3.51, 3.51]
IRT-anchor MIRT gp	5.57 [5.57, 5.57]	3.77 [3.77, 3.77]	2.74 [2.74, 2.74]

Table 65: GSM: RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.00%, 3.00%, 10.00% of $N = 1,000$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	24.79 [24.79, 24.79]	24.79 [24.79, 24.79]	24.79 [24.79, 24.79]
Random mean	11.85 [10.02, 13.68]	7.29 [6.14, 8.45]	3.82 [3.47, 4.16]
Source difference	10.59 [9.93, 11.25]	6.31 [5.72, 6.90]	3.16 [2.90, 3.41]
Random-subset ridge	10.11 [9.28, 10.94]	6.93 [6.33, 7.53]	3.73 [3.43, 4.02]
Paper AIPW (tuned)	11.41 [9.91, 12.92]	6.24 [5.75, 6.72]	3.04 [2.78, 3.30]
Code AIPW (alpha 10)	10.86 [9.94, 11.77]	6.25 [5.77, 6.74]	3.18 [2.94, 3.42]
Crossfit difference	10.33 [9.62, 11.04]	6.19 [5.71, 6.67]	3.02 [2.77, 3.27]
Crossfit clipped	10.21 [9.50, 10.92]	6.12 [5.67, 6.56]	3.01 [2.76, 3.25]
Half-fit impute (tuned)	10.45 [9.79, 11.11]	6.47 [6.03, 6.90]	3.78 [3.48, 4.08]
Half-fit impute (matched)	13.19 [12.59, 13.79]	7.49 [7.20, 7.77]	8.85 [8.72, 8.98]
Full-fit impute	10.11 [9.28, 10.94]	6.93 [6.33, 7.53]	3.15 [2.92, 3.38]
Correctness-anchor mean	11.26 [11.26, 11.26]	4.89 [4.89, 4.89]	3.97 [3.97, 3.97]
Correctness-anchor ridge	8.84 [8.84, 8.84]	6.23 [6.23, 6.23]	3.96 [3.96, 3.96]
Random MIRT p	15.47 [14.77, 16.17]	9.12 [8.80, 9.44]	4.14 [3.93, 4.35]
Random MIRT gp	10.92 [9.22, 12.61]	6.72 [5.77, 7.68]	3.45 [3.18, 3.72]
IRT-anchor mean	9.24 [9.24, 9.24]	4.78 [4.78, 4.78]	4.00 [4.00, 4.00]
IRT-anchor MIRT p	12.79 [12.79, 12.79]	7.82 [7.82, 7.82]	4.31 [4.31, 4.31]
IRT-anchor MIRT gp	7.61 [7.61, 7.61]	4.41 [4.41, 4.41]	3.69 [3.69, 3.69]

Table 66: GSM: Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.00%, 3.00%, 10.00% of $N = 1,000$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.867 [0.834, 0.899]	0.945 [0.931, 0.960]	0.982 [0.978, 0.986]
Source difference	0.867 [0.834, 0.899]	0.945 [0.931, 0.960]	0.982 [0.978, 0.986]
Random-subset ridge	0.868 [0.835, 0.902]	0.928 [0.915, 0.942]	0.974 [0.968, 0.979]
Paper AIPW (tuned)	0.871 [0.838, 0.903]	0.940 [0.929, 0.951]	0.982 [0.978, 0.986]
Code AIPW (alpha 10)	0.870 [0.838, 0.902]	0.940 [0.929, 0.951]	0.981 [0.976, 0.985]
Crossfit difference	0.873 [0.843, 0.903]	0.939 [0.928, 0.951]	0.982 [0.978, 0.986]
Crossfit clipped	0.873 [0.843, 0.903]	0.940 [0.928, 0.951]	0.982 [0.978, 0.986]
Half-fit impute (tuned)	0.868 [0.833, 0.904]	0.940 [0.929, 0.951]	0.973 [0.966, 0.980]
Half-fit impute (matched)	0.872 [0.838, 0.906]	0.945 [0.935, 0.956]	0.980 [0.976, 0.984]
Full-fit impute	0.868 [0.835, 0.902]	0.928 [0.914, 0.942]	0.981 [0.976, 0.985]
Correctness-anchor mean	0.919 [0.919, 0.919]	0.965 [0.965, 0.965]	0.977 [0.977, 0.977]
Correctness-anchor ridge	0.921 [0.921, 0.921]	0.947 [0.947, 0.947]	0.973 [0.973, 0.973]
Random MIRT p	0.877 [0.852, 0.902]	0.945 [0.934, 0.956]	0.982 [0.978, 0.986]
Random MIRT gp	0.876 [0.847, 0.905]	0.946 [0.934, 0.959]	0.982 [0.978, 0.986]
IRT-anchor mean	0.953 [0.953, 0.953]	0.976 [0.976, 0.976]	0.969 [0.969, 0.969]
IRT-anchor MIRT p	0.950 [0.950, 0.950]	0.945 [0.945, 0.945]	0.964 [0.964, 0.964]
IRT-anchor MIRT gp	0.953 [0.953, 0.953]	0.977 [0.977, 0.977]	0.968 [0.968, 0.968]

Table 67: GSM: Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.00%, 3.00%, 10.00% of $N = 1,000$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.743 [0.705, 0.781]	0.842 [0.820, 0.863]	0.917 [0.905, 0.929]
Source difference	0.743 [0.705, 0.781]	0.842 [0.820, 0.863]	0.917 [0.905, 0.929]
Random-subset ridge	0.724 [0.685, 0.763]	0.797 [0.774, 0.820]	0.889 [0.872, 0.906]
Paper AIPW (tuned)	0.727 [0.687, 0.767]	0.815 [0.797, 0.834]	0.912 [0.900, 0.924]
Code AIPW (alpha 10)	0.724 [0.685, 0.764]	0.815 [0.797, 0.834]	0.909 [0.896, 0.922]
Crossfit difference	0.726 [0.689, 0.764]	0.814 [0.795, 0.833]	0.915 [0.902, 0.927]
Crossfit clipped	0.726 [0.689, 0.764]	0.815 [0.796, 0.833]	0.915 [0.902, 0.927]
Half-fit impute (tuned)	0.726 [0.683, 0.769]	0.815 [0.793, 0.837]	0.888 [0.871, 0.906]
Half-fit impute (matched)	0.727 [0.685, 0.769]	0.827 [0.807, 0.847]	0.907 [0.895, 0.918]
Full-fit impute	0.724 [0.684, 0.763]	0.797 [0.774, 0.820]	0.910 [0.895, 0.924]
Correctness-anchor mean	0.792 [0.792, 0.792]	0.866 [0.866, 0.866]	0.896 [0.896, 0.896]
Correctness-anchor ridge	0.792 [0.792, 0.792]	0.825 [0.825, 0.825]	0.871 [0.871, 0.871]
Random MIRT p	0.733 [0.699, 0.767]	0.823 [0.802, 0.843]	0.912 [0.899, 0.924]
Random MIRT gp	0.734 [0.696, 0.771]	0.828 [0.806, 0.850]	0.914 [0.901, 0.926]
IRT-anchor mean	0.839 [0.839, 0.839]	0.891 [0.891, 0.891]	0.865 [0.865, 0.865]
IRT-anchor MIRT p	0.826 [0.826, 0.826]	0.838 [0.838, 0.838]	0.858 [0.858, 0.858]
IRT-anchor MIRT gp	0.839 [0.839, 0.839]	0.891 [0.891, 0.891]	0.865 [0.865, 0.865]

Table 68: LegalBench: MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 0.49%, 1.47%, 4.89% of $N = 2,047$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	7.46 [7.46, 7.46]	7.46 [7.46, 7.46]	7.46 [7.46, 7.46]
Random mean	12.18 [9.73, 14.64]	6.46 [5.84, 7.08]	3.99 [3.29, 4.70]
Source difference	8.76 [8.07, 9.45]	5.38 [5.03, 5.74]	3.02 [2.78, 3.25]
Random-subset ridge	5.12 [4.71, 5.53]	3.43 [3.20, 3.65]	2.65 [2.50, 2.80]
Paper AIPW (tuned)	7.95 [6.86, 9.03]	4.72 [4.30, 5.14]	2.66 [2.43, 2.89]
Code AIPW (alpha 10)	8.59 [7.53, 9.65]	4.72 [4.30, 5.14]	2.67 [2.45, 2.89]
Crossfit difference	8.56 [7.79, 9.34]	5.13 [4.71, 5.55]	2.78 [2.56, 3.01]
Crossfit clipped	8.56 [7.79, 9.34]	5.13 [4.71, 5.55]	2.78 [2.56, 3.01]
Half-fit impute (tuned)	5.01 [4.52, 5.51]	3.54 [3.29, 3.79]	2.58 [2.39, 2.77]
Half-fit impute (matched)	5.79 [5.49, 6.10]	4.06 [3.83, 4.29]	2.49 [2.34, 2.65]
Full-fit impute	5.11 [4.70, 5.53]	3.43 [3.20, 3.65]	2.65 [2.50, 2.80]
Correctness-anchor mean	7.14 [7.14, 7.14]	4.43 [4.43, 4.43]	1.95 [1.95, 1.95]
Correctness-anchor ridge	3.87 [3.87, 3.87]	3.44 [3.44, 3.44]	2.32 [2.32, 2.32]
Random MIRT p	6.06 [5.57, 6.56]	4.08 [3.86, 4.31]	2.52 [2.36, 2.68]
Random MIRT gp	6.06 [5.57, 6.56]	3.89 [3.60, 4.18]	2.38 [2.16, 2.60]
IRT-anchor mean	5.50 [5.50, 5.50]	4.55 [4.55, 4.55]	1.88 [1.88, 1.88]
IRT-anchor MIRT p	5.31 [5.31, 5.31]	3.85 [3.85, 3.85]	2.23 [2.23, 2.23]
IRT-anchor MIRT gp	4.59 [4.59, 4.59]	3.13 [3.13, 3.13]	1.91 [1.91, 1.91]

Table 69: LegalBench: RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 0.49%, 1.47%, 4.89% of $N = 2,047$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	8.60 [8.60, 8.60]	8.60 [8.60, 8.60]	8.60 [8.60, 8.60]
Random mean	14.58 [12.01, 17.16]	7.94 [7.20, 8.69]	4.86 [4.08, 5.63]
Source difference	10.81 [9.98, 11.64]	6.71 [6.24, 7.19]	3.78 [3.47, 4.09]
Random-subset ridge	6.12 [5.63, 6.62]	4.32 [4.03, 4.60]	3.28 [3.07, 3.49]
Paper AIPW (tuned)	10.09 [8.80, 11.38]	5.89 [5.36, 6.41]	3.32 [3.03, 3.61]
Code AIPW (alpha 10)	10.72 [9.51, 11.93]	5.88 [5.36, 6.41]	3.33 [3.05, 3.62]
Crossfit difference	10.65 [9.67, 11.63]	6.41 [5.89, 6.93]	3.48 [3.18, 3.78]
Crossfit clipped	10.65 [9.67, 11.63]	6.41 [5.89, 6.93]	3.48 [3.18, 3.78]
Half-fit impute (tuned)	6.07 [5.50, 6.63]	4.50 [4.19, 4.81]	3.24 [3.03, 3.44]
Half-fit impute (matched)	6.83 [6.49, 7.18]	4.92 [4.65, 5.19]	3.03 [2.85, 3.21]
Full-fit impute	6.12 [5.62, 6.62]	4.31 [4.02, 4.60]	3.28 [3.07, 3.49]
Correctness-anchor mean	8.73 [8.73, 8.73]	5.32 [5.32, 5.32]	2.50 [2.50, 2.50]
Correctness-anchor ridge	5.32 [5.32, 5.32]	4.33 [4.33, 4.33]	2.98 [2.98, 2.98]
Random MIRT p	6.98 [6.48, 7.48]	4.97 [4.70, 5.24]	3.12 [2.93, 3.31]
Random MIRT gp	6.98 [6.48, 7.48]	4.85 [4.50, 5.21]	2.94 [2.66, 3.22]
IRT-anchor mean	6.75 [6.75, 6.75]	5.29 [5.29, 5.29]	2.42 [2.42, 2.42]
IRT-anchor MIRT p	6.52 [6.52, 6.52]	4.51 [4.51, 4.51]	2.68 [2.68, 2.68]
IRT-anchor MIRT gp	5.82 [5.82, 5.82]	4.06 [4.06, 4.06]	2.30 [2.30, 2.30]

Table 70: LegalBench: Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 0.49%, 1.47%, 4.89% of $N = 2,047$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.500 [0.386, 0.613]	0.708 [0.666, 0.751]	0.851 [0.820, 0.882]
Source difference	0.500 [0.386, 0.613]	0.708 [0.666, 0.751]	0.851 [0.820, 0.882]
Random-subset ridge	0.601 [0.514, 0.688]	0.798 [0.774, 0.823]	0.875 [0.854, 0.896]
Paper AIPW (tuned)	0.564 [0.471, 0.657]	0.745 [0.714, 0.776]	0.867 [0.840, 0.893]
Code AIPW (alpha 10)	0.538 [0.448, 0.627]	0.746 [0.714, 0.777]	0.865 [0.839, 0.890]
Crossfit difference	0.531 [0.435, 0.626]	0.726 [0.688, 0.764]	0.867 [0.842, 0.893]
Crossfit clipped	0.531 [0.435, 0.626]	0.726 [0.688, 0.764]	0.867 [0.842, 0.893]
Half-fit impute (tuned)	0.590 [0.500, 0.679]	0.780 [0.757, 0.803]	0.867 [0.846, 0.888]
Half-fit impute (matched)	0.591 [0.512, 0.670]	0.800 [0.782, 0.818]	0.900 [0.882, 0.919]
Full-fit impute	0.601 [0.514, 0.688]	0.796 [0.772, 0.821]	0.874 [0.852, 0.895]
Correctness-anchor mean	0.474 [0.474, 0.474]	0.761 [0.761, 0.761]	0.910 [0.910, 0.910]
Correctness-anchor ridge	0.631 [0.631, 0.631]	0.787 [0.787, 0.787]	0.901 [0.901, 0.901]
Random MIRT p	0.609 [0.533, 0.685]	0.787 [0.764, 0.810]	0.907 [0.884, 0.930]
Random MIRT gp	0.609 [0.533, 0.685]	0.767 [0.741, 0.793]	0.897 [0.871, 0.923]
IRT-anchor mean	0.547 [0.547, 0.547]	0.745 [0.745, 0.745]	0.923 [0.923, 0.923]
IRT-anchor MIRT p	0.513 [0.513, 0.513]	0.741 [0.741, 0.741]	0.880 [0.880, 0.880]
IRT-anchor MIRT gp	0.560 [0.560, 0.560]	0.753 [0.753, 0.753]	0.892 [0.892, 0.892]

Table 71: LegalBench: Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 0.49%, 1.47%, 4.89% of $N = 2,047$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.394 [0.303, 0.485]	0.553 [0.512, 0.594]	0.704 [0.668, 0.740]
Source difference	0.394 [0.303, 0.485]	0.553 [0.512, 0.594]	0.704 [0.668, 0.740]
Random-subset ridge	0.459 [0.391, 0.527]	0.632 [0.609, 0.656]	0.722 [0.693, 0.751]
Paper AIPW (tuned)	0.428 [0.352, 0.503]	0.573 [0.544, 0.602]	0.718 [0.687, 0.750]
Code AIPW (alpha 10)	0.405 [0.333, 0.477]	0.574 [0.545, 0.603]	0.716 [0.685, 0.746]
Crossfit difference	0.393 [0.319, 0.467]	0.554 [0.520, 0.589]	0.711 [0.681, 0.742]
Crossfit clipped	0.393 [0.319, 0.467]	0.554 [0.520, 0.589]	0.711 [0.681, 0.742]
Half-fit impute (tuned)	0.442 [0.371, 0.513]	0.604 [0.579, 0.628]	0.711 [0.687, 0.736]
Half-fit impute (matched)	0.446 [0.385, 0.507]	0.626 [0.606, 0.647]	0.755 [0.732, 0.778]
Full-fit impute	0.459 [0.390, 0.527]	0.630 [0.606, 0.654]	0.721 [0.690, 0.751]
Correctness-anchor mean	0.345 [0.345, 0.345]	0.606 [0.606, 0.606]	0.783 [0.783, 0.783]
Correctness-anchor ridge	0.479 [0.479, 0.479]	0.628 [0.628, 0.628]	0.768 [0.768, 0.768]
Random MIRT p	0.457 [0.397, 0.517]	0.614 [0.592, 0.637]	0.763 [0.735, 0.790]
Random MIRT gp	0.457 [0.397, 0.517]	0.593 [0.569, 0.617]	0.752 [0.721, 0.782]
IRT-anchor mean	0.432 [0.432, 0.432]	0.580 [0.580, 0.580]	0.775 [0.775, 0.775]
IRT-anchor MIRT p	0.394 [0.394, 0.394]	0.558 [0.558, 0.558]	0.703 [0.703, 0.703]
IRT-anchor MIRT gp	0.439 [0.439, 0.439]	0.594 [0.594, 0.594]	0.725 [0.725, 0.725]

Table 72: MATH (HELM): MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 2.29%, 6.86%, 22.88% of $N = 437$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	23.15 [23.15, 23.15]	23.15 [23.15, 23.15]	23.15 [23.15, 23.15]
Random mean	9.46 [8.43, 10.49]	5.92 [5.36, 6.49]	2.97 [2.70, 3.24]
Source difference	8.36 [7.60, 9.11]	4.86 [4.58, 5.14]	2.57 [2.44, 2.69]
Random-subset ridge	7.45 [6.97, 7.93]	5.28 [4.92, 5.64]	2.63 [2.49, 2.77]
Paper AIPW (tuned)	8.47 [7.62, 9.32]	4.77 [4.41, 5.13]	2.61 [2.47, 2.75]
Code AIPW (alpha 10)	8.47 [7.62, 9.32]	4.77 [4.41, 5.13]	2.76 [2.60, 2.92]
Crossfit difference	8.35 [7.59, 9.11]	4.72 [4.40, 5.04]	2.54 [2.38, 2.71]
Crossfit clipped	8.11 [7.41, 8.82]	4.70 [4.38, 5.02]	2.54 [2.38, 2.71]
Half-fit impute (tuned)	7.75 [7.35, 8.15]	5.07 [4.72, 5.43]	3.19 [2.96, 3.42]
Half-fit impute (matched)	22.32 [22.27, 22.37]	5.86 [5.62, 6.10]	2.87 [2.73, 3.01]
Full-fit impute	7.45 [6.96, 7.93]	4.76 [4.48, 5.04]	2.57 [2.42, 2.71]
Correctness-anchor mean	8.24 [8.24, 8.24]	3.98 [3.98, 3.98]	3.08 [3.08, 3.08]
Correctness-anchor ridge	7.96 [7.96, 7.96]	4.63 [4.63, 4.63]	3.10 [3.10, 3.10]
Random MIRT p	11.16 [10.62, 11.71]	6.28 [5.97, 6.59]	2.86 [2.69, 3.03]
Random MIRT gp	8.16 [7.31, 9.02]	5.36 [4.82, 5.90]	2.69 [2.47, 2.91]
IRT-anchor mean	7.71 [7.71, 7.71]	5.62 [5.62, 5.62]	1.96 [1.96, 1.96]
IRT-anchor MIRT p	10.73 [10.73, 10.73]	6.24 [6.24, 6.24]	2.06 [2.06, 2.06]
IRT-anchor MIRT gp	8.24 [8.24, 8.24]	5.27 [5.27, 5.27]	1.63 [1.63, 1.63]

Table 73: MATH (HELM): RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 2.29%, 6.86%, 22.88% of $N = 437$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	26.56 [26.56, 26.56]	26.56 [26.56, 26.56]	26.56 [26.56, 26.56]
Random mean	11.80 [10.60, 13.00]	7.42 [6.73, 8.11]	3.67 [3.36, 3.97]
Source difference	10.28 [9.48, 11.08]	6.15 [5.78, 6.52]	3.19 [3.04, 3.34]
Random-subset ridge	9.33 [8.72, 9.94]	6.74 [6.30, 7.19]	3.28 [3.11, 3.46]
Paper AIPW (tuned)	10.60 [9.57, 11.62]	6.07 [5.63, 6.51]	3.33 [3.17, 3.49]
Code AIPW (alpha 10)	10.59 [9.56, 11.62]	6.06 [5.62, 6.50]	3.54 [3.36, 3.71]
Crossfit difference	10.27 [9.47, 11.07]	6.03 [5.59, 6.47]	3.23 [3.05, 3.40]
Crossfit clipped	10.01 [9.25, 10.78]	5.99 [5.55, 6.43]	3.23 [3.05, 3.40]
Half-fit impute (tuned)	9.39 [8.89, 9.89]	6.35 [5.92, 6.79]	3.96 [3.70, 4.22]
Half-fit impute (matched)	25.62 [25.57, 25.66]	6.90 [6.60, 7.20]	3.46 [3.29, 3.62]
Full-fit impute	9.33 [8.72, 9.94]	5.91 [5.57, 6.24]	3.23 [3.05, 3.40]
Correctness-anchor mean	11.09 [11.09, 11.09]	5.38 [5.38, 5.38]	3.66 [3.66, 3.66]
Correctness-anchor ridge	10.04 [10.04, 10.04]	6.05 [6.05, 6.05]	3.71 [3.71, 3.71]
Random MIRT p	12.87 [12.34, 13.39]	7.29 [6.93, 7.65]	3.46 [3.25, 3.67]
Random MIRT gp	10.39 [9.43, 11.35]	6.69 [6.05, 7.33]	3.33 [3.07, 3.58]
IRT-anchor mean	10.74 [10.74, 10.74]	6.64 [6.64, 6.64]	2.60 [2.60, 2.60]
IRT-anchor MIRT p	12.29 [12.29, 12.29]	7.12 [7.12, 7.12]	2.65 [2.65, 2.65]
IRT-anchor MIRT gp	10.57 [10.57, 10.57]	6.28 [6.28, 6.28]	2.27 [2.27, 2.27]

Table 74: MATH (HELM): Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 2.29%, 6.86%, 22.88% of $N = 437$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.872 [0.851, 0.893]	0.942 [0.931, 0.952]	0.977 [0.973, 0.981]
Source difference	0.872 [0.851, 0.893]	0.942 [0.931, 0.952]	0.977 [0.973, 0.981]
Random-subset ridge	0.853 [0.832, 0.874]	0.916 [0.898, 0.934]	0.976 [0.972, 0.980]
Paper AIPW (tuned)	0.859 [0.828, 0.889]	0.940 [0.929, 0.952]	0.975 [0.970, 0.980]
Code AIPW (alpha 10)	0.858 [0.828, 0.889]	0.940 [0.928, 0.952]	0.975 [0.970, 0.980]
Crossfit difference	0.862 [0.836, 0.889]	0.938 [0.925, 0.950]	0.974 [0.970, 0.979]
Crossfit clipped	0.862 [0.836, 0.889]	0.938 [0.925, 0.950]	0.974 [0.970, 0.979]
Half-fit impute (tuned)	0.863 [0.842, 0.883]	0.928 [0.914, 0.942]	0.968 [0.962, 0.974]
Half-fit impute (matched)	0.882 [0.864, 0.899]	0.942 [0.931, 0.952]	0.976 [0.972, 0.980]
Full-fit impute	0.854 [0.834, 0.875]	0.940 [0.929, 0.951]	0.976 [0.972, 0.980]
Correctness-anchor mean	0.886 [0.886, 0.886]	0.926 [0.926, 0.926]	0.957 [0.957, 0.957]
Correctness-anchor ridge	0.905 [0.905, 0.905]	0.912 [0.912, 0.912]	0.957 [0.957, 0.957]
Random MIRT p	0.859 [0.832, 0.887]	0.942 [0.930, 0.953]	0.976 [0.972, 0.979]
Random MIRT gp	0.866 [0.838, 0.894]	0.943 [0.932, 0.953]	0.977 [0.973, 0.981]
IRT-anchor mean	0.891 [0.891, 0.891]	0.907 [0.907, 0.907]	0.991 [0.991, 0.991]
IRT-anchor MIRT p	0.905 [0.905, 0.905]	0.916 [0.916, 0.916]	0.986 [0.986, 0.986]
IRT-anchor MIRT gp	0.909 [0.909, 0.909]	0.907 [0.907, 0.907]	0.991 [0.991, 0.991]

Table 75: MATH (HELM): Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 2.29%, 6.86%, 22.88% of $N = 437$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.751 [0.724, 0.778]	0.839 [0.823, 0.854]	0.906 [0.894, 0.918]
Source difference	0.751 [0.724, 0.778]	0.839 [0.823, 0.854]	0.906 [0.894, 0.918]
Random-subset ridge	0.702 [0.677, 0.726]	0.789 [0.766, 0.811]	0.898 [0.889, 0.907]
Paper AIPW (tuned)	0.712 [0.677, 0.746]	0.824 [0.809, 0.838]	0.895 [0.883, 0.908]
Code AIPW (alpha 10)	0.712 [0.677, 0.746]	0.822 [0.807, 0.838]	0.895 [0.883, 0.908]
Crossfit difference	0.718 [0.688, 0.748]	0.820 [0.802, 0.839]	0.894 [0.884, 0.905]
Crossfit clipped	0.717 [0.687, 0.747]	0.820 [0.802, 0.839]	0.894 [0.884, 0.905]
Half-fit impute (tuned)	0.716 [0.691, 0.742]	0.808 [0.789, 0.828]	0.878 [0.866, 0.890]
Half-fit impute (matched)	0.743 [0.718, 0.767]	0.830 [0.814, 0.846]	0.901 [0.890, 0.911]
Full-fit impute	0.703 [0.679, 0.727]	0.826 [0.810, 0.841]	0.901 [0.892, 0.910]
Correctness-anchor mean	0.714 [0.714, 0.714]	0.789 [0.789, 0.789]	0.870 [0.870, 0.870]
Correctness-anchor ridge	0.756 [0.756, 0.756]	0.783 [0.783, 0.783]	0.865 [0.865, 0.865]
Random MIRT p	0.714 [0.686, 0.743]	0.831 [0.816, 0.846]	0.898 [0.890, 0.907]
Random MIRT gp	0.724 [0.693, 0.755]	0.830 [0.815, 0.846]	0.902 [0.890, 0.913]
IRT-anchor mean	0.732 [0.732, 0.732]	0.769 [0.769, 0.769]	0.945 [0.945, 0.945]
IRT-anchor MIRT p	0.753 [0.753, 0.753]	0.803 [0.803, 0.803]	0.918 [0.918, 0.918]
IRT-anchor MIRT gp	0.759 [0.759, 0.759]	0.776 [0.776, 0.776]	0.945 [0.945, 0.945]

Table 76: MedQA: MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.00%, 3.00%, 10.00% of $N = 1,000$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	12.78 [12.78, 12.78]	12.78 [12.78, 12.78]	12.78 [12.78, 12.78]
Random mean	11.59 [10.56, 12.62]	6.21 [5.14, 7.27]	3.88 [3.36, 4.40]
Source difference	9.43 [8.89, 9.98]	5.42 [4.96, 5.87]	2.85 [2.61, 3.09]
Random-subset ridge	7.12 [6.55, 7.69]	5.25 [4.88, 5.61]	2.67 [2.50, 2.84]
Paper AIPW (tuned)	9.85 [9.04, 10.66]	5.18 [4.74, 5.62]	2.73 [2.52, 2.95]
Code AIPW (alpha 10)	9.85 [9.04, 10.66]	5.18 [4.74, 5.63]	2.63 [2.48, 2.78]
Crossfit difference	9.07 [8.42, 9.71]	5.42 [4.97, 5.87]	2.65 [2.48, 2.81]
Crossfit clipped	9.03 [8.39, 9.66]	5.42 [4.97, 5.87]	2.65 [2.48, 2.81]
Half-fit impute (tuned)	7.16 [6.66, 7.67]	4.99 [4.65, 5.33]	3.08 [2.91, 3.25]
Half-fit impute (matched)	7.33 [6.84, 7.82]	9.86 [9.69, 10.02]	2.90 [2.70, 3.10]
Full-fit impute	7.12 [6.55, 7.69]	5.24 [4.88, 5.61]	2.61 [2.45, 2.77]
Correctness-anchor mean	6.53 [6.53, 6.53]	3.00 [3.00, 3.00]	3.06 [3.06, 3.06]
Correctness-anchor ridge	6.79 [6.79, 6.79]	3.67 [3.67, 3.67]	3.06 [3.06, 3.06]
Random MIRT p	9.29 [8.84, 9.75]	6.34 [6.02, 6.66]	3.17 [2.96, 3.39]
Random MIRT gp	8.00 [7.33, 8.68]	5.17 [4.50, 5.84]	2.92 [2.71, 3.13]
IRT-anchor mean	7.99 [7.99, 7.99]	3.92 [3.92, 3.92]	2.73 [2.73, 2.73]
IRT-anchor MIRT p	9.47 [9.47, 9.47]	5.10 [5.10, 5.10]	3.06 [3.06, 3.06]
IRT-anchor MIRT gp	8.43 [8.43, 8.43]	3.46 [3.46, 3.46]	2.54 [2.54, 2.54]

Table 77: MedQA: RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.00%, 3.00%, 10.00% of $N = 1,000$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	15.97 [15.97, 15.97]	15.97 [15.97, 15.97]	15.97 [15.97, 15.97]
Random mean	14.06 [12.82, 15.29]	7.64 [6.54, 8.74]	4.60 [4.06, 5.14]
Source difference	11.66 [10.93, 12.38]	6.85 [6.33, 7.36]	3.59 [3.29, 3.90]
Random-subset ridge	8.90 [8.26, 9.55]	6.64 [6.19, 7.09]	3.39 [3.16, 3.62]
Paper AIPW (tuned)	12.19 [11.15, 13.24]	6.54 [6.04, 7.04]	3.41 [3.16, 3.66]
Code AIPW (alpha 10)	12.19 [11.15, 13.23]	6.55 [6.05, 7.05]	3.29 [3.10, 3.47]
Crossfit difference	11.18 [10.31, 12.04]	6.85 [6.36, 7.35]	3.31 [3.10, 3.51]
Crossfit clipped	11.10 [10.26, 11.94]	6.85 [6.36, 7.35]	3.31 [3.10, 3.51]
Half-fit impute (tuned)	8.91 [8.30, 9.51]	6.28 [5.79, 6.76]	3.87 [3.57, 4.16]
Half-fit impute (matched)	9.12 [8.55, 9.69]	12.50 [12.31, 12.70]	3.77 [3.52, 4.02]
Full-fit impute	8.90 [8.25, 9.55]	6.64 [6.19, 7.08]	3.30 [3.07, 3.52]
Correctness-anchor mean	8.52 [8.52, 8.52]	3.64 [3.64, 3.64]	3.67 [3.67, 3.67]
Correctness-anchor ridge	9.04 [9.04, 9.04]	5.41 [5.41, 5.41]	4.01 [4.01, 4.01]
Random MIRT p	11.16 [10.69, 11.63]	7.67 [7.31, 8.03]	4.06 [3.80, 4.31]
Random MIRT gp	9.64 [8.95, 10.33]	6.39 [5.70, 7.09]	3.60 [3.38, 3.83]
IRT-anchor mean	10.71 [10.71, 10.71]	4.77 [4.77, 4.77]	3.56 [3.56, 3.56]
IRT-anchor MIRT p	12.21 [12.21, 12.21]	6.10 [6.10, 6.10]	3.75 [3.75, 3.75]
IRT-anchor MIRT gp	10.83 [10.83, 10.83]	4.09 [4.09, 4.09]	3.25 [3.25, 3.25]

Table 78: MedQA: Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.00%, 3.00%, 10.00% of $N = 1,000$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.805 [0.765, 0.845]	0.914 [0.897, 0.931]	0.973 [0.969, 0.978]
Source difference	0.805 [0.765, 0.845]	0.914 [0.897, 0.931]	0.973 [0.969, 0.978]
Random-subset ridge	0.794 [0.756, 0.832]	0.898 [0.882, 0.914]	0.972 [0.967, 0.978]
Paper AIPW (tuned)	0.789 [0.745, 0.833]	0.910 [0.893, 0.928]	0.973 [0.969, 0.978]
Code AIPW (alpha 10)	0.790 [0.746, 0.833]	0.911 [0.894, 0.929]	0.968 [0.963, 0.972]
Crossfit difference	0.790 [0.755, 0.826]	0.912 [0.896, 0.928]	0.971 [0.967, 0.976]
Crossfit clipped	0.790 [0.755, 0.826]	0.912 [0.896, 0.928]	0.971 [0.967, 0.976]
Half-fit impute (tuned)	0.798 [0.755, 0.842]	0.910 [0.893, 0.927]	0.960 [0.953, 0.968]
Half-fit impute (matched)	0.804 [0.763, 0.844]	0.919 [0.904, 0.934]	0.973 [0.968, 0.979]
Full-fit impute	0.794 [0.756, 0.832]	0.898 [0.882, 0.914]	0.972 [0.967, 0.977]
Correctness-anchor mean	0.861 [0.861, 0.861]	0.955 [0.955, 0.955]	0.985 [0.985, 0.985]
Correctness-anchor ridge	0.858 [0.858, 0.858]	0.959 [0.959, 0.959]	0.976 [0.976, 0.976]
Random MIRT p	0.811 [0.769, 0.853]	0.912 [0.896, 0.928]	0.971 [0.965, 0.976]
Random MIRT gp	0.810 [0.767, 0.852]	0.915 [0.899, 0.930]	0.974 [0.970, 0.978]
IRT-anchor mean	0.808 [0.808, 0.808]	0.971 [0.971, 0.971]	0.962 [0.962, 0.962]
IRT-anchor MIRT p	0.890 [0.890, 0.890]	0.957 [0.957, 0.957]	0.976 [0.976, 0.976]
IRT-anchor MIRT gp	0.874 [0.874, 0.874]	0.969 [0.969, 0.969]	0.964 [0.964, 0.964]

Table 79: MedQA: Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.00%, 3.00%, 10.00% of $N = 1,000$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.671 [0.630, 0.712]	0.789 [0.759, 0.818]	0.890 [0.877, 0.903]
Source difference	0.671 [0.630, 0.712]	0.789 [0.759, 0.818]	0.890 [0.877, 0.903]
Random-subset ridge	0.634 [0.594, 0.674]	0.751 [0.729, 0.773]	0.883 [0.868, 0.897]
Paper AIPW (tuned)	0.631 [0.587, 0.676]	0.764 [0.736, 0.793]	0.886 [0.873, 0.899]
Code AIPW (alpha 10)	0.633 [0.588, 0.678]	0.766 [0.737, 0.795]	0.871 [0.857, 0.885]
Crossfit difference	0.623 [0.587, 0.659]	0.770 [0.741, 0.800]	0.882 [0.869, 0.895]
Crossfit clipped	0.623 [0.587, 0.659]	0.770 [0.741, 0.800]	0.882 [0.869, 0.895]
Half-fit impute (tuned)	0.644 [0.600, 0.688]	0.766 [0.738, 0.794]	0.854 [0.837, 0.871]
Half-fit impute (matched)	0.647 [0.605, 0.689]	0.782 [0.758, 0.806]	0.887 [0.873, 0.901]
Full-fit impute	0.634 [0.594, 0.674]	0.752 [0.730, 0.774]	0.883 [0.869, 0.897]
Correctness-anchor mean	0.698 [0.698, 0.698]	0.849 [0.849, 0.849]	0.916 [0.916, 0.916]
Correctness-anchor ridge	0.698 [0.698, 0.698]	0.858 [0.858, 0.858]	0.891 [0.891, 0.891]
Random MIRT p	0.656 [0.609, 0.702]	0.771 [0.749, 0.793]	0.879 [0.864, 0.894]
Random MIRT gp	0.655 [0.611, 0.700]	0.776 [0.752, 0.800]	0.887 [0.876, 0.898]
IRT-anchor mean	0.645 [0.645, 0.645]	0.871 [0.871, 0.871]	0.858 [0.858, 0.858]
IRT-anchor MIRT p	0.746 [0.746, 0.746]	0.831 [0.831, 0.831]	0.884 [0.884, 0.884]
IRT-anchor MIRT gp	0.726 [0.726, 0.726]	0.864 [0.864, 0.864]	0.864 [0.864, 0.864]

Table 80: MMLU: MAE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.76%, 5.29%, 17.64% of $N = 567$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	9.21 [9.21, 9.21]	9.21 [9.21, 9.21]	9.21 [9.21, 9.21]
Random mean	11.41 [9.46, 13.36]	7.03 [5.93, 8.12]	3.73 [3.18, 4.28]
Source difference	8.45 [7.73, 9.17]	4.80 [4.38, 5.23]	2.54 [2.32, 2.77]
Random-subset ridge	5.85 [5.28, 6.42]	4.20 [3.86, 4.54]	2.58 [2.34, 2.83]
Paper AIPW (tuned)	8.64 [6.95, 10.33]	4.60 [4.10, 5.10]	2.49 [2.29, 2.68]
Code AIPW (alpha 10)	8.58 [6.99, 10.18]	4.57 [4.09, 5.05]	2.56 [2.35, 2.77]
Crossfit difference	8.41 [7.70, 9.11]	4.81 [4.30, 5.32]	2.50 [2.30, 2.69]
Crossfit clipped	8.41 [7.70, 9.11]	4.81 [4.30, 5.32]	2.50 [2.30, 2.69]
Half-fit impute (tuned)	6.17 [5.67, 6.66]	4.44 [4.12, 4.77]	2.78 [2.58, 2.97]
Half-fit impute (matched)	8.59 [8.48, 8.71]	5.01 [4.72, 5.30]	2.78 [2.58, 2.97]
Full-fit impute	5.84 [5.27, 6.41]	4.19 [3.86, 4.53]	2.56 [2.32, 2.80]
Correctness-anchor mean	5.74 [5.74, 5.74]	4.08 [4.08, 4.08]	2.31 [2.31, 2.31]
Correctness-anchor ridge	5.38 [5.38, 5.38]	3.47 [3.47, 3.47]	3.32 [3.32, 3.32]
Random MIRT p	7.42 [6.95, 7.89]	4.96 [4.65, 5.27]	2.70 [2.51, 2.88]
Random MIRT gp	6.94 [6.14, 7.75]	4.61 [4.14, 5.08]	2.65 [2.37, 2.94]
IRT-anchor mean	6.80 [6.80, 6.80]	4.54 [4.54, 4.54]	1.78 [1.78, 1.78]
IRT-anchor MIRT p	7.99 [7.99, 7.99]	4.26 [4.26, 4.26]	2.39 [2.39, 2.39]
IRT-anchor MIRT gp	6.55 [6.55, 6.55]	3.95 [3.95, 3.95]	1.68 [1.68, 1.68]

Table 81: MMLU: RMSE in percentage points. Entries are supplied means [95% t interval] across 20 item orderings. Lower is better. The $k = 10, 30, 100$ budgets use 1.76%, 5.29%, 17.64% of $N = 567$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	11.58 [11.58, 11.58]	11.58 [11.58, 11.58]	11.58 [11.58, 11.58]
Random mean	13.70 [11.66, 15.74]	8.38 [7.29, 9.48]	4.43 [3.87, 4.99]
Source difference	10.52 [9.61, 11.44]	6.03 [5.50, 6.57]	3.21 [2.93, 3.49]
Random-subset ridge	7.40 [6.70, 8.09]	5.25 [4.81, 5.68]	3.27 [3.01, 3.53]
Paper AIPW (tuned)	10.98 [9.10, 12.87]	5.71 [5.15, 6.28]	3.08 [2.85, 3.30]
Code AIPW (alpha 10)	10.63 [8.85, 12.41]	5.69 [5.13, 6.25]	3.17 [2.93, 3.40]
Crossfit difference	10.48 [9.57, 11.38]	6.01 [5.42, 6.60]	3.10 [2.86, 3.34]
Crossfit clipped	10.48 [9.57, 11.38]	6.01 [5.42, 6.60]	3.10 [2.86, 3.34]
Half-fit impute (tuned)	7.70 [7.13, 8.27]	5.48 [5.08, 5.88]	3.47 [3.26, 3.69]
Half-fit impute (matched)	10.85 [10.74, 10.96]	6.27 [5.92, 6.61]	3.47 [3.26, 3.69]
Full-fit impute	7.39 [6.70, 8.08]	5.24 [4.80, 5.67]	3.24 [2.97, 3.50]
Correctness-anchor mean	7.81 [7.81, 7.81]	6.23 [6.23, 6.23]	3.02 [3.02, 3.02]
Correctness-anchor ridge	6.99 [6.99, 6.99]	4.00 [4.00, 4.00]	4.39 [4.39, 4.39]
Random MIRT p	9.04 [8.54, 9.53]	6.08 [5.77, 6.39]	3.32 [3.11, 3.53]
Random MIRT gp	8.44 [7.57, 9.32]	5.53 [5.04, 6.01]	3.23 [2.91, 3.54]
IRT-anchor mean	8.91 [8.91, 8.91]	5.83 [5.83, 5.83]	2.36 [2.36, 2.36]
IRT-anchor MIRT p	9.38 [9.38, 9.38]	5.39 [5.39, 5.39]	2.95 [2.95, 2.95]
IRT-anchor MIRT gp	8.12 [8.12, 8.12]	5.01 [5.01, 5.01]	1.93 [1.93, 1.93]

Table 82: MMLU: Spearman ρ in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.76%, 5.29%, 17.64% of $N = 567$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.701 [0.615, 0.786]	0.864 [0.827, 0.901]	0.964 [0.958, 0.969]
Source difference	0.701 [0.615, 0.786]	0.864 [0.827, 0.901]	0.964 [0.958, 0.969]
Random-subset ridge	0.734 [0.665, 0.803]	0.882 [0.858, 0.906]	0.959 [0.950, 0.969]
Paper AIPW (tuned)	0.675 [0.592, 0.759]	0.873 [0.844, 0.902]	0.961 [0.955, 0.967]
Code AIPW (alpha 10)	0.708 [0.626, 0.791]	0.873 [0.843, 0.902]	0.958 [0.951, 0.964]
Crossfit difference	0.705 [0.626, 0.784]	0.862 [0.831, 0.893]	0.964 [0.959, 0.968]
Crossfit clipped	0.705 [0.626, 0.784]	0.862 [0.831, 0.893]	0.964 [0.959, 0.968]
Half-fit impute (tuned)	0.747 [0.680, 0.814]	0.877 [0.853, 0.901]	0.969 [0.964, 0.975]
Half-fit impute (matched)	0.755 [0.695, 0.814]	0.908 [0.890, 0.927]	0.969 [0.964, 0.975]
Full-fit impute	0.737 [0.672, 0.802]	0.883 [0.859, 0.906]	0.960 [0.951, 0.968]
Correctness-anchor mean	0.896 [0.896, 0.896]	0.964 [0.964, 0.964]	0.982 [0.982, 0.982]
Correctness-anchor ridge	0.871 [0.871, 0.871]	0.952 [0.952, 0.952]	0.957 [0.957, 0.957]
Random MIRT p	0.738 [0.675, 0.800]	0.919 [0.900, 0.938]	0.972 [0.969, 0.976]
Random MIRT gp	0.730 [0.652, 0.808]	0.916 [0.895, 0.937]	0.971 [0.966, 0.975]
IRT-anchor mean	0.884 [0.884, 0.884]	0.885 [0.885, 0.885]	0.974 [0.974, 0.974]
IRT-anchor MIRT p	0.857 [0.857, 0.857]	0.937 [0.937, 0.937]	0.986 [0.986, 0.986]
IRT-anchor MIRT gp	0.872 [0.872, 0.872]	0.933 [0.933, 0.933]	0.982 [0.982, 0.982]

Table 83: MMLU: Kendall τ_b in correlation units. Entries are supplied means [95% t interval] across 20 item orderings. Higher is better; undefined (20) means all 20 draws lack a defined correlation. Ranks retain ties using the documented post-scoring 12-decimal numerical convention; raw predictions and error metrics are unchanged. The $k = 10, 30, 100$ budgets use 1.76%, 5.29%, 17.64% of $N = 567$ items. The source prior observes zero items in every nominal-budget block. Intervals are conditional Monte Carlo mean uncertainty. Source-selected anchors use one fixed fit and selection; copied repeats do not measure anchor-fit uncertainty.

Method	$k = 10$	$k = 30$	$k = 100$
Source prior (0 items)	undefined (20)	undefined (20)	undefined (20)
Random mean	0.572 [0.495, 0.649]	0.730 [0.689, 0.771]	0.874 [0.862, 0.887]
Source difference	0.572 [0.495, 0.649]	0.730 [0.689, 0.771]	0.874 [0.862, 0.887]
Random-subset ridge	0.576 [0.512, 0.640]	0.728 [0.699, 0.757]	0.854 [0.832, 0.876]
Paper AIPW (tuned)	0.528 [0.453, 0.603]	0.722 [0.685, 0.758]	0.860 [0.845, 0.876]
Code AIPW (alpha 10)	0.555 [0.482, 0.627]	0.723 [0.686, 0.760]	0.853 [0.836, 0.870]
Crossfit difference	0.546 [0.476, 0.616]	0.707 [0.670, 0.744]	0.864 [0.852, 0.877]
Crossfit clipped	0.546 [0.476, 0.616]	0.707 [0.670, 0.744]	0.864 [0.852, 0.877]
Half-fit impute (tuned)	0.592 [0.531, 0.654]	0.717 [0.686, 0.749]	0.874 [0.861, 0.888]
Half-fit impute (matched)	0.590 [0.536, 0.644]	0.762 [0.736, 0.789]	0.874 [0.861, 0.888]
Full-fit impute	0.577 [0.515, 0.639]	0.728 [0.700, 0.756]	0.855 [0.834, 0.875]
Correctness-anchor mean	0.725 [0.725, 0.725]	0.858 [0.858, 0.858]	0.916 [0.916, 0.916]
Correctness-anchor ridge	0.691 [0.691, 0.691]	0.831 [0.831, 0.831]	0.831 [0.831, 0.831]
Random MIRT p	0.573 [0.515, 0.631]	0.775 [0.746, 0.804]	0.882 [0.873, 0.892]
Random MIRT gp	0.571 [0.503, 0.639]	0.775 [0.745, 0.805]	0.884 [0.871, 0.898]
IRT-anchor mean	0.705 [0.705, 0.705]	0.722 [0.722, 0.722]	0.888 [0.888, 0.888]
IRT-anchor MIRT p	0.692 [0.692, 0.692]	0.783 [0.783, 0.783]	0.925 [0.925, 0.925]
IRT-anchor MIRT gp	0.698 [0.698, 0.698]	0.796 [0.796, 0.796]	0.912 [0.912, 0.912]

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    Answer: [Yes]

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    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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